# SoFAME (Soil-Forage-Animal Management and Economics): Soil Organic Carbon in Grazing Decisions and Policy Incentives

Yining Wu[1,2,*], Dale Manning[1,2], Yao Zhang[3], Frank Lupi[4], Nathan DeLay[5], Dabit Bista[4]

[1]Department of Agricultural & Resource Economics, University of Tennessee
[2]Baker School of Public Policy, University of Tennessee
[3]Department of Soil and Crop Sciences, Colorado State University
[4]Department of Agricultural, Food, and Resource Economics, Michigan State University
[5]Department of Agricultural and Resource Economics, Colorado State University

[*]Corresponding author: ywu96@utk.edu

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**Abstract.** Economic models of soil organic carbon (SOC) on grazing land rely on soil responses fitted offline and miss the daily feedback among management, forage, animals, and soil. We develop SoFAME (Soil-Forage-Animal Management and Economics), an integrated cow-calf farm model in which a neural surrogate of a biogeochemical process-based model (MEMS) updates each patch's soil and forage daily. After ten years of recursive forecasts, the mean absolute error of the surrogate's SOC is 0.4 percent of the MEMS stock, and the surrogate simulates ten farm years in 80 to 85 seconds, against 3.1 hours with MEMS. We apply SoFAME to the profit maximization problem of a representative Michigan cow-calf farm. Five management histories over the past 100 years give the starting soil states, and for each we simulate 3,315 management systems, each a combination of planned stocking rate, paddock count, grazing period, and feed-gap supplementation. Profit and SOC move in opposite directions up to the most profitable stocking rate, and both fall beyond it. Taking the most profitable management system as the farm's current management, we simulate three policies: an outcome payment per Mg C, and a practice payment and a calf-price premium for rotation. At prices of $44 per Mg C or more, within the observed range, an outcome payment against this baseline moves every farm to a management system with more SOC at the least private cost. More broadly, payments for ecosystem services should be judged against each farm's whole management system and soil history, which SoFAME makes practical.

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## 1. Introduction

Grazing land feeds livestock and stores soil organic carbon (SOC), and the way it is grazed changes both (Briske et al. 2023; Teague et al. 2016). Across 115 studies of improved grassland management and conversion to grassland, SOC rose by 0.54 Mg C per hectare per year on average (Conant, Paustian, and Elliott 2001). Programs now pay cow-calf producers to change how they graze. The Environmental Quality Incentives Program (EQIP) and the Conservation Stewardship Program (CSP) pay per hectare for prescribed grazing (USDA-NRCS 2024b,a), and carbon protocols credit SOC gains against a modeled baseline (Verra 2025; Climate Action Reserve 2022). Grazing labels set standards for qualifying beef (American Grassfed Association 2024; National Audubon Society 2023), and calves from verified programs sell at a premium (Zimmerman et al. 2012; Odde et al. 2019). These designs pay for a practice, a measured outcome, or a label. Which of them changes what a producer does, and how much SOC it buys, depends on how the farm responds.

Economic analyses have represented the soil and forage process in two ways. The older approach writes the process as one or two stocks that follow a few simple equations. In grazing models, forage grows logistically and the herd eats part of it (Noy-Meir 1975; Huffaker and Cooper 1995). Torell, Lyon, and Godfrey (1991) and Ritten et al. (2010) choose the annual stocking rate given a forage state that follows one equation, and Wang et al. (2018) follow two grasses and steer weight each day at constant stocking rates. In soil-capital models, soil is a stock with a shadow value (Burt 1981; McConnell 1983). In Berazneva et al. (2019), SOC changes each year by a residue input less a first-order loss, with two parameters fitted to equilibria of the RothC model, and an estimated yield function links SOC to profit. These models are fast and transparent, but they let a few parameters stand in for the plant and soil processes. The soil models treat a field as one uniform stock, and the grazing models carry no soil.

A second approach runs a process model offline and carries its output into the economic model. In the econometric-process approach, a biophysical model maps management to soil carbon, and an economic model chooses the management (Antle and Capalbo 2001). Antle et al. (2003) used carbon rates from the Century model to compare contracts per hectare and per tonne, and Antle et al. (2007) built supply curves of soil carbon for the central United States. Learned surrogates now emulate process models at low cost (Reichstein et al. 2019; Liu et al. 2024). Nguyen, Nong, and Paustian (2019) trained a neural network on output of the DayCent model to map fixed annual rates of nitrogen, irrigation, and residue removal to SOC after 30 years and to mean corn grain and stover yields, and they used the network to maximize farm profit subject to SOC constraints. This approach brings the process science into the decision, but it has two problems. It fixes the response before the decision problem is solved, at an annual step or for the whole horizon, whereas

on a grazing farm the feedback among management, forage, animals, and soil works at the daily scale. The herd moves to the forage it prefers, hay fills the daily gap between forage and demand, cow condition sets conception, and local grazing changes the soil of each patch. A fixed response averages these feedbacks out. Process models are also slow, so studies vary a few decisions and hold the others fixed, and grazing studies usually vary the stocking rate or the rotation. A cow-calf producer, by contrast, decides jointly on the stocking rate, the paddocks, the grazing period, and how much of the grazing-season feed gap to fill with supplemental hay, and the value of each choice depends on the others.

We ask whether a soil process model can run inside the daily decision loop of a farm cheaply enough to analyze whole management systems and, if it can, which decisions matter for profit and SOC and which payment designs change the management system that a producer chooses.

We develop SoFAME (Soil-Forage-Animal Management and Economics), an integrated model of a cow-calf farm in which a recurrent neural surrogate of the Microbial Efficiency-Matrix Stabilization (MEMS) model, a biogeochemical process-based model of plant and soil carbon and nitrogen (Zhang et al. 2024; Santos et al. 2024), advances the soil and forage state of each of 12 patches every day. Each day, the farm model moves the herd by a calendar rule and by the forage that the animals prefer, and it fills a chosen share of the feed gap, the herd demand minus the grazeable forage, with supplemental hay on pasture. We call this share the feed-gap supplementation. The surrogate receives the local grazing density and the hay of each patch and returns the next soil and forage state, and the forage then sets the energy balance, body weight, and conception of each animal class. On 18 test simulations, each for a combination of site and start history that training never sees, the mean absolute error of total SOC after ten years of recursive forecasts is 0.40 percent of the MEMS stock. In the farm loop, a ten-year simulation of the farm runs 130 to 140 times faster with the surrogate than with MEMS, because a daily loop must restart MEMS, which cannot save and resume its state.

We apply the model to a 60-hectare Michigan farm with five start histories, which come from a century of different grazing simulated in MEMS. For each history, we evaluate 3,315 management systems, every combination of 13 planned stocking rates, 51 paddock designs, and 5 levels of feed-gap supplementation. The planned stocking rate sets the size of the year-round herd, and a paddock design pairs the paddock count with the grazing period. The surrogate runs all 16,575 simulations in 197 CPU-hours, against about 52,000 CPU-hours with MEMS. We then add an outcome payment per Mg C, a practice payment per hectare, and a calf-price premium at observed rates.

The experiment yields four main findings. First, the starting soil, which the past 100 years of grazing set, determines whether a farm gains or loses SOC, while management determines how much. The farm with a history of heavy continuous grazing gains SOC

under every management system, and the ungrazed farm loses SOC under every management system. The history explains 75 percent of the variance of the ten-year SOC change. Second, the decisions interact. The best paddock design depends on the stocking rate, with continuous grazing earning the most at low stocking rates and rotation at high ones. Third, the least costly way to gain SOC changes the whole management system. For the first 150 Mg C on a 60-hectare farm, moving all decisions together gives up $39 of net present value per Mg C, half the cost of lowering the stocking rate alone. Fourth, the qualifying rule of a payment decides what the payment buys. A practice payment buys SOC only when the profit optimum of the farm does not already use the practice. On this farm, the profit optimum of every start history already rotates, so a payment for rotation alone buys no SOC, and a calf premium for rotation raises the stocking rate of three farms. In contrast, at the same budget, either an outcome payment against the farm's own baseline or a practice payment with a stocking cap of 1.6 LSU per hectare moves every farm onto its profit and SOC frontier.

The paper contributes to three literatures. First, it adds a method to the bioeconomic modeling of soil (Antle and Capalbo 2001; Berazneva et al. 2019) by replacing the fitted soil response with a learned emulator of a process model that runs inside the daily decision loop. Herd movement, nutrition, and accounting stay in the economic model, where a researcher can change them. Earlier learned surrogates serve a static optimization (Nguyen, Nong, and Paustian 2019) or a regional estimate (Liu et al. 2024), or they emulate the farm model itself (Shang et al. 2024), whereas ours advances the daily soil state in step with the decisions. Whole-farm simulators such as APEX (Gassman et al. 2010) and GrassGro (Moore, Donnelly, and Freer 1997) also advance soil, pasture, and livestock together each day, but they rely on their own built-in soil and pasture models. SoFAME brings a detailed soil biogeochemical model into the daily loop through its surrogate, fast enough to compare every management system of a grid under several payment designs. Second, it adds to grazing economics by treating management as a system of interacting decisions. While earlier studies vary the stocking rate (Torell, Lyon, and Godfrey 1991; Ritten et al. 2010) or compare rotation with continuous grazing (Briske et al. 2008; Wang et al. 2018), we show that the value of rotation depends on the stocking rate and that the cheapest SOC comes from moving several decisions together. Third, it adds to the design of payments for ecosystem services (Engel, Pagiola, and Wunder 2008; Ferraro and Pattanayak 2006) by showing that additionality depends on where the private optimum lies among the management systems. A payment for a practice that the optimum already uses buys nothing and becomes the windfall that Chabé-Ferret and Subervie (2013) measure for cover crops. A premium on output can lower the SOC that it targets. As the offset literature predicts (Mason and Plantinga 2013), the baseline of an outcome payment decides who is paid.

The analysis has a deliberately narrow scope. It covers one site, one weather path, and ten years. The producer chooses among fixed management rules and does not update the rule as SOC changes, and the surrogate inherits the structure and the errors of MEMS. Because one ten-year simulation runs in less than 90 seconds on one CPU core, however, the same model can extend the analysis to climate scenarios, many sites, and rules that respond to the farm state, and ultimately to the dynamic optimization of grazing with SOC as a state. Section 2 states the producer's problem and the payment incentives, and Section 3 describes the surrogate in the farm loop and the experiment. Section 4 reports the accuracy and speed of the surrogate, Section 5 reports the results, and Section 6 concludes.

## 2. The cow-calf producer's problem and payment incentives

We model the cow-calf producer's ten-year choice of a management system and add three payment designs in order to derive when a payment changes that choice and what the change buys.

### 2.1. The producer's problem

A cow-calf producer keeps beef cows and sells their calves at weaning weight. The farm has $A = 60$ hectares in $K = 12$ patches of $A_k = 5$ hectares, and years $t = 0, \ldots, 9$ run from 2026 to 2035. A start history is the state of the site at the end of 2025 after one of five grazing histories simulated in MEMS. We treat each start history as one farm and drop its index.

At the start, the producer chooses one management system $a = (\mathrm{SR}, N, \mathrm{gp}, \phi)$ from the grid of 3,315 management systems in Section 3 and keeps it for ten years. The planned stocking rate SR is the year-round herd that the producer plans to keep, in livestock units (LSU) per hectare. Because a cow counts as 1 LSU, a heifer as 0.7, and a calf as 0.2, at fixed herd shares of 0.50, 0.10, and 0.40, SR sets the cow target $0.50\,\mathrm{SR}\,A/0.65$. The paddock design pairs the paddock count $N$, where one paddock is continuous grazing, with the grazing period gp, the days that the herd stays in one paddock. The feed-gap supplementation $\phi \in [0, 1]$ is the share of the daily feed gap that the producer fills with supplemental hay on pasture.

In the grazing season, from day of year 120 to 270, the daily feed gap is the herd demand $D_{t,d}$ minus the forage $F_{k_{t,d},t,d-1}$ above a residual of 20 grams of dry matter (DM) per square meter on the occupied patch $k_{t,d}$. Herd demand follows the current body weights, at 15 kilograms of DM per day for each demand-equivalent LSU, $L_{t,d} = D_{t,d}/15$. The producer feeds $h_{t,d} = \phi \max\{0, D_{t,d} - F_{k_{t,d},t,d-1}\}$ of hay on pasture, and the herd eats forage first, then hay. Unmet demand makes the animals lose weight. Outside the grazing season, the herd eats a fixed ration of hay and corn grain off pasture, the same in every management

system.

A simulation runs one management system from one start history. A management system is complete when the herd of its simulation runs the ten years without reaching an animal weight floor, and the set $\mathcal{M}$ contains the complete management systems.

The states are the soil and forage of each patch, $\mathbf{X}_t$, including its SOC, and the count and body weight of each animal class, $\mathbf{n}_t$. Farm SOC, $\mathrm{SOC}_t$, is the mean of the patch values in Mg C per hectare. Given the weather $W_t$, the management system advances both states,

$$\mathbf{X}_{t+1} = G_N(\mathbf{X}_t, \mathbf{n}_t, a, W_t), \qquad \mathbf{n}_{t+1} = \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, a, W_t). \tag{1}$$

Inside $G_N$, the recurrent surrogate of MEMS advances each patch every day.

The producer values a management system by its ten-year net present value (NPV),

$$\mathrm{NPV}(a) = \sum_{t=0}^{9} \delta^t \pi_t(a) + \delta^9 p_{\mathrm{cull}} \sum_g n_g(a)\big[w_g(a) - w_g^5\big], \qquad \delta = \frac{1}{1.05}. \tag{2}$$

The first year is undiscounted, and the terminal herd value prices the weight of each cow cohort $g$ at the end of 2035 above or below its weight at body condition score (BCS) 5, at the cull price $p_{\mathrm{cull}}$. Annual profit is

$$\pi_t(a) = R_{\mathrm{calf},t} + R_{\mathrm{cull},t} - C_{\mathrm{feed},t} - C_{\mathrm{herd},t} - C_{\mathrm{min},t} - C_{\mathrm{labor},t}(N, \mathrm{gp}) - C_{\mathrm{fence}}(N) - C_{\mathrm{repl},t}. \tag{3}$$

Feed cost covers the hay on pasture and the off-season hay and corn, labor cost includes the scheduled moves, and fencing is an annual equivalent.

The profit optimum $a^0 = \arg\max_{a \in \mathcal{M}} \mathrm{NPV}(a)$ is the choice without a payment. Let $\Delta\mathrm{SOC}(a) = \mathrm{SOC}_{10}(a) - \mathrm{SOC}_0$ be the change of farm SOC from 1 January 2026 to 31 December 2035. The least-loss management system is the one in $\mathcal{M}$ with the largest $\Delta$SOC. The profit and SOC frontier is the set of complete management systems for which no other complete management system has both a higher NPV and a larger $\Delta$SOC, and it runs from the profit optimum to the least-loss management system (Figure 4).

The experiment compares a grid of fixed rules that do not respond to SOC or the herd, and SOC left in the soil at the end of year 10 has no value in the objective. Dynamic optimization with SOC as a state is a future use of the platform.

### 2.2. How SOC reaches profit

SOC has no price in equation (2) and changes profit only through the forage, whose quantity and quality can rise with SOC in MEMS. More forage on the occupied patch narrows the feed gap, so the producer buys less hay when $\phi > 0$ and $C_{\mathrm{feed}}$ falls. More and better forage also keeps the cows in condition. Because the BCS of a cow cohort on the

calving date sets its conception rate, $\rho_g = \rho(\mathrm{BCS}_g)$ with $\rho$ increasing, better condition raises the next calf crop and $R_{\mathrm{calf}}$. Feed-gap supplementation buys this condition with hay. On each patch, grazing and the excreta from forage and hay in turn change SOC and forage in later years.

### 2.3. The calendar rule

Under rotation ($N \geq 2$), the herd stays gp days in one paddock and then moves to the next in a fixed order, and within the open paddock it grazes the patch with the highest grazing value. On a grazing day,

$$\text{rest} = (N-1)\,\text{gp}, \qquad u_{k,t,d} = \mathbf{1}\{k = k_{t,d}\}\,\frac{L_{t,d}}{A_k}. \tag{4}$$

Each paddock thus rests $(N-1)$gp days between visits, and the occupied patch carries the whole herd at the local density $u_{k,t,d}$ in LSU per hectare while the other patches carry none. More paddocks force the herd onto more of the patches and so spread grazing across the farm. Under continuous grazing, every patch is open all season, and the grazing value alone selects the patch. Appendix B gives the selection rule, the herd demand, and the surrogate inputs.

### 2.4. Payment incentives

A payment adds to NPV the present value $P(a)$ of a transfer under a contract that covers 2026 to 2035. Profit is computed after the simulation, so a payment does not change the herd or SOC path of a simulation. A payment for SOC uses the annual change of farm SOC, $\Delta_t(a) = \mathrm{SOC}_{t+1}(a) - \mathrm{SOC}_t(a)$, times the 60 hectares.

The outcome payment pays $p$ dollars per Mg C. In the main case, its baseline is the farm's own no-payment path:

$$P^{\mathrm{out}}(a) = p\,S(a), \qquad S(a) = A\sum_{t=0}^{9}\delta^t\big[\Delta_t(a) - \Delta_t(a^0)\big]. \tag{5}$$

The payment is two-sided, and the profit optimum earns zero. This baseline follows VM0042 and the Soil Enrichment Protocol (Verra 2025; Climate Action Reserve 2022), with the no-payment path known exactly. Appendix B.13 defines two contrast cases against the start stock and a threshold variant.

The practice payment pays $r$ dollars per hectare and year to a management system that meets the qualifying rule $\mathcal{Q}$:

$$P^{\mathrm{pr}}(a) = r\,A\,\Gamma\,\mathbf{1}\{a \in \mathcal{Q}\}, \qquad \Gamma = \sum_{t=0}^{9}\delta^t = 8.11. \tag{6}$$

The qualifying rule is rotation, $N \geq 2$, or rotation with a stocking cap SR $\leq \overline{\text{SR}}$ of 2.0 or 1.6 LSU per hectare, after the Prescribed Grazing standard (USDA-NRCS 2017). The calf premium, which the buyer of the calves pays, raises the calf price of a qualifying management system by the share $q$:

$$P^{\text{prem}}(a) = q\,\overline{R}_{\text{calf}}(a)\,\mathbf{1}\{a \in \mathcal{Q}\}, \qquad \overline{R}_{\text{calf}}(a) = \sum_{t=0}^{9} \delta^t R_{\text{calf},t}(a). \tag{7}$$

Its qualifying rule is rotation, rotation with a cap, or the outcome-verified rule $\Delta\text{SOC}(a) \geq \Delta\text{SOC}(a^0)$.

With a payment, the producer chooses

$$a^* = \arg\max_{a \in \mathcal{M}} \text{NPV}(a) + P(a). \tag{8}$$

The additional SOC is $A[\Delta\text{SOC}(a^*) - \Delta\text{SOC}(a^0)]$ in Mg C per farm, and the transfer $P(a^*)$ is the budget cost. The windfall $P(a^0)$, the payment for no change, lies between 0 and $P(a^*)$. The private cost is $\text{NPV}(a^0) - \text{NPV}(a^*)$, and the switch price is the lowest level that moves the farm. The cost per additional Mg C is the mean transfer divided by the mean additional SOC of the five farms. The equal budget is \$29,527 per farm, the present value of a five-year CSP contract at \$108.25 per hectare per year (USDA-NRCS 2024a), and each design takes the one level for all five farms that buys the largest mean additional SOC with a mean transfer within this budget.

### 2.5. Implications for payment design

The propositions below depend only on the NPV and SOC of the complete management systems, and Section 5.4 reports each of them in the experiment.

PROPOSITION 1 (A payment for a practice already in use). *If the profit optimum meets the qualifying rule, a practice payment at any rate leaves the farm at $a^0$, so it adds no SOC and the whole transfer is windfall.*

ARGUMENT. The payment adds the same amount to every qualifying management system, whereas a management system outside $\mathcal{Q}$ gets nothing and has no more NPV than $a^0$. □

PROPOSITION 2 (Switch price of an outcome payment). *Against the farm's own path, the outcome payment moves the farm when $p$ exceeds*

$$\hat{p} = \min_{a \in \mathcal{M}:\, S(a) > 0} \frac{\text{NPV}(a^0) - \text{NPV}(a)}{S(a)}, \tag{9}$$

*the least NPV given up per discounted Mg C gained. No complete management system then beats the new choice on both NPV and S.*

ARGUMENT. Below $\hat{p}$, every management system with $S(a) > 0$ loses more NPV than it earns in payment, and a management system with $S(a) \le 0$ has no more NPV than $a^0$ and earns no positive payment. A management system that another one beats on both NPV and $S$ cannot maximize NPV + $pS$. □

PROPOSITION 3 (A premium for a practice already in use). *If the profit optimum qualifies and the calf premium moves the farm, the new management system earns more discounted calf revenue than $a^0$.*

ARGUMENT. The new management system qualifies, because a management system outside $\Omega$ gets no premium and has less NPV than $a^0$. The inequalities

$$\mathrm{NPV}(a^*) + q\overline{R}_{\mathrm{calf}}(a^*) \ge \mathrm{NPV}(a^0) + q\overline{R}_{\mathrm{calf}}(a^0) \quad \text{and} \quad \mathrm{NPV}(a^*) < \mathrm{NPV}(a^0)$$

then give $\overline{R}_{\mathrm{calf}}(a^*) > \overline{R}_{\mathrm{calf}}(a^0)$. □

The premium thus rewards more calves, and it can lower SOC unless the rule caps the stocking rate.

PROPOSITION 4 (A stocking cap reaches the frontier). *Suppose that the capped rule excludes $a^0$, and let $a^\Omega$ be the complete management system with the highest NPV under the cap. The capped practice payment then either leaves the farm at $a^0$ or moves it to $a^\Omega$. If $a^\Omega$ lies on the profit and SOC frontier, no management system gains as much SOC at a lower private cost, and every design that moves the farm to $a^\Omega$ differs only in its transfer.*

ARGUMENT. Because the payment is the same for every qualifying management system, it selects the qualifying one with the highest NPV, and outside $\Omega$ the best management system is $a^0$. On the frontier, no management system with at least the SOC of $a^\Omega$ has a higher NPV. □

## 3. The surrogate in the farm loop and the experiment

Earlier economic models represent the soil process in one of two ways, by a few equations or by a response fitted offline, whereas in our model the surrogate, the herd, and the accounts exchange state every day for ten years (Figure 1). Table 1 compares the paper with these earlier models.

TABLE 1. **Position against earlier models.** Each column has a predecessor, but no earlier study has all of them. "SOC as a decision state" asks whether the decision problem carries SOC as a state, and "Daily feedback" asks whether decisions and the simulated state exchange information each day. The row for this paper covers both the theory and the experiment. Because the experiment compares a grid of fixed rules, none of its decisions responds to SOC.

| Study | SOC as a decision state | How the soil process enters | Daily feedback from decisions | Animal condition responds to feed | Within-farm grazing pattern | Payment design |
|---|---|---|---|---|---|---|
| Antle and Capalbo (2001), Antle et al. (2003) | No. Century carbon rate per practice change | Century carbon rates by practice and zone, as a lookup table | No. Fall and spring crop choice | No | Across fields and zones only | Per hectare against per tonne, with measurement cost |
| Berazneva et al. (2019) | Yes, optimal control | Annual SOC equation with two parameters fitted to RothC, econometric yield | No, annual | No, maize cropland | No | No. Shadow value against carbon prices |
| Torell, Lyon, and Godfrey (1991), Ritten et al. (2010) | No. Forage state | Forage growth or production function, no soil | No. Annual stocking given the forage state | Partly. Steer gain falls with stocking or forage | No. Homogeneous range | No |
| Ritten, Bastian, and Rashford (2012) | No. Fixed regional credit rate | Regional offset rates, no process or forage model | No decision model | No livestock model | Between four regions | Offsets at fixed rates per hectare |
| Wang, Teague, and Park (2016); Wang et al. (2018) | No. Two-grass biomass state | No soil. Two-grass Lotka-Volterra forage model | Daily forage and grazing loop, fixed calendar rotation | 2016 fixed cow weight. 2018 daily steer gain | 30 paddocks, selective patch and species grazing | No |
| Nguyen, Nong, and Paustian (2019) | No. SOC after 30 years is a constraint in a static profit maximization | One-hidden-layer neural network trained on DayCent. It maps fixed annual rates to SOC after 30 years and mean corn grain and stover yields | No. Constant annual rates of nitrogen, irrigation, and residue removal | No livestock. Irrigated corn | No. Fields and a landscape in eastern Colorado | No. GHG is an objective of a regional planner, not priced |
| Liu et al. (2024) | No decision | Machine learning pretrained on *ecosys*, tuned to observations, cropland | No decisions, daily fluxes | No, cropland | No. Regional 250 m grid | No |
| Shang et al. (2024) | No soil carbon | Neural surrogate of the FarmDyn farm model, no soil | No. Static input-output map | No, arable farms | No | No |
| Zhang et al. (2026) | No decision. SOC per spatial unit | MEMS 3.0 process model, run directly | Daily. Herd moves by forage availability | No. Fixed 500 kg animal units | Habitat-suitability spatial units | No |
| **This paper** | **SOC per patch. A state in the theory, a grid of fixed rules in the experiment** | **Recurrent surrogate of MEMS, full daily state** | **Daily patch choice and hay, calendar rotation** | **Body weight, condition, conception** | **Selective grazing over 12 patches** | **Outcome, practice, premium** |

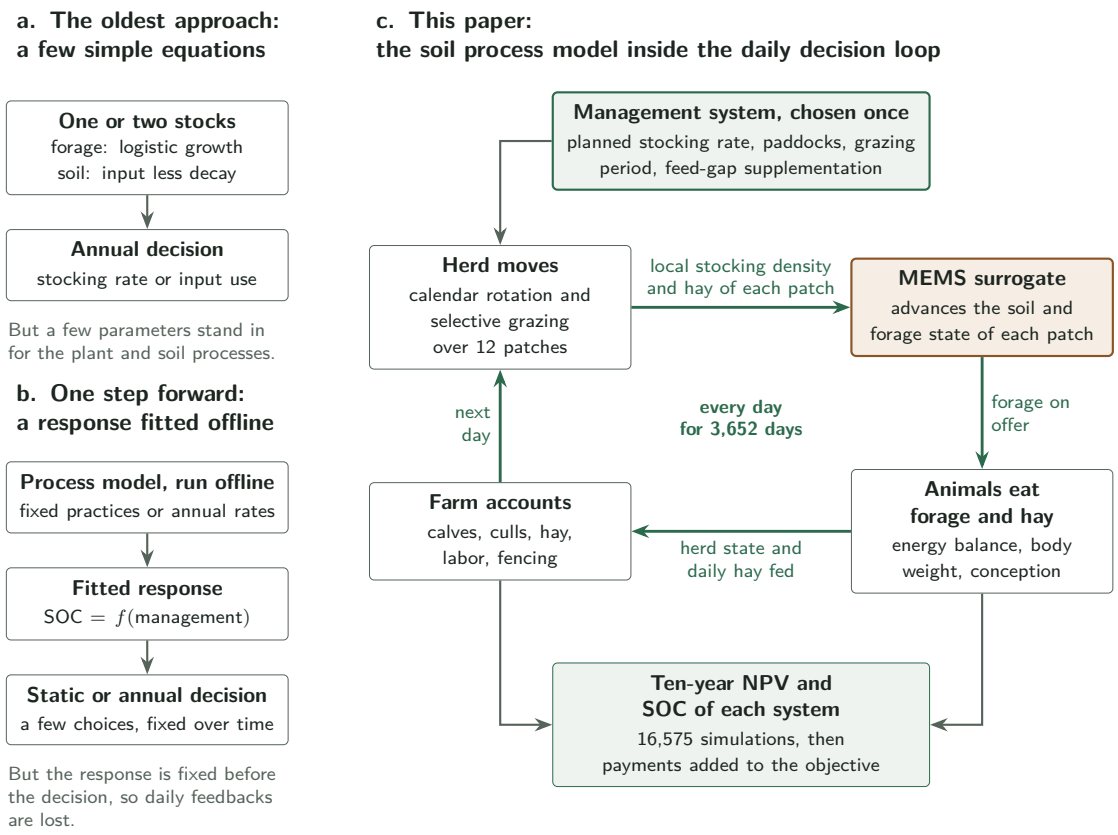



FIGURE 1. **Three ways to represent the soil process in an economic model.** (a) The oldest approach writes forage or soil as one or two stocks with a few simple equations, for example logistic forage growth or an SOC input less a first-order loss (Noy-Meir 1975; Ritten et al. 2010; Berazneva et al. 2019). (b) A later approach runs a process model offline for fixed practices or rates and carries the fitted response into the decision model (Antle et al. 2003; Nguyen, Nong, and Paustian 2019). (c) In this paper, a management system chosen once sets the rules, and the herd, the surrogate of the MEMS soil model, the animals, and the farm accounts exchange state every day for ten years.

### 3.1. Study site, weather, and start states

We demonstrate the model at a single cow-calf site in Michigan, one of the sites of the surrogate training data, whose 16-layer soil profile comes from SSURGO (Soil Survey Staff 2026). The representative farm has 60 hectares, which corresponds in scale to the 50-to-99-head herd class reported for Michigan in the 2022 USDA Census of Agriculture (USDA-NASS 2024), and we divide it into twelve 5-hectare patches. We treat grazing within a patch as uniform, because grazing is far less uneven in paddocks of a few hectares than in large pastures (Barnes et al. 2008). The farm simulation is in-sample for the surrogate, which trained on every MEMS simulation at this site.

The management decade 2026 to 2035 replays the daily gridMET weather of 1996 to

2005 at the site (Abatzoglou 2013), year by year, at a constant atmospheric carbon dioxide concentration of 427 parts per million, which is the forcing of the surrogate training data. Every simulation receives the same forcing, so the comparison is conditional on one common forcing path. Each simulation starts on 1 January 2026 with all twelve patches at the MEMS state of the site at the end of 2025 under one of five management histories (Section 3.3). Restricting the demonstration to one site isolates management effects from between-site variation.

### 3.2. The surrogate

The surrogate emulates MEMS. For one patch, it predicts the complete MEMS state at the end of a day from the state at the end of the previous day, the fixed site characteristics, the weather and calendar of the day, and the management of the day. The state holds 425 values, and its layered variables keep the 16 soil layers of MEMS, although the surrogate does not predict the internal fluxes of MEMS.

A convolutional encoder over the 16 soil layers and an encoder for the other inputs feed a two-layer gated recurrent unit (GRU), a recurrent network layer that carries a learned memory of recent weather, vegetation, soil, and management from one day to the next. The two management inputs of a patch are its local grazing density, in demand-equivalent LSU per hectare, and the hay fed on it. Paddock count, farm geometry, the calendar, and selective grazing remain in the farm model, outside the neural network. On each day of the grazing season, the farm model chooses the occupied patch, sets its hay, and passes the two inputs to the surrogate, which advances all 12 patches. The realized forage and hay then determine the energy balance and the body weights of the next day. Appendix D specifies the surrogate in full, and Appendix C gives the herd, nutrition, movement, and accounting rules.

Training uses 1,667 thirty-year MEMS simulations (2026 to 2055) at 39 sites, 36 of them in Michigan, after five management histories at each site. From 2026, each simulation follows one of 90 synthetic grazing schedules, with daily local densities from 0.5 to 36 LSU per hectare and hay of up to 54 grams of dry matter per square meter per day. These schedules do not come from the farm model of this paper, so the surrogate learns the response of MEMS to the two management inputs without a herd model. The training objective combines the error of the one-day prediction with a penalty for drift over long forecasts and a penalty for persistent level error in seven forage and SOC outputs (Appendix D.3).

Validation tests recursive prediction over the ten-year horizon of the management analysis. The test holds out one start history at each of two sites, a low-pressure rotational history at test site 1 and a history of heavy continuous grazing with supplement at test site 2, while training sees these two sites only under their other histories. Each held-out history has nine MEMS simulations under different synthetic grazing schedules, two

continuous and seven rotational, and we call these 18 simulations the test simulations. For each test simulation, the surrogate starts from the MEMS state at the end of 2025 and then advances from its own predictions for 3,652 days with the recorded weather and management. Because the same test simulations guided the choice of the final training checkpoint, the test accuracy is slightly optimistic. Section 4 reports the comparison, and Appendix D.4 defines the measures.

### 3.3. Management experiment design

We simulate every management system in a finite grid. The planned stocking rate ranges from 0.8 to 3.2 LSU per hectare in increments of 0.2. At each of the 13 stocking rates, continuous grazing is represented by $N = 1$, with all patches eligible throughout the grazing season. Calendar-based rotation uses $N \in \{2, 3, 4\}$ with $\text{gp} \in \{1, 2, 3, 4, 5, 6, 7, 10, 15, 20, 25, 30\}$ days and $N \in \{6, 12\}$ with $\text{gp} \in \{1, 2, 3, 4, 5, 6, 7\}$ days, and rest remains the derived quantity $(N - 1)\text{gp}$. Within the active paddock, equation (C1) applies the same animal selective-grazing rule as under continuous grazing. Each stocking rate therefore has one continuous-grazing design and $3(12) + 2(7) = 50$ calendar-based-rotation designs, for 51 paddock designs. Each paddock design runs at five levels of feed-gap supplementation, $\phi \in \{0, 0.25, 0.5, 0.75, 1\}$, and from five start states. The start states are the MEMS states of the site at the end of 2025 after the five management histories of the surrogate training data (ungrazed, low-pressure continuous grazing, low-pressure rotational grazing, heavy continuous grazing with supplement, and intensive rotational grazing with supplement), so the experiment has $13 \times 51 \times 5 \times 5 = 16{,}575$ simulations.

The farm and financial accounts cover the ten management years 2026 to 2035, and the SOC change compares 1 January 2026 with 31 December 2035. Appendix C.10 defines the feed, energy, and weight calculation. Of the 16,575 simulations, 10,373 complete the ten years, and 6,202 end early when animals reach a weight floor (Appendix E.1). Because each simulation holds management fixed, the profit optimum is the highest-NPV management system in the tested grid, and solving the Bellman problem lies beyond this paper. The speed of the surrogate makes the comparison of the full grid practical and provides a basis for later optimization over mixed discrete and continuous decisions.

### 3.4. Payment experiment

We add the payments of Section 2.4 to the complete management systems after the simulation. The practice payment and the premium use three qualifying rules, rotation ($N \geq 2$) alone and rotation with a planned stocking rate of at most 2.0 or 1.6 LSU per hectare, and the premium also uses an outcome-verified rule. The cap represents the balance of stocking rate and forage supply that the NRCS Prescribed Grazing standard and grazing

labels require (USDA-NRCS 2017; American Grassfed Association 2024; National Audubon Society 2023). The Michigan cost scenarios for prescribed grazing describe the situation before the practice as an inefficient use of forage, with stocking rates that are likely higher than the current level of production (USDA-NRCS 2024c). The buyer of the calves pays the premium, so it is not a public expense.

We evaluate each instrument over a band of observed payment levels. Outcome rates run from $28 per Mg C, the 2024 average price of agricultural carbon credits ($7.66 per t $CO_2e$, Procton 2025), to $128 per Mg C, the highest price that U.S. farmer programs paid, which is $35 "per metric ton of carbon sequestered" in Cargill's RegenConnect (Plastina, Wongpiyabovorn, and Jo 2024). Three other programs in the same review state their price per t $CO_2e$, so we read this ton as one credit of 1 t $CO_2e$. The social cost of carbon is far higher, $190 per t $CO_2$ for 2020 emissions at a 2 percent near-term discount rate, or $697 per Mg C (U.S. Environmental Protection Agency 2023), and we cite it only for comparison. Practice rates run from $76.55 to $175.84 per hectare per year, the Michigan EQIP rates of fiscal year 2025 for Pasture Standard and Pasture Intensive prescribed grazing (USDA-NRCS 2024b). Premiums run from 0.3 to 3.0 percent of the calf price, which expresses the auction premiums of $0.73 to $7.92 per hundredweight in Appendix A as shares of the model calf price of $264 per hundredweight, with 10 percent as a stress test. The Michigan schedule does not state for how many years prescribed grazing is paid, so the ten-year practice contract is an assumption, and a three-year contract would need 2.84 times the rate for the same present value.

For each design and level, we report the number of farms that change management, the additional SOC, the present value of the payment, the payment per additional Mg C, and the windfall share, which is the part of the payment that the farm would receive at its profit optimum.

## 4. The surrogate: accuracy over ten years and speed in the loop

### 4.1. Accuracy over ten years

We score three outputs that carry forage and soil carbon. Plant carbon (C) and nitrogen (N), each the sum of the five MEMS plant pools (live leaf and stem, dead leaf and stem, and seed), are scored on the grazing-season days (day of year 120 to 270) of the ten test years, and the total SOC stock is scored at the end of the tenth year. Table 2 reports the relative error and the mean absolute error (MAE) and compares the MAE with that of the site climatology, which is the mean MEMS value of each day of the year in the training simulations at the test site.

Over the ten years, the relative error is 7.95 percent for plant C and 8.84 percent for plant N, with MAEs of 36.5 kilograms of C and 1.58 kilograms of N per hectare against

TABLE 2. **Surrogate accuracy over ten years and speed in the farm loop.** Panel A compares the surrogate with MEMS on the 18 test simulations. Each forecast starts from the MEMS state at the end of 2025 and advances from its own predictions for ten years. Plant C and N are scored on the grazing-season days of all ten years, and total SOC on the stock at the end of the tenth year. The relative error is the sum of the absolute errors divided by the sum of the absolute MEMS values, and MAE is the mean absolute error. The last column divides the surrogate MAE by the MAE of the site climatology, the mean MEMS value of each day of the year in the training simulations at the test site. For total SOC, the climatology adds the mean change of these simulations to the MEMS stock of the test simulation on 1 January 2026. Panel B gives the time of one ten-year farm simulation on one CPU core and the CPU time of all 16,575 simulations, each including the farm, herd, and animal model. The surrogate advances 51 simulations together on one core, and the MEMS value multiplies the time of one simulation by 16,575.

*A. Ten-year recursive forecast, 18 test simulations*

| Output | Relative error (%) | Ten-year MAE | MAE divided by site-climatology MAE |
|---|---|---|---|
| Total SOC, end of year 10 | 0.40 | 0.48 Mg C $ha^{-1}$ | 0.013 |
| Plant C | 7.95 | 36.5 kg C $ha^{-1}$ | 0.062 |
| Plant N | 8.84 | 1.58 kg N $ha^{-1}$ | 0.070 |

*B. Time in the same farm loop*

| | MEMS in the loop | Surrogate in the loop | Speed factor |
|---|---|---|---|
| One ten-year farm simulation, one CPU core | 3.1 h | 80 to 85 s | 132 to 140 |
| All 16,575 simulations (CPU-hours) | 52,000 | 197 | 262 to 263 |

MEMS means of 459 and 17.9 kilograms per hectare. At the end of the tenth year, the total SOC MAE is 0.48 Mg C $ha^{-1}$ against a mean stock of 119 Mg C $ha^{-1}$, a relative error of 0.40 percent. On all 18 test simulations and for all three outputs, the surrogate is more accurate than the site climatology and than persistence, which holds the MEMS value of 1 January 2026, with skill scores from 0.91 to 0.99 (Appendix Table D1). Because the relative error divides by the whole stock, the more informative measure is its skill against persistence, 0.98, which compares the SOC error with the SOC change in MEMS. The plant errors do not grow with the forecast horizon, and the SOC error stays at 1.3 to 2.3 percent of the site-climatology error in every year (Appendix D.4). Figure 2 shows one test simulation at each test site.

### 4.2. Speed in the farm loop

In the same farm loop, one ten-year simulation of the farm takes 80 to 85 seconds on one CPU core with the surrogate and 3.1 hours with MEMS (Table 2). Both times include the farm, herd, and animal model. Because MEMS cannot save and resume its state, a loop with MEMS must restart MEMS for a patch from 1 January 2020, with all actions so far, on each day on which the patch is grazed or fed hay. One farm simulation therefore restarts MEMS 1,511 times, at about 7 seconds per restart, and this restart coupling is the least costly one that MEMS permits. If the daily actions were known in advance, MEMS would need one simulation for each patch, about 83 seconds for the farm, so the difference comes from the restarts that a daily decision loop requires.

In the experiment, the surrogate advances the 51 paddock designs of one start history, planned stocking rate, and level of feed-gap supplementation together on one CPU core, which halves its time to 42.8 seconds per simulation. All 16,575 simulations took 197 CPU-hours, or 1 hour 42 minutes on a computer cluster, whereas with MEMS they would take about 52,000 CPU-hours.

Over the 10,373 complete simulations, the local grazing density and the hay offer never exceed the training ranges of the surrogate. The crude protein of the plant forage eaten is clipped into its training band on 24.4 percent of the grazing days, and because the clip removes 3.1 times as much nitrogen as it adds, it mostly lowers crude protein above the band. The outcomes inherit the process representation and limitations of MEMS and await independent field validation.

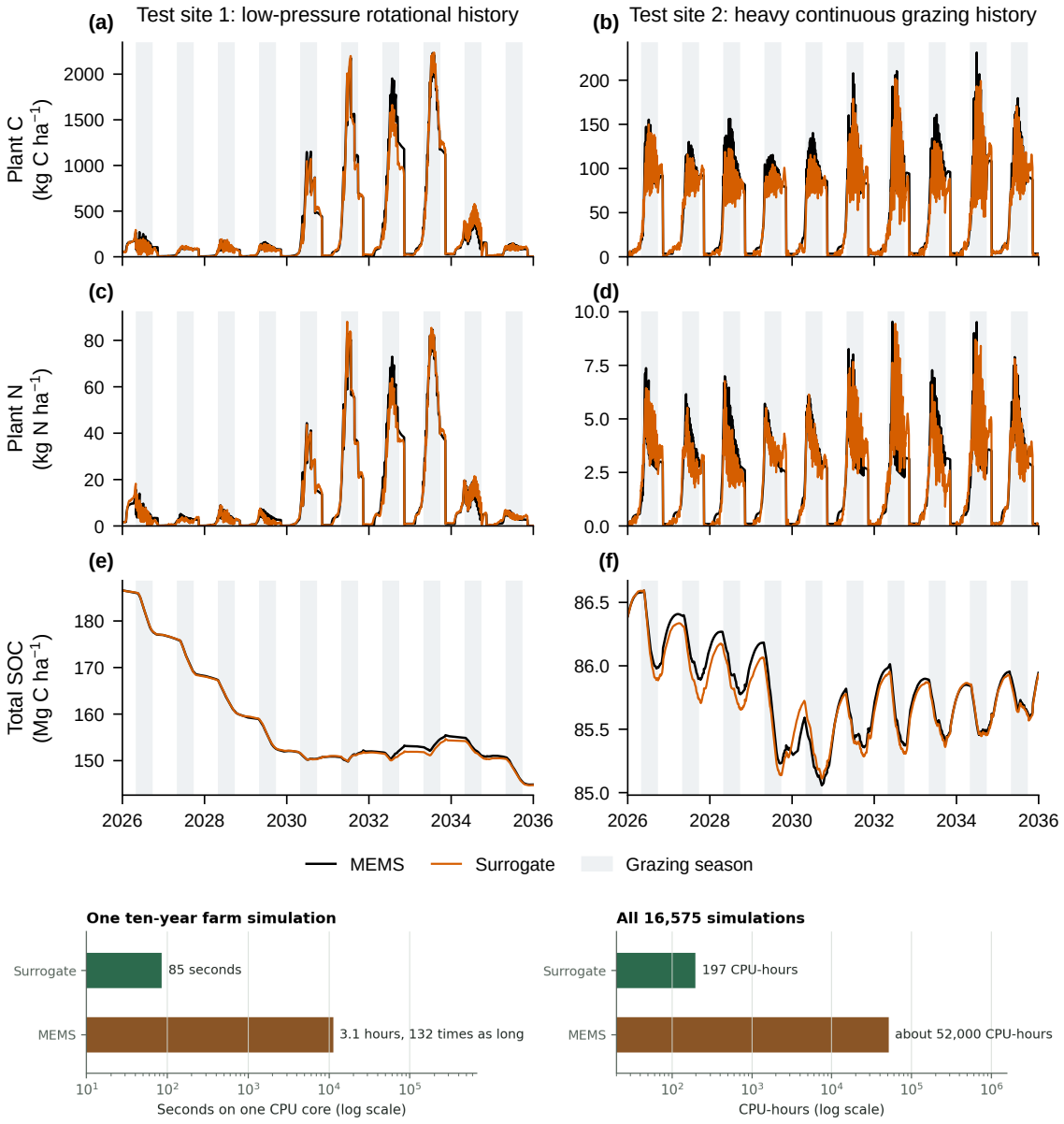



FIGURE 2. **Accuracy of the surrogate over ten years and its speed in the farm loop.** Top: plant C, plant N, and total SOC in MEMS and in the ten-year recursive forecast of the surrogate, for one test simulation (a MEMS simulation held out from training) at each of the two test sites, the one with the median plant C error. Shaded bands mark the grazing season. Bottom left: time of one ten-year farm simulation on one CPU core, with the surrogate and with MEMS in the same farm loop. Bottom right: CPU-hours of all 16,575 simulations.

# 5. Results

All results in this section come from the farm loop with the surrogate. Of the 16,575 simulations, 10,373 (62.6 percent) complete the ten years, and the other 6,202 end early at an animal weight floor (Appendix C.5). Completion falls with the planned stocking rate, from 92.1 percent of the simulations at 0.8 LSU per hectare to 9.7 percent at 3.2 LSU per hectare. Because a simulation that ends early has no ten-year NPV, every outcome below refers to the complete simulations, over which Section 2.1 defines the profit optimum, the least-loss management system, and the profit and SOC frontier of each start history. Appendix E maps which simulations complete (Figure E1) and compares the simulated animal weights with a research herd (Figure E2).

## 5.1. Past grazing decides whether SOC rises or falls, and management decides by how much

All five histories start from one MEMS spin-up near 197 Mg C per hectare in 1926 (Figure 3a). A century of different grazing then leaves them far apart at the end of 2025, from 84.1 Mg C per hectare after heavy continuous grazing through 158.5 after intensive rotational grazing, 196.5 after low-pressure continuous grazing, and 212.7 after low-pressure rotational grazing, to 218.9 without grazing.

Over the ten experiment years, the heavy continuous history gains SOC under every complete management system, by 4.3 to 48.8 Mg C per hectare (Figure 3b), whereas the ungrazed history loses 5.1 to 50.6 Mg C per hectare and the low-pressure rotational history loses 2.6 to 47.0 Mg C per hectare under every complete management system. Only the low-pressure continuous and the intensive rotational histories have complete simulations on both sides of zero. Across the 10,373 complete simulations, the start history explains 75 percent of the variance of the ten-year SOC change, and the planned stocking rate explains 21 percent (Figure 4c).

The soil of each history is still adjusting to its past in 2026, and the path with no grazing and no hay from 2026 shows this adjustment. On that path, SOC rises in every history, by 1.3 Mg C per hectare in the ungrazed history and by 52.6 Mg C per hectare in the heavy continuous history. Every complete simulation ends below the no-grazing path of its history. History therefore sets the path that the soil follows without grazing, and management sets how far below that path the farm ends.

Within a history, management moves the ten-year change over a wide range, 45.5 Mg C per hectare in the ungrazed history. Because the histories start from different stocks, we compare SOC only within a history, and for the same reason a payment that measures SOC against the start stock pays mainly for the history (Section 5.4).

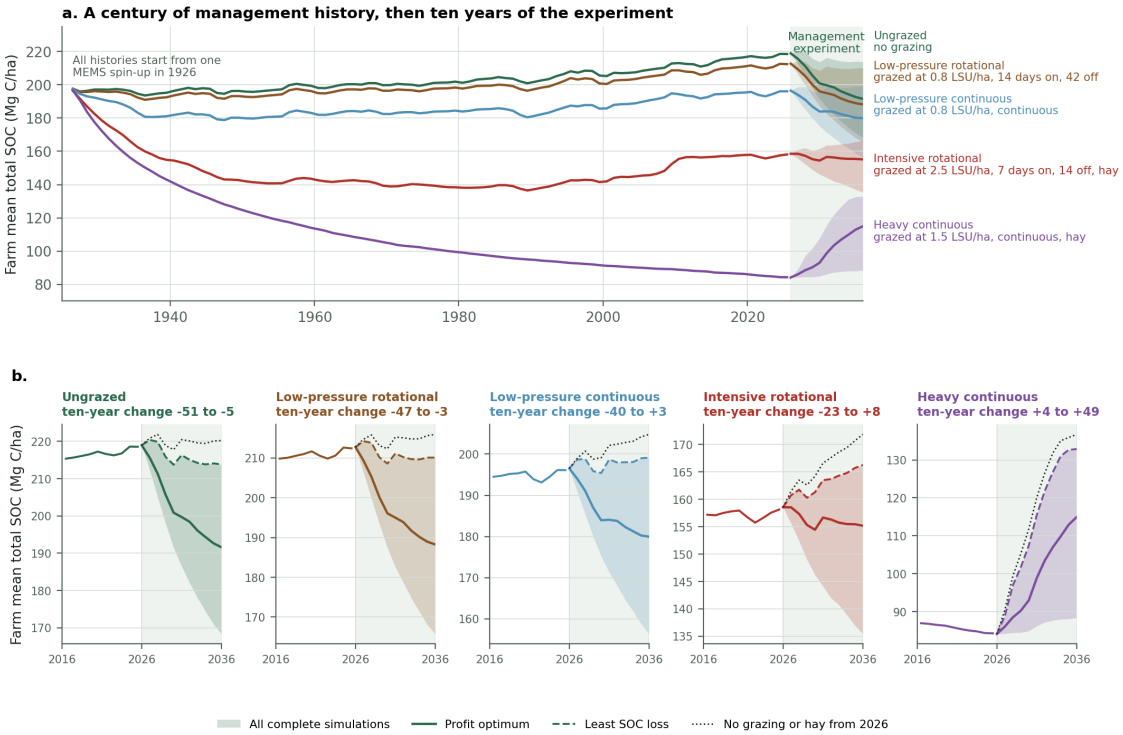



FIGURE 3. **Past grazing decides whether SOC rises or falls, and management decides by how much.** (a) Farm mean total SOC of the five start histories in the MEMS simulations that define them, from the end of the spin-up in 1926 to the end of 2025, and then over the ten experiment years. The rate in each label is the grazing density on pasture of the MEMS schedule that the history repeats every year, not a planned stocking rate. (b) 2016 to 2035 for each history. The band spans all complete simulations. The solid line is the profit optimum, the dashed line the least-loss management system, and the dotted line the path with no grazing and no hay from 2026. Each panel title gives the range of the ten-year SOC change in Mg C per hectare.

## 5.2. The decisions interact

The value of each decision depends on the other decisions, and the best paddock design depends most of all on the planned stocking rate. Figure 4c splits the variance of the ten-year NPV across the complete simulations into Shapley shares, where the Shapley share of a factor is its mean gain in $R^2$ when it joins a regression on indicator variables, over all orders of entry. The planned stocking rate explains 70 percent of the NPV variance, the start history 6 percent, and the feed-gap supplementation and the paddock design 4 percent each. These main effects explain 84 percent together, and the pairwise interactions add 15 percent, which a model that adds up the effect of each decision on its own leaves out.

The interaction shows in the best management system at each stocking rate (Appendix Table E1). Continuous grazing earns the most at low planned stocking rates, and rotation

earns the most at high ones. At 0.8 to 1.4 LSU per hectare, continuous grazing earns more than the best rotation in 15 of the 20 pairs of history and stocking rate. From 1.6 LSU per hectare, the best rotation earns more in every pair in which a continuous-grazing simulation completes, and from 2.6 LSU per hectare its lead is $24,197 to $124,760. The profit optimum of every history uses two paddocks and full feed-gap supplementation ($\phi = 1$), at 2.0 or 2.2 LSU per hectare.

Herd condition links these decisions to profit. Every management system feeds the same off-season ration, which returns the mature cows to BCS 5 before calving, whereas the first-calf cows carry the condition of the grazing season into calving, at a median BCS of 4.40. Between BCS 4 and 5, the conception rate rises by 0.25 per score, against 0.07 per score between BCS 5 and 6 (equation (B4)). Because every open cow is culled, thinner first-calf cows give fewer calves and more cull sales in the next year.

The feed gap carries the stocking rate into herd condition. The planned stocking rate sets most of its size, the paddock design changes where the herd grazes, and the feed-gap supplementation fills a share $\phi$ of the gap with hay on pasture. We match management systems that differ only in $\phi$, in the four histories that complete without hay. At 0.8 and 1.0 LSU per hectare, there is little gap to fill, and the median NPV difference between $\phi = 1$ and $\phi = 0$ is zero. At 2.2 LSU per hectare, $\phi = 1$ costs a median $12,848 of hay over ten years, but it raises the first-calf BCS at calving by 0.52 and the first-calf conception rate by 0.117 and earns $18,586 more NPV.

The paddock design acts on the feed gap through selective grazing. Within the eligible area, the herd occupies the patch with the highest grazing value, which rises with biomass and live-leaf share and favors the patches near water (equation (C1)). Under continuous grazing, the herd can stay on a preferred patch after it grazes the patch down, so the feed gap opens while other patches still hold forage, whereas a calendar rotation moves the herd at the end of each grazing period. In the ungrazed history at 1.0 LSU per hectare, the move barely matters, and continuous grazing leads by $2,867 because the rotation pays more for fencing and labor. At 2.2 LSU per hectare, two paddocks with a four-day grazing period feed 3.0 tonnes of hay per year on pasture, against 8.1 tonnes under continuous grazing, and lead by $7,100 after fencing and labor.

### 5.3. The least costly SOC gains change the whole management system

Profit and SOC move in opposite directions up to the most profitable stocking rate, and both fall beyond it. In every history, the highest NPV at each planned stocking rate rises up to 2.0 or 2.2 LSU per hectare and falls at higher rates, while the mean ten-year SOC change falls at every step from 0.8 to 2.8 LSU per hectare. A farm at its profit optimum can therefore gain SOC only by giving up profit.

SOC is cheapest when all four decisions move together, at half the NPV given up when

the planned stocking rate moves alone. In every history, the profit optimum and the least-loss management system lie far apart, by $60,949 to $88,027 of NPV and 665 to 1,336 Mg C per 60-hectare farm. The least-loss management system uses 0.8 LSU per hectare in every history, with 12 paddocks and a 3 to 5 day grazing period in four histories and three paddocks with a 25-day period in the intensive rotational history. In the ungrazed and both low-pressure histories, the least-loss management system feeds no hay on pasture at any level of $\phi$, so all five levels give the same outcome.

Figure 4a follows the frontier of the low-pressure rotational history from its profit optimum. The first steps keep the stocking rate and change the paddock design, and four paddocks with a five-day period add 120 Mg C for $2,712 of NPV. The stocking rate then falls in steps to 0.8 LSU per hectare while the other decisions change with it. Most management systems on the frontier use a grazing period of seven days or less, and the paddock count rises to twelve at the high-SOC end.

At a high stocking rate, the paddock design alone moves SOC, but only at a high cost in NPV. At 2.2 LSU per hectare with $\phi = 1$, the paddock design moves the ten-year SOC change by 9 to 18 Mg C per hectare, depending on the history, and twelve paddocks with a seven-day period lose the least SOC in every history. The SOC ranking of the paddock designs follows the rest days, because each paddock rests $N - 1$ grazing periods between visits (equation (4)). Twelve paddocks with a seven-day period give 77 days of rest, whereas the profit optimum of the low-pressure rotational history, two paddocks with a 15-day period, gives 15 days.

Twelve paddocks with a seven-day period, however, are costly at 2.2 LSU per hectare. In the low-pressure rotational history, this design earns $54,144, against $109,205 at the profit optimum. Each of its paddocks is one patch, so the calendar holds the whole herd on one 5-hectare patch for seven days, and selective grazing cannot move the herd to more forage. Hay must then fill a larger feed gap, and the first-calf cows calve in poorer condition. At a lower stocking rate the feed gap shrinks and the cost of the design falls, to $15,292 of NPV against the best design at 0.8 LSU per hectare.

The cheapest path therefore combines the decisions. It lowers the stocking rate, which shrinks the feed gap, and it adds paddocks and rest days once the herd no longer runs short of forage. For the first 150 Mg C per farm, moving all decisions together gives up $39 of NPV per Mg C on average over the five histories (Figure 4b), whereas lowering the stocking rate alone gives up $77 per Mg C and changing the paddock design alone gives up $135 per Mg C. Feed-gap supplementation alone never adds 150 Mg C. For the first 300 Mg C, the three costs are $42, $60, and $147 per Mg C.

### 5.4. The qualifying rule decides what a payment buys

A payment buys SOC only when the profit optimum of the farm does not qualify for it. We treat each start history as one 60-hectare farm, which chooses the complete management system with the highest NPV plus the present value of the payment (Section 2.4). The added SOC is the ten-year SOC change of the chosen management system minus that of the profit optimum, times 60 hectares. To compare the designs, we fix a budget of $29,527 per farm, the present value of a five-year Conservation Stewardship Program contract for a prescribed-grazing enhancement (USDA-NRCS 2024a), and each design uses the one level for all five farms that buys the largest mean added SOC within this budget. Table 3 and Figure 5 report the results.

A practice payment for rotation alone buys no SOC. Every profit optimum already uses two paddocks, so every farm qualifies without a change, and no rate up to $300 per hectare per year moves a farm. At the Michigan EQIP rate of $76.55 per hectare per year (USDA-NRCS 2024b), the payment transfers $37,240 per farm and adds no SOC. The whole transfer is windfall, as Proposition 1 states and as Chabé-Ferret and Subervie (2013) measure for cover crops.

This result depends on where the profit optimum lies. In the heavy continuous history, the best management system with continuous grazing earns only $758 less than the profit optimum, so a cost of rotation higher by that amount would make continuous grazing the optimum. A payment for rotation alone would then move the farm, but it would buy little SOC. The move from the best management system with continuous grazing to the best one with rotation, which is the profit optimum, adds 42 to 124 Mg C per farm in four histories. In the intensive rotational history, it lowers SOC by 119 Mg C, because rotation also raises the most profitable stocking rate from 1.8 to 2.0 LSU per hectare.

A calf premium for rotation alone lowers SOC. The profit optimum qualifies, and because the rule sets no limit on the stocking rate, a premium that raises the revenue of every calf rewards a larger herd. From 1.65 percent of the calf price, the low-pressure rotational farm moves from 2.2 to 2.4 LSU per hectare and loses 205 Mg C (Figure 5a), with more calf revenue, as Proposition 3 requires. The ungrazed and low-pressure continuous farms make the same move from 2.85 and 1.95 percent and lose 264 and 317 Mg C. All three rates lie inside the observed band of 0.3 to 3.0 percent of the calf price.

An outcome payment against the farm's own baseline pays nothing at the profit optimum, so it pays only for change. A farm leaves its profit optimum when the price exceeds the least NPV given up per discounted Mg C gained (Proposition 2). On a grid with steps of $2, these switch prices run from $14 per Mg C for the heavy continuous farm to $44 per Mg C for the intensive rotational farm, and the observed prices run from $28 to $128 per Mg C. At the budget level of $72 per Mg C, the five farms move to 1.4 to 2.0 LSU per hectare with two or four paddocks and add 423 Mg C per farm on average. Each chosen

TABLE 3. **Added SOC per farm at the equal budget of $29,527 per farm.**

| Design | Level at the budget | Profit optimum paid | Added SOC (Mg C per farm) |
|---|---|---|---|
| Practice payment for rotation alone | No rate up to $300 per hectare per year moves a farm | Yes | 0 |
| Outcome payment against the farm's own baseline | $72 per Mg C | No | 423 |
| Practice payment for rotation with a 1.6 LSU per hectare cap | $58 per hectare per year | No | 460 |

*Notes:* Each start history is one 60-hectare farm that chooses the complete management system with the highest NPV plus the present value of the payment. Each design uses the one level for all five farms with the largest mean added SOC whose mean present value of payment per farm is within the budget. Added SOC is the ten-year SOC change of the chosen management system minus that of the profit optimum, times 60 hectares, as a mean over the five farms. "Profit optimum paid" shows whether the farm receives the payment without a change. The mean present value of payment per farm is $26,375 for the outcome payment and $28,215 for the capped practice payment. Every dollar of the practice payment for rotation alone is windfall. The budget is the present value of a five-year Conservation Stewardship Program contract for a prescribed-grazing enhancement at $108.25 per hectare per year (USDA-NRCS 2024a).

management system lies on the profit and SOC frontier of its history, so it buys its SOC at the least private cost (Figure 5b).

A practice payment for rotation with a stocking cap of 1.6 LSU per hectare excludes the profit optimum. At $58 per hectare per year, all five farms move to 1.6 LSU per hectare with two paddocks and $\phi = 1$ and add 251 to 580 Mg C, 460 Mg C per farm on average. Each chosen management system also lies on the frontier of its history, so the cap buys its SOC at the least private cost (Proposition 4). Within the same budget, the cap buys 460 Mg C against 423 Mg C for the outcome payment, because the capped payment buys one discrete move and needs to cover only the private cost of that move, whereas the outcome payment pays the same price on every added Mg C, including the Mg C that the farm would add for less.

A baseline at the start stock pays mainly for history. Against the start stock, the heavy continuous farm receives payment for SOC that it gains under every management system, and the two start-stock designs buy 134 Mg C per farm within the budget (Appendix B.13).

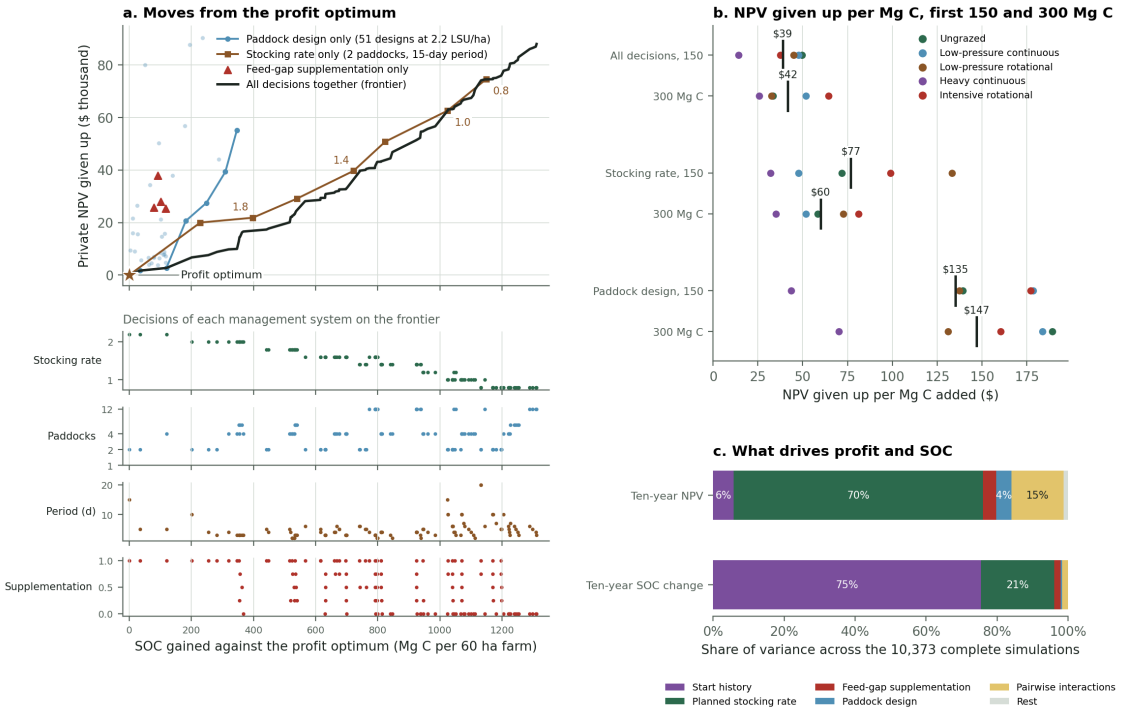



FIGURE 4. **Management is a system: one decision at a time costs more.** (a) Low-pressure rotational history. From the profit optimum, each colored path moves one decision and holds the others fixed. The brown path moves the planned stocking rate (labels in LSU per hectare), the blue path the paddock design over the 51 designs at 2.2 LSU per hectare (the line joins the designs that no other design beats on both NPV and SOC), and the red path the feed-gap supplementation. The black line moves all decisions together and is the profit and SOC frontier, and the four strips under panel (a) give the decisions of each management system on the frontier. (b) NPV given up per Mg C for the first 150 and 300 Mg C per 60-hectare farm, by decision and start history, computed as the smallest NPV loss among the management systems that add at least the target, divided by the target. Vertical bars mark the means over the five histories. Feed-gap supplementation alone never adds 150 Mg C. (c) Shapley shares of the variance of the ten-year NPV and of the ten-year SOC change across the complete simulations, for the start history, the three decisions (paddock count and grazing period together as the paddock design), all pairwise interactions, and the rest.

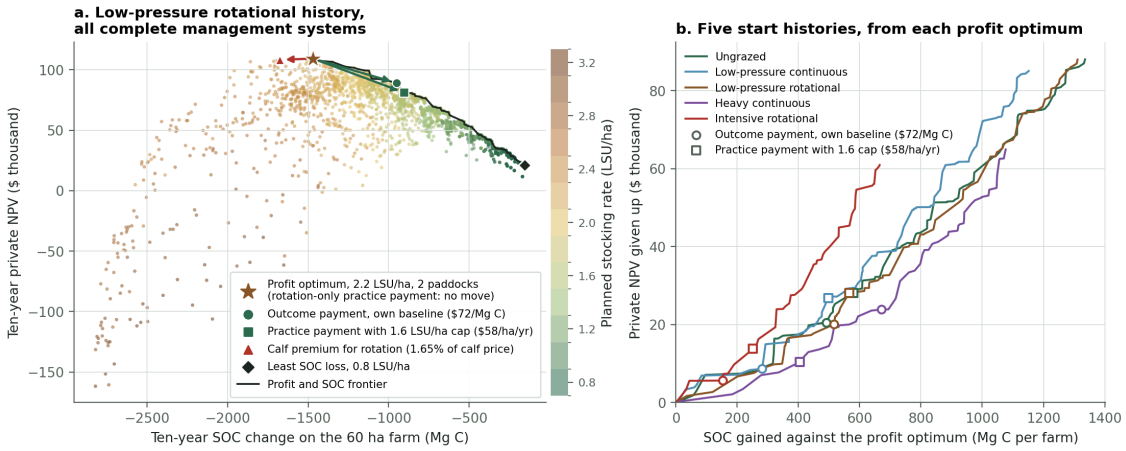



FIGURE 5. **The profit and SOC frontier, and where each payment moves the farm.** (a) Every complete management system of the low-pressure rotational history, colored by planned stocking rate. The black line is the profit and SOC frontier, and the diamond marks the least-loss management system. The outcome payment against the farm's own baseline ($72 per Mg C) and the practice payment for rotation with a 1.6 LSU per hectare cap ($58 per hectare per year) move the farm down the frontier. The practice payment for rotation alone does not move it, and the calf premium for rotation alone (1.65 percent of the calf price) moves it to 2.4 LSU per hectare. (b) The frontier of each start history, measured from its profit optimum, as SOC gained against NPV given up. Circles and squares mark the management systems that the outcome payment and the capped practice payment buy at the equal-budget levels of Table 3.

## 6. Conclusion

We develop SoFAME, an integrated model of a cow-calf farm, and place a learned surrogate of MEMS, a biogeochemical process-based model, inside its daily management loop. The surrogate advances the soil and forage of 12 patches from the local grazing density and the hay of each day, while herd movement, animal nutrition, and the accounts remain in the farm model, open to change by the researcher. On 18 test simulations, each for a combination of site and start history that training never sees, the mean absolute error of its total SOC forecasts after ten years is 0.40 percent of the MEMS stock. In the same farm loop, the surrogate runs a ten-year farm simulation 130 to 140 times faster than MEMS. At this cost, a study can evaluate every management system of a grid, here 3,315 systems under each of five start histories, in 197 CPU-hours, against about 52,000 CPU-hours with MEMS.

Across these management systems, the soil that 100 years of past grazing leave behind determines whether a farm gains or loses SOC over the next decade, and management determines the size of the change. The decisions also interact. Although the planned stocking rate explains 70 percent of the variance of NPV, the best paddock design shifts with it, and the pairwise interactions of the decisions explain another 15 percent. The least costly route to more SOC therefore changes the whole management system, and on a 60-hectare farm it gives up $39 of NPV per Mg C over the first 150 Mg C, half of what a lower stocking rate alone gives up.

For program design, the lessons hold for one site and one weather path, and they assume that the program knows the no-payment choice of each farm. Under these conditions, paying for a practice that the profit optimum already uses buys no SOC. On this farm, the profit optimum of every start history rotates, so a practice payment for rotation alone is a pure windfall, and a calf premium for rotation alone, within the observed range of 0.3 to 3.0 percent of the calf price, raises the stocking rate of the ungrazed farm and the two low-pressure farms and adds to their SOC loss. If continuous grazing were the profit optimum, the same payment would move the farm, but the move from the best management system with continuous grazing to the best one with rotation adds at most 124 Mg C per farm and lowers SOC in the intensive rotational history. The SOC goal must therefore enter the qualifying rule, because no payment rate moves a farm whose profit optimum already qualifies.

A stocking cap turns a practice payment into a contract that binds. At the budget of $29,527 per farm, a practice payment with a cap of 1.6 LSU per hectare buys 460 Mg C per farm, against 423 Mg C for an outcome payment against the farm's own baseline. Both designs move every farm onto its profit and SOC frontier and so buy their SOC at the least private cost. The cap buys more within the budget because it pays for one discrete move, while the outcome payment pays the same price on every added Mg C. To set such a cap,

however, a program must know where the profit optimum lies. The choice of baseline for an outcome payment, in turn, determines who receives the money. Against the start stock, most of the money goes to the start history that gains SOC under any management, the self-selection that the additionality literature predicts (Ferraro and Pattanayak 2006; Chabé-Ferret and Subervie 2013). The payments of the experiment run for ten years and are stewardship payments, not offsets that meet a permanence rule (Climate Action Reserve 2022; Integrity Council for the Voluntary Carbon Market 2025).

These results come from one Michigan site, one weather path, five start histories, and ten years. The cow-calf producer chooses among fixed rules and does not change a rule as SOC changes, so the profit optimum is the best rule in the grid rather than the solution of the dynamic problem. The rankings count private value and exclude the social value of carbon, and the experiment has no transaction, measurement, or verification costs and no SOC loss after a contract ends. The surrogate also inherits the structure of MEMS and has a known bias at high stocking rates. In a check that reruns 1,095 farm simulations of an earlier specification of the experiment in MEMS with the same daily actions, the surrogate's farm SOC at the end of 2035 exceeds the MEMS value by a median of 3.9 Mg C per hectare at 2.0 to 2.2 LSU per hectare, 2.3 at 1.6 to 1.8, and −0.3 at 0.8 to 1.0. The surrogate therefore understates the SOC gain of a move to a lower stocking rate. We expect this bias to make the switch prices too high, but we have not tested it. Appendix E.9 lists the limits of the animal, grazing, and payment models.

At less than 90 seconds per ten-year simulation on one CPU core, the same model can address questions that a fitted response cannot. Evaluating the management systems across sites, soils, and climate scenarios would show where the rankings hold and would give a program the heterogeneity that it needs to target payments (Stanley et al. 2026). The fixed rules can give way to rules that respond to the farm state, such as a rule that moves the herd when the occupied patch has no grazeable forage or a rule that sets the stocking rate each year from the forage. Within the conditions on which the surrogate was trained, the model can also support the dynamic optimization of grazing with SOC as a state. It can likewise test payment designs under the frictions of a real program, including a baseline estimated with error, the costs of measurement and verification, and contracts long enough to meet a permanence rule. The structure is not specific to grazing, and a farm decision model that needs the daily state of a soil process model, for cover crops, nitrogen, or irrigation, can use it once a surrogate is trained for that model and those decisions.

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## Appendix A. Related literature and program details

This appendix gives the literature and the program details behind Section 1.

Soil organic carbon is the slow-moving stock at the center of the grazing system. It changes little from one year to the next, yet over a decade it governs both the forage a pasture can grow (Lal 2004; Stanley et al. 2024) and the climate benefit a grazing operation can claim (Conant, Paustian, and Elliott 2001; Teague et al. 2016). The grazing case differs from the cropland settings of the first soil-as-capital models in the mechanism that links carbon to net value. In cropping, soil organic carbon raises crop yield and revenue through an agronomic response. In grazing the link occurs through SOC's impact on forage quantity and quality and, in the present model, on the hay that fills the feed gap on pasture and on the body condition of the animals, which sets the calf crop. Since hay substitutes economically for pasture-supplied energy, the value of SOC depends on its ability to produce more forage within a season, reduce the annual feed bill, and keep the herd in condition.

Treating soil as a depletable capital stock whose value should enter farm-level decisions began with Burt (1981) and McConnell (1983), the latter deriving the shadow value of soil depth. In discrete time, with SOC as the stock, the steady-state shadow value $\lambda$ satisfies $\lambda^{ss} = (\partial\pi/\partial\mathrm{SOC})/(1 - \delta\,\partial\mathrm{SOC}_{t+1}/\partial\mathrm{SOC}_t)$. We follow Berazneva et al. (2019), who paired an annual SOC equation, with two parameters fitted to equilibria of the Rothamsted Carbon Model, and an econometric maize-yield function for smallholder Kenyan farms and solved the steady-state optimality conditions simultaneously for shadow values of $95 to $168 per Mg C. We share their state variable but not their solution strategy, because we compare fixed management systems over ten years instead of solving the steady-state conditions. The production system also differs. Their carbon-to-revenue link is the direct response of crop yield to SOC, whereas ours runs indirectly through forage quantity and quality and through the annual stored-feed bill that more abundant forage reduces.

Pairing a calibrated biophysical model with an estimated economic one, as in Berazneva et al. (2019), is an instance of the Econometric-Process Model (EPM) strategy of Antle and Capalbo (2001), in which a biophysical model maps management to yields and stocks and an economic model chooses management in response. Antle et al. (2003) used this strategy to design carbon-sequestration contracts, and Antle et al. (2007) applied it to the central United States to estimate agricultural carbon supply curves. Our coupling of a MEMS surrogate with a comparison of stocking and rotation rules adapts this EPM strategy to a process-based grassland model.

Grazing land (or grassland) soil carbon accumulates differently from cropland. It enters through root turnover and the manure the herd returns to the pasture, and because pasture is perennial and untilled it avoids the tillage that accelerates cropland carbon loss (Stanley et al. 2024; Teague et al. 2016). That improved management can build the SOC

stock is well documented: Conant, Paustian, and Elliott (2001) synthesized 115 studies that found a mean gain of 0.54 Mg C per hectare per year. We use this estimate as empirical context for the single-site model. Lal (2004) listed improved grazing among the strategies that increase the soil carbon pool. In cropland settings, soil-health practices capitalize into farmland prices (Chen et al. 2023, 2026) and build resilience to climate stress (Aglasan et al. 2024), and Chen et al. (2025) solve a cover-crop adoption problem with soil fertility as the state variable, the closest cropland analog to ours and a method we draw on.

A parallel grazing-bioeconomics literature supplies the dynamic structure. Grazing systems can exhibit nonmonotonic responses: Noy-Meir (1975) (generalized by May (1977)) showed that moderate stocking admits both a productive and a degraded stable state separated by a threshold across which a small stocking change can tip the system into collapse. Huffaker and Cooper (1995) separated rangeland into a fast forage process and a slow successional one and formalized the split as a slow-manifold decomposition, a structure we keep with SOC as the slow stock, while Huffaker and Wilen (1991) cast within-season forage decline as a fishery-style control problem. Whether the slow stock matters is what divides long-horizon from short-horizon analyses. Torell, Lyon, and Godfrey (1991) found the two nearly identical, because the falling weight gain of the steers stops the stocking before the range is damaged, and Ritten et al. (2010) solved the annual stocking problem by stochastic dynamic programming, with logistic forage growth whose rate depends on precipitation. Ritten, Bastian, and Rashford (2012) found carbon payments at prevailing prices well below the management costs they were meant to cover. We therefore track production and SOC within one dynamic system and exclude standalone carbon revenue from the private objective. The value of rotation itself is contested: Wang et al. (2018) found that multi-paddock rotation raises ranch net present value and Bosch et al. (2008) report ranch-scale gains, whereas Briske et al. (2008) found no advantage in controlled plot-scale experiments, and Derner et al. (2021) found that adaptive management recovered part of the steer gain lost at high stocking density in a multipaddock rotation, but not enough to cover its cost. Those experiments are too short to resolve the decade-scale SOC feedbacks that our simulation captures, and Section 5.2 and Appendix E.5 return to that debate.

Grazing further differs from cropping because cattle use space unevenly within a farm. By GPS tracking, Gersie, Augustine, and Derner (2019) found that 4 to 16 percent of a pasture went ungrazed while a small fraction received more than four times the average use, the pattern sharpening as forage quality differentiated through the season, and Adler, Raff, and Lauenroth (2001) showed that such preferential use reshapes vegetation into a mosaic of heavily used lawns and lightly used roughs that carries through to the soil. Zhang et al. (2026) embedded explicit livestock movement in MEMS to capture this within-pasture variation, the closest modeling predecessor to our patch-based representation. Our focus

on within-farm heterogeneity complements the between-site heterogeneity Antle et al. (2003) used for contract design. A representative-paddock model cannot express this spatial pattern, so we represent the farm as patches grazed preferentially and let SOC diverge across them under management.

The optimization template comes from renewable-resource theory. Gordon (1954) showed that open access drives a stock below the social optimum, Schaefer (1954) supplied the logistic growth backbone, and Clark (1976) synthesized fishery economics into the modified golden rule that is the direct ancestor of our shadow-value formula, with Smith (1969) adding the effort dynamics that parallel herd expansion and contraction. Homans and Wilen (1997) showed that a season-length catch limit lets fishing capacity expand until rents dissipate, which maps onto practice-based grazing payments that may invite overstocking within a rotation. Our within-season problem chooses the grazing period conditional on an installed paddock design. The resulting rest interval is nested inside the across-year Bellman equation for SOC. The daily dynamics of the within-season problem resemble the drawdown of fish stocks in Homans and Wilen (1997), with the farmer as the resource manager and with multiple spatially distinct drawdown periods per year.

Process-based models take a long time to run. Scaling this single-site analysis to many sites and repeatedly evaluating management therefore motivates a surrogate trained on process-model output. Reichstein et al. (2019) argued that hybrid mechanistic-and-learning models suit Earth-system problems, and Liu et al. (2024) realized this for cropland carbon, training networks on the *ecosys* model to $R^2 > 0.9$ against its output. Such surrogates can be trained from process simulations (Lu and Ricciuto 2019) and serve as a bridge from mechanistic models to economic integrated assessment (Blanc 2017; Franke et al. 2020). On the economic side, Shang et al. (2024) built a deep-learning surrogate of a farm optimization model. Our structured recurrent surrogate emulates the complete daily MEMS state and accepts two management inputs directly: local LSU per hectare and the hay fed on the patch. This design lets the same farm-management and animal modules evaluate thousands of management systems without explicitly constructing MEMS's internal action fluxes.

We also use the framework to compare three payments for SOC: an outcome payment per Mg C, a practice payment per hectare, and a premium on the calf price for qualifying management. This application connects to the literature on payment design for ecosystem services. Antle et al. (2003) found per-hectare (practice-based) carbon contracts to be as much as five times more costly than per-tonne (outcome-based) contracts in the dryland Northern Great Plains, with the gap driven by spatial heterogeneity in soil carbon rates across sub-regions and in the opportunity cost of changing practices across fields. Rouhi Rad et al. (2026) compare an incremental abatement subsidy with land retirement for groundwater conservation. The subsidy has the lower budgetary cost

for small reductions in water use and the higher cost for large reductions, because a high uniform threshold also pays users who already use less water. Their subsidy pays for use below a threshold, and a threshold set for each well from its own past use tends to give the lowest cost to the policymaker. Engel, Pagiola, and Wunder (2008) laid out the three dimensions of payment for environmental services design: conditionality on outcomes, additionality of conservation, and targeting of payments to high-impact sites. Melkonyan and Taylor (2013) compared input regulation, cost-sharing, and performance regulation for public rangelands and found performance regulation theoretically efficient but operationally costly. The additionality problem received its sharpest formulation in Ferraro and Pattanayak (2006) ("money for nothing") and was extended by Pates and Hendricks (2020) to dynamic settings where technology diffusion erodes additionality over time. Wongpiyabovorn, Plastina, and Crespi (2023) documented the practical barriers to outcome-based payments (fixed certification costs of $5,000 or more and ongoing verification expenses) that may exclude small cow-calf operations regardless of the theoretical efficiency advantage. Wang, Isengildina-Massa, and Stewart (2023) found that spending on food away from home is the main driver of U.S. retail premiums for grass-fed beef and that media information on climate change has no significant effect. This pattern complicates the use of market premiums as ecosystem-service payments. These retail premiums also differ from the premiums that a cow-calf producer receives for weaned calves. At video auctions, in the years with a significant premium, calves eligible for natural programs earned $0.73 to $1.09 per hundredweight more, age-and-source-verified calves $0.99 to $3.23, and calves of the non-hormone-treated cattle program $1.02 to $4.04. Organic calves earned $7.92 per hundredweight more in 2017 (Zimmerman et al. 2012; Odde et al. 2019; Saitone 2018). We found no auction estimate of a calf premium for a grazing standard.

The baseline is the central design choice of an outcome payment. Offset protocols for agricultural soils take two approaches. Verra VM0042 and the Soil Enrichment Protocol of the Climate Action Reserve model the business-as-usual counterfactual with the weather of the project and credit avoided loss (Verra 2025; Climate Action Reserve 2022). The Australian soil carbon method of 2021 and the Gold Standard methodology credit only a gain above the stock at the start of the project (Commonwealth of Australia 2021; Gold Standard 2020). The difference matters for land that loses SOC: a relative SOC gain often comes from a reduction of losses rather than an increase in stocks (Sanderman and Baldock 2010). Mason and Plantinga (2013) write an offset contract as a per-unit payment plus a lump-sum transfer, which separates the incentive from the payment to landowners who would supply carbon without it. Payments to such landowners are large in practice: a French subsidy for cover crops added 10 hectares of cover crops on the average recipient farm and paid for almost 7 hectares that would have been covered without it (Chabé-Ferret and Subervie 2013).

These literatures have largely developed separately. Bioeconomic SOC models focus on cropland, grazing models generally omit detailed SOC dynamics, and grazing-land biogeochemistry remains largely absent from existing surrogate applications to crop-process and economic models. Studies of carbon-payment design likewise concentrate on cropland. One contribution of this paper is to integrate these strands in a coupled private-profit management framework that treats SOC as productive natural capital while also reporting it as a separate environmental outcome.

# Appendix B. Model details

This appendix states the full farm model behind Section 2. Appendix B.1 writes the dynamic problem of a forward-looking cow-calf producer, and Appendix B.9 derives the shadow value of SOC that it implies. The experiment restricts the producer to fixed rules, and it neither solves this problem nor computes the shadow value.

The framework runs from the whole farm to the local patch and back. At the whole-farm level, the producer selects a planned annual herd target, a calendar grazing rule conditional on installed infrastructure, and a feed-gap supplementation, the fraction of the daily feed gap on pasture that the producer fills with hay. The herd controller converts the herd target into realized cow, heifer, and calf counts. The animal demand schedule converts those counts and the current body weights into a daily feed target and a demand-equivalent herd input. The calendar restricts the herd to a management paddock, animal selectivity determines the patch actually used, and the resulting local grazing pressure and hay drive daily forage and soil transitions. The realized forage and hay set the energy balance of each animal class and therefore its body weight on the next day. At weaning and the year boundary, the herd-composition transition promotes animals between classes, records sales and replacements, and sets the next year's calf crop from cow body condition at calving. Aggregating the daily transitions and herd outcomes yields annual SOC, feed cost, and profit for the producer's dynamic problem.

## B.1. Whole-farm management and dynamic decision problem

Years are indexed by $t$ and calendar days within a year by $d$. Let $\mathcal{C} = \{\text{cow}, \text{heifer}, \text{calf}\}$ index animal classes. Let $\mathbf{X}_t$ collect the complete farm biophysical state at the start of year $t$, including the spatial distributions of vegetation, soil water and temperature, nutrients, surface organic matter, and SOC. Here $X_{k,t,d}$ denotes the vector state of patch $k$ at the end of day $d$. Let $\mathbf{n}_t = (n_{c,t}, w_{c,t})_{c\in\mathcal{C}}$ collect the opening herd state: the animal count $n_{c,t}$ and the representative body weight $w_{c,t}$ of each class. Finally, let $W_t = (w_{t,d})_d$ denote the year's exogenous daily forcing path, including weather, atmospheric carbon dioxide, and calendar-indexed inputs. Farm area $A$, the grazing-season start and end dates $d_{\text{on}}$ and $d_{\text{off}}$, and the herd-demographic parameters are fixed features of the production setting. The producer's annual operating controls are the planned whole-farm herd target $H_t$, measured in 500-kilogram reference LSU, the feed-gap supplementation $\phi_t \in [0, 1]$, and, for rotational designs, the calendar grazing-period length $\text{gp}_t$, measured in days spent in each management paddock before advancing. Because area is fixed, planned stocking rate is $\text{SR}_t = H_t/A$. This planned target differs from the demand-equivalent daily MEMS input defined below. The feed-gap supplementation is fixed for the year, like the herd target, and Section B.6 defines the daily hay that it implies. When $N = 1$, the timing control

is inactive and the maximization below is only over $H_t$ and $\phi_t$.

The number of management paddocks $N$ is a discrete, long-lived design choice selected from feasible set $\mathcal{N}$. Changing $N$ requires fencing, which precludes costless annual adjustment. The operating problem is therefore solved conditional on installed $N$, fence investment is annualized in the farm account, and designs are compared separately. The exogenous forcing affects forage and soil transitions and, through the stored-feed requirement, current profit. Let $\mathcal{W}_{t:\infty} = (W_t, W_{t+1}, \ldots)$ be a specified forcing sequence and let $\delta \in (0,1)$ be the annual discount factor. Conditional on that sequence, the producer's time-dependent operating value is

$$
\begin{aligned}
V_{N,t}(\mathbf{X}_t, \mathbf{n}_t; \mathcal{W}_{t:\infty}) = \max_{H_t,\, \mathrm{gp}_t,\, \phi_t} \Big\{ & \\
& \pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t) \\
& + \delta V_{N,t+1}\Big( G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \\
& \qquad \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t); \mathcal{W}_{t+1:\infty} \Big) \Big\}.
\end{aligned} \tag{B1}
$$

The preferred installed design is $N^* \in \arg\max_{N \in \mathcal{N}} V_{N,0}(\mathbf{X}_0, \mathbf{n}_0; \mathcal{W}_{0:\infty})$, with annualized fencing costs included in $\pi_N$. The function $G_N$ advances the biophysical state and $\mathcal{H}$ advances the herd state. The herd transition depends on the pasture and the management because forage and hay set body weights, and cow condition sets the next calf crop. This formulation evaluates management conditional on a deterministic forcing scenario, exactly as in the numerical replay below. Under uncertain forcing, the same equation is written with an expectation conditional on the producer's information when $H_t$, $\mathrm{gp}_t$, and $\phi_t$ are chosen. The following subsections specify the mechanisms summarized by $G_N$, $\mathcal{H}$, and $\pi_N$.

### B.2. Herd composition and transition model

The planned herd target is expressed through class shares. These shares construct target class counts. Observed opening proportions may differ, and the demographic controller does not reimpose the shares each day or year. Let $\alpha_c$ be the planned share of class $c \in \mathcal{C}$, let $\ell_c$ be its representative LSU coefficient, and let $\bar{\ell} = \sum_{c \in \mathcal{C}} \alpha_c \ell_c$. The planned class target associated with $H_t$ is

$$
\widetilde{n}_{c,t}(H_t) = \alpha_c \frac{H_t}{\bar{\ell}}. \tag{B2}
$$

We use $\boldsymbol{\alpha} = (0.50, 0.10, 0.40)$ and $\boldsymbol{\ell} = (1.00, 0.70, 0.20)$, ordered as cows, replacement heifers, and the annual calf cohort. The vector $\widetilde{\mathbf{n}}_t$ gives the planned management target. Realized

herd composition evolves through the demographic controller. At numerical initialization, the planned cow target is translated into attrition-consistent opening cow and heifer counts, while the calf cohort remains pending until the calving date. Within year $t$, $n_{c,t,d}$ denotes the realized count of class $c$ on day $d$ after any scheduled demographic event.

The demographic balance for each class can be written generally as

$$n_{c,t+1} = (1 - \mu_c) n_{c,t}^{\text{end}} + b_{c,t} + p_{c,t}^{\text{buy}} - v_{c,t}^{\text{sale}} - q_{c,t}^{\text{cull}} + \sum_{c' \neq c} \tau_{c' \to c,t} - \sum_{c' \neq c} \tau_{c \to c',t}. \tag{B3}$$

Here $n_{c,t}^{\text{end}}$ is the class count immediately before year-boundary flows, $\mu_c$ is mortality, $b$ denotes births, $p^{\text{buy}}$ purchases, $v^{\text{sale}}$ sales, $q^{\text{cull}}$ culls, and $\tau$ promotions between classes. In the implemented three-class rule, pending calves enter the live class at calving. At weaning, surviving calves are split by sex. Male calves and female calves not needed for replacement are sold, existing heifers are promoted to cows, and retained female calves become the new heifer cohort. Replacement heifers are purchased only if retained females cannot fill the cohort needed to return the herd toward its target. The cow class holds two cohorts with their own counts and body weights. The promoted heifers form the first-calf cohort until their first calf is weaned, and they then join the mature cohort. On the calving date, the cows of cohort $g$ conceive at the rate

$$\rho_{g,t} = \rho\big(\text{BCS}_{g,t}\big), \tag{B4}$$

where $\text{BCS}_{g,t}$ is the body condition score of the cohort on the calving date of year $t$, a linear function of its body weight, and $\rho$ is an increasing function. At the year boundary, mortality applies and every open cow is culled. When fewer than 12 percent of the cows are open, pregnant cows are also culled up to 12 percent. The cows that remain are all pregnant, so they set the next pending calf cohort. A herd that loses condition before calving therefore produces fewer calves in the next year and sells more cull cows. These rules define $\mathcal{H}$, supply sales and replacement flows to annual accounting, and pass $\mathbf{n}_{t+1}$ to the next annual cycle. The same realized class counts and body weights drive animal demand and farm accounting. All management systems use the same demographic rates and event dates. Although equation (B1) permits annual retargeting of $H_t$, the reported numerical comparison holds $H_t \equiv H$ within each management system.

### B.3. Calendar movement rule

Let the farm contain $K$ biophysical patches indexed by $k$, with areas $A_k$ satisfying $\sum_k A_k = A$. The installed design partitions those patches into $N$ non-overlapping management

paddocks,

$$\mathcal{G}(N) = \{\mathcal{P}_1, \ldots, \mathcal{P}_N\}, \qquad \bigcup_{j=1}^{N} \mathcal{P}_j = \{1, \ldots, K\}.$$

The biophysical patches are the units at which local state is simulated, and management paddocks are groups of patches used by the farm calendar.

The fixed grazing season has length $L_g = d_{\text{off}} - d_{\text{on}} + 1$. Under equal-duration sequential rotation, the full calendar cycle and the scheduled rest interval of a management paddock are

$$c_t = N\,\text{gp}_t, \qquad r_t = (N-1)\,\text{gp}_t. \tag{B5}$$

The producer's primitive timing decision is grazing-period length. Once $N$ and $\text{gp}_t$ are set, the calendar determines rest. Farm area and paddock count alone are insufficient to determine a duration. Grazing-season dates remain fixed throughout the analysis.

Let $j_{t,0}$ denote the first management paddock used in year $t$. During the grazing season, the scheduler holds the herd in one management paddock for $\text{gp}_t$ days and then advances sequentially to the next. The first paddock may rotate across years to distribute incomplete end-of-season cycles, but residual forage is an outcome and never triggers a management-paddock advance. When $N = 1$, the calendar places no within-farm restriction on the herd. Appendix B.14.1 gives the exact day-index rule and boundary bookkeeping.

### B.4. Animal demand schedule

The animal demand schedule converts the realized class counts and the current body weights into a daily feed target. Let $w_{c,t,d}$ be the representative body weight of class $c$ at the start of day $d$, and let $D_{c,t,d}$ be the dry matter per head that the class can eat on that day. For cows and heifers, $D_{c,t,d}$ is the dry-matter intake capacity at the current weight, physiological stage, and air temperature. For calves, it is the allowance of solid feed at the current age and weight, which the calf eats in addition to milk. The herd feed target is

$$D_{t,d} = \sum_{c \in \mathcal{C}} n_{c,t,d} D_{c,t,d}.$$

The MEMS grazing interface measures grazing pressure in demand-equivalent LSU. One demand-equivalent LSU removes $\kappa_{\text{MEMS}} = 15$ kilograms of forage dry matter per day, the conversion inside MEMS, so the daily herd input is

$$L_{t,d} = \frac{D_{t,d}}{\kappa_{\text{MEMS}}}. \tag{B6}$$

The annual target $H_t$ therefore represents intended whole-farm herd size, whereas $L_{t,d}$ is the demand-equivalent herd that drives the daily biophysical model. Because $D_{t,d}$ depends on the current weights, a herd that loses weight on day $d$ eats less and places a smaller demand on the pasture on day $d+1$ (Section B.8).

### B.5. Animal selective-grazing model

Within the active management paddock, cattle may use patches selectively. Let $\mathcal{E}_{t,d} = \mathcal{P}_{j_{t,d}}$ be the eligible set and let $\Omega(\cdot)$ be the grazing value of a patch as a function of its preceding state, the herd's preceding location, and fixed spatial attributes. The occupied patch is

$$k_{t,d} = \arg\max_{k \in \mathcal{E}_{t,d}} \Omega\Big(X_{k,t,d-1}, k_{t,d-1}, k\Big). \tag{B7}$$

This abstract selection rule is sufficient for the theory. Section C.4 specifies the numerical score. Under continuous grazing, all $K$ patches remain eligible throughout the grazing season, and animal selective grazing determines the occupied patch through equation (B7). Spatial use can therefore remain uneven under continuous grazing, which is represented numerically by $N = 1$. Under calendar-based rotation, $N > 1$ and the active management paddock restricts the eligible set. When $N = K$, every management paddock contains one patch and the calendar fully determines location.

### B.6. Local grazing-pressure mapping

The annual management choices produce daily local grazing pressure

$$u_{k,t,d} = \mathbf{1}\{d_{\text{on}} \le d \le d_{\text{off}}\}\mathbf{1}\{k = k_{t,d}\}\frac{L_{t,d}}{A_k}, \tag{B8}$$

measured in LSU per hectare. The occupied patch receives the herd and every other patch receives zero.

The feed-gap supplementation sets the hay fed on pasture. Let $F_{k,t,d-1}$ be the grazeable forage on patch $k$ at the end of day $d-1$, the forage dry matter above a retained residual, in kilograms. On a grazing day, the feed gap of the herd is $\max\{0, D_{t,d} - F_{k_{t,d},t,d-1}\}$, and the producer fills the fixed share $\phi_t$ of it with hay on the occupied patch:

$$h_{t,d} = \phi_t \max\big\{0,\ D_{t,d} - F_{k_{t,d},t,d-1}\big\}, \qquad s_{k,t,d} = \mathbf{1}\{k = k_{t,d}\}\frac{h_{t,d}}{10A_k}. \tag{B9}$$

Here $h_{t,d}$ is in kilograms of dry matter, and the local hay input $s_{k,t,d}$ is in grams per square meter per day when $A_k$ is in hectares. The rule has no minimum amount. With $\phi_t = 0$, the herd eats forage only. With $\phi_t = 1$, forage and hay together meet the feed target.

### B.7. Biophysical transition model

Hay fed on pasture enters the patch transition through the local input $s_{k,t,d}$ of equation (B9). Hay fed off pasture in winter does not enter any patch. Local grazing pressure enters the transition as a management input and carries no carbon or nutrient flux directly. Let

$$\mathbf{a}_{k,t,d} = \mathcal{A}\Big(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d}\Big) \tag{B10}$$

denote the causal action map from the preceding patch state and current management inputs to realized forage removal, trampling, fecal carbon and nitrogen, urine carbon and nitrogen, and hay intake. The animals eat forage first, up to the feed target, and then hay. Realized pasture intake is therefore $I^{\text{past}}_{t,d} = \min\{D_{t,d}, F_{k_{t,d},t,d-1}\}$, and the hay of equation (B9) is eaten in full. Realized removal is capped by forage available at the start of the day, so all action components are known before the patch state is advanced. The hay changes the patch state through the excreta of the animals that eat it.

Let $z_k$ denote fixed site characteristics and $w_{t,d}$ the day's exogenous forcing and calendar context. The generic local transition is

$$X_{k,t,d} = F\Big(X_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}\Big), \tag{B11}$$

where the arguments are all predetermined at the transition boundary. This representation deliberately leaves the transition technology unspecified. Section C.3 describes its process-model and surrogate implementations.

### B.8. Annual farm accounting

*Feed account and body weight.* The feed account sums the feed that the producer buys. On a grazing day, the hay is $h_{t,d}$ from equation (B9). Off pasture, the herd eats a fixed winter ration of hay $h^{\text{off}}_{t,d}$ and corn grain $m_{t,d}$, the same rule in every management system. The ration depends on the body weights, so its quantity responds to management through the condition in which the herd leaves the pasture. The annual quantities are

$$S^{g}_{t} = \sum_{d=d_{\text{on}}}^{d_{\text{off}}} h_{t,d}, \qquad S^{\text{off}}_{t} = \sum_{d \notin [d_{\text{on}}, d_{\text{off}}]} h^{\text{off}}_{t,d}, \qquad M_t = \sum_{d \notin [d_{\text{on}}, d_{\text{off}}]} m_{t,d}, \tag{B12}$$

and the annual hay quantity priced as purchased feed is

$$S^{\text{store}}_{t} = S^{\text{off}}_{t} + S^{g}_{t}. \tag{B13}$$

The realized forage and feed are divided among the animal classes, and the diet of each class sets its energy balance. Let $E_{c,t,d}$ be the net energy that the daily diet of one animal

of class $c$ supplies minus its fixed obligations, maintenance, pregnancy for a pregnant cow, and lactation for a nursing cow, in megacalories per day. When $E_{c,t,d} < 0$, the animal mobilizes body reserves and loses weight:

$$w_{c,t,d+1} = w_{c,t,d} + \frac{E_{c,t,d}}{r_c}, \tag{B14}$$

where $r_c$ is the net energy that one kilogram of body weight supplies. When $E_{c,t,d} \geq 0$, the animal gains the weight that the surplus energy supports. The new weight sets the next day's feed target through Section B.4. A feed-gap supplementation below one therefore lowers the hay bill but, in a forage shortage, also lowers body weight, the next day's grazing pressure, and, through equation (B4), the next calf crop. Appendix C.10 gives the allocation among classes, the energy terms, and the weight limits.

*Farm state and profit accounts.* Let $\mathbf{X}_t = (X_{1,t}, \ldots, X_{K,t})$ collect the patch states at the start of year $t$, and let $\mathrm{SOC}(X_{k,t})$ extract total SOC from patch $k$. Farm-average SOC is

$$\mathrm{SOC}_t = \sum_{k=1}^{K} \frac{A_k}{A} \, \mathrm{SOC}(X_{k,t}). \tag{B15}$$

Composing equation (B11) over all days and patches under the calendar, selection, feed, and local-pressure rules defines the annual farm transition

$$\mathbf{X}_{t+1} = G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \qquad \mathrm{SOC}_{t+1} = g_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t), \tag{B16}$$

with $\mathbf{n}_{t+1} = \mathcal{H}(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t)$, where $W_t$ is the annual exogenous forcing path. The complete transition depends on the spatial distribution of state and on the herd counts and weights in addition to farm-average SOC. The scalar function $g_N$ used below is therefore a reduced-form SOC projection of the full daily system, conditional on its other state components, the herd state, and forcing.

Annual profit conditional on installed design $N$ is

$$\begin{aligned} \pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t; W_t) = {} & R_{\mathrm{calf}} + R_{\mathrm{cull}} - C_{\mathrm{feed}} - C_{\mathrm{herd}} - C_{\mathrm{min}} \\ & - C_{\mathrm{labor}}(N, \mathrm{gp}_t) - C_{\mathrm{fence}}(N) - C_{\mathrm{repl}}, \end{aligned} \tag{B17}$$

with every term evaluated for year $t$ under forcing path $W_t$. Calf revenue is the number of calves sold times their realized weaning weight, and cull revenue is the number of culled cows times their realized weight. Feed cost $C_{\mathrm{feed}}$ prices the hay quantity in equation (B13) and the corn $M_t$ of equation (B12). Herd nonfeed cost and mineral cost are recorded separately as $C_{\mathrm{herd}}$ and $C_{\mathrm{min}}$. Rotation affects labor through scheduled management-paddock moves and fencing through installed $N$, with the durable fence cost converted to

an annual equivalent. Higher SOC can raise private profit by increasing forage quantity and quality, which lowers $C_{\text{feed}}$ when $\phi_t > 0$ and raises animal weights and the calf crop when the feed gap is not filled. Terminal SOC and the carbon benchmark are reported as separate environmental criteria and are excluded from $\pi_N$.

The model distinguishes forage, SOC, myopic-profit, and forward-looking management benchmarks. Their formal definitions appear in Appendix B.14.2. A longer calendar grazing period simultaneously lengthens scheduled rest, changes movement frequency, and changes how animal selective grazing distributes use across patches. The resulting ranking is an empirical property of $G_N$ and $\pi_N$, and the calendar identity alone leaves it indeterminate.

### B.9. Dynamic optimization and SOC valuation

Equation (B1) gives the private-profit operating problem conditional on installed infrastructure. The outer comparison over $N$ respects the discreteness and durability of paddock design and avoids treating fencing as a costless annual adjustment.

For a transparent scalar shadow-value expression, condition on the non-SOC components of $\mathbf{X}_t$, the herd state $\mathbf{n}_t$, and a stationary reference forcing environment $\overline{W}$, and evaluate both profit and the transition under that same environment. Suppressing these conditioned arguments gives the autonomous reduced operating problem

$$\widetilde{V}_N(\mathrm{SOC}_t) = \max_{H_t,\, \mathrm{gp}_t,\, \phi_t} \left\{ \pi_N(\mathrm{SOC}_t, H_t, \mathrm{gp}_t, \phi_t) + \delta \widetilde{V}_N(g_N(\mathrm{SOC}_t, H_t, \mathrm{gp}_t, \phi_t)) \right\}.$$

Let $\lambda_{t+1} = \widetilde{V}'_N(\mathrm{SOC}_{t+1})$ denote the private marginal continuation value of farm-average SOC. For a rotational design $N \geq 2$, an interior solution satisfies

$$\frac{\partial \pi_N}{\partial H_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial H_t} = 0, \tag{B18}$$

$$\frac{\partial \pi_N}{\partial \mathrm{gp}_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial \mathrm{gp}_t} = 0, \tag{B19}$$

$$\frac{\partial \pi_N}{\partial \phi_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial \phi_t} = 0. \tag{B20}$$

The herd condition balances the current marginal return to another planned LSU against its discounted SOC effect. The calendar condition balances the current feed, labor, and movement consequences of changing the grazing period against its discounted SOC effect. The supplementation condition balances the current hay cost against the weight and calf crop that the hay preserves and against its discounted SOC effect. Hay changes SOC through the excreta that it adds to the occupied patch and through the herd weight, which sets later grazing pressure. Because $\phi_t \in [0, 1]$, the supplementation condition holds with equality

only at an interior level. At $\phi_t = 0$ its left side is not positive, and at $\phi_t = 1$ it is not negative. Holding the planned herd target and installed $N$ fixed, let $\mathrm{gp}^m(H;N)$ denote a unique interior myopic timing optimum, so $\partial\pi_N/\partial\mathrm{gp} = 0$. If $\lambda_{t+1} > 0$, $\partial g_N/\partial\mathrm{gp} > 0$ there, and the dynamic objective is strictly concave in gp over the feasible interval, the forward-looking timing choice satisfies $\mathrm{gp}^d(H;N) > \mathrm{gp}^m(H;N)$. By equation (4), scheduled rest is then also longer. Joint adjustment of the herd target, timing, and feed-gap supplementation requires the full system in equations (B18) to (B20). The comparative result applies only to timing conditional on fixed $H$, $\phi$, and $N$. The forage, profit, and SOC optima may follow different rankings.

Applying the envelope theorem and imposing a stationary scalar equilibrium yields the private steady-state shadow value of SOC:

$$\lambda_N^{ss} = \frac{\partial\pi_N/\partial\mathrm{SOC}}{1 - \delta\,\partial g_N/\partial\mathrm{SOC}}. \tag{B21}$$

It rises with the marginal profit of SOC, with the persistence of SOC, and with the producer's patience. The scalar expression is exact for the autonomous reduced problem under $\overline{W}$. The numerical experiment uses a nonstationary finite forcing path: it replays ten years of historical weather. The resulting paths support finite-horizon management comparison. Direct computation of the steady-state shadow value requires the stationary reference environment. The numerical model retains the complete patch-state vector and can evaluate how the reduction depends on the distribution of state across the farm.

For each installed $N$ and a fixed feed-gap supplementation, Appendix B.14.4 gives four stationary equations for planned herd target, grazing period, SOC, and the SOC shadow value. A free feed-gap supplementation adds equation (B20) to the system. Eliminating $\lambda_N^{ss}$ from the herd and calendar conditions gives the operating-management isoquant

$$\frac{\partial\pi_N/\partial H}{\partial\pi_N/\partial\mathrm{gp}} = \frac{\partial g_N/\partial H}{\partial g_N/\partial\mathrm{gp}}. \tag{B22}$$

It compares the two annual operating margins conditional on $N$. Installed paddock designs require a direct comparison of their values because the design choice is discrete.

### B.10. Payments for SOC

A payment adds a transfer $P_t$ to annual profit, so the producer maximizes the discounted sum of $\pi_N + P_t$ in equation (B1). We consider three instruments. Let $\Omega$ be a set of qualifying management, for example rotation ($N \geq 2$) with a planned stocking rate at or below a cap $\overline{\mathrm{SR}}$, and let $\mathbb{1}_\Omega$ equal one when the management of year $t$ belongs to $\Omega$. The outcome payment at rate $p$ per Mg C, the practice payment at rate $r$ per hectare, and the premium

at share $q$ of the calf price are

$$P_t^{\text{out}} = p\,A\left(\text{SOC}_{t+1} - \text{SOC}_t - b_t\right), \qquad P_t^{\text{pr}} = r\,A\,\mathbb{1}_{\mathcal{Q}}, \qquad P_t^{\text{prem}} = q\,R_{\text{calf}}\,\mathbb{1}_{\mathcal{Q}}, \tag{B23}$$

where $b_t$ is the baseline SOC change of year $t$.

The outcome payment is linear in the SOC transition $g_N$, so it adds $pA$ to the discounted shadow value in each first-order condition. For the herd target, equation (B18) becomes

$$\frac{\partial \pi_N}{\partial H_t} + \left(\delta \lambda_{t+1} + pA\right)\frac{\partial g_N}{\partial H_t} = 0, \tag{B24}$$

and the grazing-period and supplementation conditions change in the same way. The rate therefore sets the response. If the producer cannot affect $b_t$, the baseline drops out of these conditions: it changes the size of the transfer and, in a voluntary program, the decision to join, but not the management of a participant. The contracts of Mason and Plantinga (2013) use the same split into a per-unit payment and a lump-sum transfer. A producer joins if the highest NPV with the payment exceeds the highest NPV without it. Two baselines bracket the offset protocols of Appendix A. Let the no-payment management be the no-payment optimum of the farm, with SOC path $\text{SOC}_t^B$. With the no-payment change as the baseline, $b_t = \text{SOC}_{t+1}^B - \text{SOC}_t^B$, the no-payment management earns nothing, every payment is for additional SOC, and the producer never loses by joining. With the start stock as the baseline, $b_t = 0$, a farm whose SOC falls under its no-payment management would owe money there, so it joins only if the payment at some other management exceeds the NPV that the change gives up. A one-sided rule that pays $pA\max(0, \text{SOC}_{t+1} - \text{SOC}_t)$ removes the debt, but it pays nothing for a smaller loss, and the marginal value $pA$ applies only in years with a gain.

The practice payment is constant on $\mathcal{Q}$, so it leaves equations (B18) to (B20) unchanged within the qualifying set. It acts only through a discrete comparison: the producer moves to the best qualifying management if the present value of the payment exceeds the NPV given up. If the no-payment management already qualifies, the payment changes nothing and is a pure transfer. A cap $\overline{\text{SR}}$ below the stocking rate of the no-payment management makes the payment bind, and the SOC that it buys is the SOC difference between the no-payment management and the best management at the cap. The premium scales calf revenue on $\mathcal{Q}$. Within the qualifying set, it multiplies the calf-revenue term $\partial R_{\text{calf}}/\partial H_t$ of equation (B18) by $1 + q$. Where calf revenue rises with the herd target, the premium therefore raises the private return to a higher stocking rate, against the SOC goal. A stocking-rate cap in $\mathcal{Q}$ limits this effect, and the premium then acts like a practice payment whose size is proportional to calf revenue.

### B.11. Spatial aggregation

The farm-management system explicitly separates management paddocks from biophysical patches. Real cattle concentrate on preferred locations and leave others lightly used (Gersie, Augustine, and Derner 2019), and that preferential use reshapes vegetation and, through it, soil (Adler, Raff, and Lauenroth 2001). Equation (B7) therefore lets patch states diverge even when all patches start with the same soil and weather. Farm-average SOC in equation (B15) is an aggregation of those heterogeneous stocks. With a nonlinear daily transition, its shadow value may differ from the area-weighted average of patch-specific values. The reduced scalar problem provides the transparent shadow-value expression, and the full transition $G_N$ preserves the spatial distribution needed to evaluate it.

### B.12. Theory-to-numerical mapping

The steady-state system (SS-1) to (SS-4) and the operating-management isoquant (B22) are reduced-form implications conditional on an installed $N$ and stationary reference forcing. The theory establishes the decision structure and a conditional comparative result: holding the planned herd target fixed, a forward-looking producer chooses a longer calendar grazing period than the myopic producer when the marginal SOC effect of gp remains positive at the myopic optimum. The ranking of forage, profit, and SOC optima remains an empirical question, as does the magnitude of $\lambda_N^{ss}$. Both depend on the daily transition and profit functions under the chosen reference environment.

We give the framework empirical content by specifying each component of equations (B10) and (B11). The numerical animal model supplies the feed target $D_{t,d}$, the demand-equivalent daily herd input $L_{t,d}$, the class energy balances, and the weight changes. The biophysical transition supplies the grazeable forage, the realized pasture intake, and its quality. Equation (B9) sets the hay on pasture, and a fixed winter ration sets the feed off pasture. Calf weaning weights, cull weights, and the calf crop therefore respond to management through the body weights. Soil carbon's private value is realized through the forage quantity and quality that reduce purchased hay or raise animal weights.

The analytical and numerical controls coincide. Both condition on an installed paddock count $N$, use planned herd target $H$, grazing-period length gp, and feed-gap supplementation $\phi$ as the operating margins, hold grazing-season dates fixed, and derive scheduled management-paddock rest as $(N-1)$gp. Both use selective patch use within the active paddock and map the resulting location and demand-equivalent herd $L_{t,d}$ into daily local LSU per hectare. The numerical experiment restricts $H_t \equiv H$ and $\phi_t \equiv \phi$, evaluates a discrete grid of fixed management systems with the surrogate, and conditions on one replayed weather path. The Bellman problem remains a theoretical benchmark for future optimization. The numerical private objective is ten-year discounted profit, and terminal

SOC is reported separately. The purchased-feed account follows equations (B12) and (B13), and the hay on pasture enters the biophysical transition through equation (B9).

### B.13. Outcome payments against the start stock and above a threshold

Two contrast cases of the outcome payment use the start stock as the baseline (Commonwealth of Australia 2021; Gold Standard 2020). One pays only the annual gains, with present value $pA \sum_t \delta^t \max\{0, \Delta_t(a)\}$. The other pays the net change, $pA \sum_t \delta^t \Delta_t(a)$, and the farm joins only if the payment raises its NPV above $\mathrm{NPV}(a^0)$. This payment differs from equation (5) by a lump sum, so a farm that joins chooses the same management system (Mason and Plantinga 2013). A variant after Rouhi Rad et al. (2026) pays $\delta^9 pA \max\{0, \Delta\mathrm{SOC}(a) - \theta\}$ at the end of 2035, with one threshold $\theta$ for all farms or one at each farm's own no-payment change plus a margin. At the equal budget of Section 5.4, the threshold variant at each farm's own no-payment change pays \$100 per Mg C and buys 427 Mg C for \$27,541 per farm. The two start-stock designs pay \$64 per Mg C and buy 134 Mg C per farm. Most of their transfer goes to the heavy continuous farm, which gains SOC under every management system.

### B.14. Theory derivations

This subsection gives the bookkeeping and envelope steps behind Section 2.3 and Appendix B.9.

#### B.14.1. Calendar identity and local grazing input

Under the equal-duration calendar, a complete cycle contains $N$ consecutive grazing periods of $\mathrm{gp}_t$ days, so its length is $c_t = N\mathrm{gp}_t$. A management paddock is active for one such period and is inactive while the other $N - 1$ paddocks are active. Its scheduled rest is therefore

$$r_t = c_t - \mathrm{gp}_t = (N-1)\mathrm{gp}_t, \tag{B25}$$

which gives equation (4). The identity records calendar bookkeeping. Biomass and residual forage respond to the schedule, while advancement occurs solely according to the calendar.

Let $j_{t,0}$ be the first management paddock used in year $t$. During the grazing season, the exact scheduler is

$$j_{t,d} = 1 + \mathrm{mod}\left(j_{t,0} - 1 + \left\lfloor \frac{d - d_{\mathrm{on}}}{\mathrm{gp}_t} \right\rfloor, N\right). \tag{B26}$$

If the grazing season ends partway through a cycle, the final realized interval can be truncated. Rotating the starting paddock across years distributes that boundary effect while preserving the planned identity.

Management-paddock rest also differs from the biological rest of an individual patch. Equation (B26) makes every patch in the active management paddock eligible, but equation (B7) occupies only the patch with the highest grazing value. An unselected patch can therefore remain ungrazed even while its management paddock is active. Conversely, all patches outside the active paddock receive zero local LSU by construction.

The local-input mapping preserves the whole-farm herd while concentrating it spatially. Because exactly one patch is occupied during the grazing season, equation (B8) implies

$$\sum_{k=1}^{K} \frac{A_k}{A} u_{k,t,d} = \frac{L_{t,d}}{A}. \tag{B27}$$

The right-hand side is the demand-equivalent whole-farm LSU per hectare, whereas the occupied patch receives $L_{t,d}/A_k$. Thus the grazing regime determines the eligible set and its timing, animal selective grazing chooses the occupied patch, and the farm-management layer passes the resulting daily local LSU per hectare to MEMS or the surrogate. The layer also passes the hay input of equation (B9), which is colocated with the herd.

### B.14.2. Management benchmarks

Let $Y_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \mathrm{gp}_t, \phi_t)$ be pasture forage supplied to the herd during the year and let $\Delta\mathrm{SOC}_N = g_N - \mathrm{SOC}_t$ be annual SOC change. Conditional on the opening state, $H_t$, $\phi_t$, and $N$, the forage and SOC benchmarks are

$$\mathrm{gp}_F^* \in \arg\max_{\mathrm{gp}} Y_N, \qquad \mathrm{gp}_{\mathrm{SOC}}^* \in \arg\max_{\mathrm{gp}} \Delta\mathrm{SOC}_N.$$

The myopic producer chooses

$$(H_\pi^*, \mathrm{gp}_\pi^*, \phi_\pi^*) \in \arg\max_{H,\mathrm{gp},\phi} \pi_N(\mathbf{X}_t, \mathbf{n}_t, H, \mathrm{gp}, \phi),$$

whereas the forward-looking producer solves equation (B1). These definitions separate biological and current-profit benchmarks from the dynamic choice and leave their ordering unrestricted.

### B.14.3. Envelope derivation of the shadow value

The shadow value (B21) follows from the envelope theorem applied to the reduced scalar, stationary-reference-forcing version of equation (B1) defined in Section B.9. Write $S$ for

SOC, and let $H^*(S)$, $\mathrm{gp}^*(S)$, and $\phi^*(S)$ denote the maximizing controls conditional on installed $N$. Subscripts on $\pi_N$ and $g_N$ below denote partial derivatives. Differentiating $\widetilde{V}_N(S) = \pi_N(S, H^*, \mathrm{gp}^*, \phi^*) + \delta \widetilde{V}_N(g_N(S, H^*, \mathrm{gp}^*, \phi^*))$ gives

$$\begin{aligned}\widetilde{V}'_N(S) = \pi_{N,S} &+ \delta \widetilde{V}'_N(g_N) g_{N,S} \\ &+ \left[\pi_{N,H} + \delta \widetilde{V}'_N(g_N) g_{N,H}\right] \frac{dH^*}{dS} + \left[\pi_{N,\mathrm{gp}} + \delta \widetilde{V}'_N(g_N) g_{N,\mathrm{gp}}\right] \frac{d\mathrm{gp}^*}{dS} \\ &+ \left[\pi_{N,\phi} + \delta \widetilde{V}'_N(g_N) g_{N,\phi}\right] \frac{d\phi^*}{dS}. \qquad \text{(B28)}\end{aligned}$$

The three bracketed terms are the operating first-order conditions (B18) to (B20), so they vanish at an interior optimum. At a corner feed-gap supplementation, $\phi^*$ stays at the corner for small changes in $S$, so $d\phi^*/dS = 0$ and the third term also vanishes. The envelope condition is therefore

$$\widetilde{V}'_N(S) = \pi_{N,S} + \delta \widetilde{V}'_N(g_N) g_{N,S}. \qquad \text{(B29)}$$

Writing $\lambda_t = \widetilde{V}'_N(\mathrm{SOC}_t)$ and imposing a stationary scalar equilibrium gives $\lambda_N^{ss} = \pi_{N,S} + \delta \lambda_N^{ss} g_{N,S}$. Solving yields equation (B21). The value is positive and finite when $\pi_{N,S} > 0$ and $\delta g_{N,S} < 1$, that is, when SOC raises current profit, for example through avoided hay purchases or heavier animals, and its discounted persistence is less than one.

### B.14.4. Stationary operating system

Conditional on installed $N$ and a fixed feed-gap supplementation $\phi$, the stationary system has four unknowns, $(H^*, \mathrm{gp}^*, \mathrm{SOC}^{ss}, \lambda_N^{ss})$, and four equations:

$$\frac{\partial \pi_N}{\partial H} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial H} = 0, \qquad \text{(SS-1)}$$

$$\frac{\partial \pi_N}{\partial \mathrm{gp}} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial \mathrm{gp}} = 0, \qquad \text{(SS-2)}$$

$$g_N(\mathrm{SOC}^{ss}, H^*, \mathrm{gp}^*) = \mathrm{SOC}^{ss}, \qquad \text{(SS-3)}$$

$$\lambda_N^{ss} = \frac{\partial \pi_N / \partial \mathrm{SOC}}{1 - \delta\, \partial g_N / \partial \mathrm{SOC}}. \qquad \text{(SS-4)}$$

Here $H^*$ is the stationary planned herd target. The first two equations determine the operating margins, the third is the SOC fixed point, and the fourth is the envelope-based shadow value. A free feed-gap supplementation adds the unknown $\phi^*$ and the stationary form of equation (B20), with the corner conditions of Section B.9. The discrete installed-paddock designs are compared directly by value and have no additional first-order condition.

### B.14.5. Conditional dynamic extension of the grazing period

Fix the current state, installed design $N$, planned herd target $H$, and feed-gap supplementation $\phi$, and suppress $\phi$ in the notation. Let $\mathrm{gp}_m = \mathrm{gp}^m(H;N)$ be an interior maximizer of current profit, so $\pi_{N,\mathrm{gp}}(S,H,\mathrm{gp}_m) = 0$. The derivative of the dynamic objective with respect to the grazing period, evaluated at that myopic choice, is

$$\frac{\partial}{\partial \mathrm{gp}}\left[\pi_N(S,H,\mathrm{gp}) + \delta \widetilde{V}_N(g_N(S,H,\mathrm{gp}))\right]\Big|_{\mathrm{gp}=\mathrm{gp}_m} = \delta \lambda_{t+1} g_{N,\mathrm{gp}}(S,H,\mathrm{gp}_m). \tag{B30}$$

If $\lambda_{t+1} > 0$ and $g_{N,\mathrm{gp}} > 0$ at $\mathrm{gp}_m$, this derivative is positive. If the dynamic objective is strictly concave in gp over the relevant interval, its conditional maximizer lies to the right: $\mathrm{gp}^d(H;N) > \mathrm{gp}^m(H;N)$. Equation (B25) then gives $r^d(H;N) > r^m(H;N)$ for $N > 1$. This result holds conditional on $H$ and $N$. When herd target, timing, and feed-gap supplementation change together, their interaction is governed by the full system of equations (B18) to (B20), which leaves the coordinate-by-coordinate ordering unrestricted.

## Appendix C. Numerical implementation and calibration

This appendix gives the herd, movement, nutrition, coupling, and accounting rules, the assumptions, and the parameter values of the numerical model in Section 3.

### C.1. Herd composition and grazing calendar

Planned mature-cow, replacement-heifer, and annual-calf targets use shares of 0.50, 0.10, and 0.40 with representative coefficients of 1.00, 0.70, and 0.20 LSU per head against a 500-kilogram reference animal. These values convert the planned annual herd target $H$ to a cow target. Realized opening class proportions emerge from the demographic controller, which initializes attrition-consistent cow and heifer counts around that target, holds calves as a pending cohort until calving, and thereafter supplies the same daily class counts to the animal-demand and accounting modules. Cows calve on day 75, and calves are weaned on day 270 at their simulated weight. At weaning, the existing heifer cohort is promoted to cows, surviving calves are split equally by sex, and the retention controller keeps enough female calves to form the next replacement-heifer cohort. Male and surplus female calves are sold, and any replacement shortfall is purchased. The promoted heifers calve for the first time in the next spring as the first-calf cohort. At the year boundary, cow and heifer mortality is 1.5 percent, every open cow is culled, and at least 12 percent of the cows are culled. The resulting counts and the next pending calf cohort from equation (B4) initialize the next annual cycle. Cattle graze on pasture from day 120 to 270 and eat stored feed off pasture from late September through April, following the Michigan grazing calendar.

### C.2. Integrated computational architecture

Figure C1 organizes the components by time scale and makes their nesting explicit. Annual setup consists of the whole-farm management system and the realized opening herd. On each day, the calendar and the animal demand schedule provide parallel inputs, and animal selective grazing chooses the occupied patch. The local LSU per hectare and the hay of equation (B9) drive the management-responsive biophysical transition. The corresponding MEMS process constructs removal, trampling, and animal-return fluxes internally. The realized forage and hay set the body weights of the next day, and the resulting patch state is the input for the next day. At calving, weaning, and the year boundary, the herd-composition controller updates class counts and sets next year's pending calf cohort. A single annual farm account combines the two hay streams, the winter corn, herd outcomes, SOC, and all revenue and cost items. The ten-year management comparison lies outside the one-year box and aggregates repeated annual cycles.

The Microbial Efficiency-Matrix Stabilization model (MEMS) supplies the process-based transition by simulating carbon and nitrogen cycling at daily resolution (Zhang et al.

2024; Santos et al. 2024). The management-responsive surrogate replaces that transition inside the farm loop and uses the preceding patch state, site and weather context, daily local LSU per hectare, and the hay input. Every simulation uses the surrogate. Hay fed on pasture enters both the surrogate and the economic account. Winter hay and corn enter the economic account only.

### C.3. MEMS process model and patch representation

MEMS is a process-based model of carbon and nitrogen cycling among soil layers, vegetation, and the atmosphere at daily resolution (Zhang et al. 2024; Santos et al. 2024). It resolves SOC into measurable pools whose turnover responds to weather, plant inputs, and grazing offtake. We treat the pool structure as given and cite the model description for the mechanism. For this paper, the relevant outputs are the daily aboveground forage state and the annual farm-level SOC stock.

We represent the 60-hectare farm as $K$ biophysically identical patches, each advanced by the surrogate with one shared soil profile, weather sequence, and forage species mix. The patches differ only in their dynamic state, that is, in the biomass and leaf-to-stem composition that grazing history produces. With a patch area of 5 hectares, the farm contains $K = 12$ patches. This spatial resolution corresponds to the scale at which grazing within a patch is treated as approximately uniform and is held fixed across all management comparisons.

### C.4. Calendar movement and selective patch use

The numerical implementation specifies the abstract grazing value in equation (B7) as

$$\Omega_{k,t,d} = P_k\big[1 - \mathcal{T}(k_{t,d-1} \to k)\big]\mathcal{I}(B_{k,t,d-1})Q^{\gamma}_{k,t,d-1}, \tag{C1}$$

where $P_k$ is a fixed spatial-preference weight, $\mathcal{T}$ is the travel-cost term, $\mathcal{I}(B)$ is a saturating intake response to biomass $B$, $Q$ is live-leaf share, and $\gamma$ governs the strength of quality selection. The score is read from the preceding end-of-day state, so selection is causal.

The two grazing regimes share identical patches, planned herd target, and season length and differ only in how the calendar restricts the herd's eligible area. Under continuous grazing, all $K$ patches remain eligible throughout the grazing season. This regime is represented by $N = 1$. Under calendar-based rotation, equation (B26) activates one of $N > 1$ management paddocks for gp days. In both regimes, equations (C1) and (B7) implement animal selective grazing by choosing the patch with the highest grazing value within the eligible area. The value combines a saturating (Type II) intake response to biomass, live-leaf share, travel cost, and a fixed preference weight for non-forage attractors such as water. Thus the management regime determines where cattle are allowed to graze,

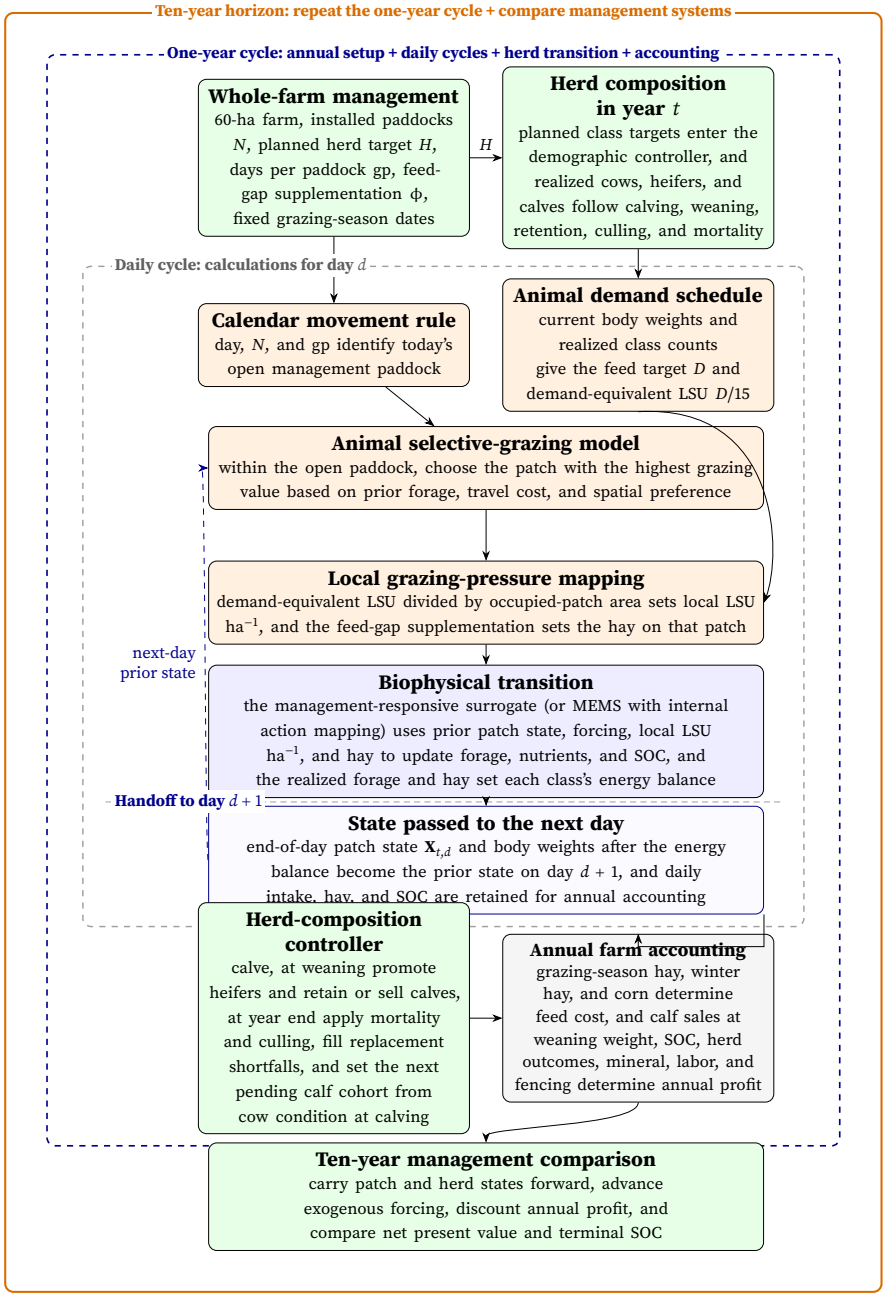



FIGURE C1. Temporal nesting and information flow. Annual setup establishes the whole-farm management system and realized herd composition. On day $d$, the calendar and animal demand determine the eligible area and the demand-equivalent herd. Animal selective grazing, local-pressure and hay mapping, and the biophysical transition produce the patch state and the body weights used on day $d + 1$. After the daily cycle has been repeated over the year, a one-way accounting handoff carries accumulated daily outcomes into the annual farm account. Annual accounting remains downstream of the daily transition. The herd controller and farm account close the year. The outer loop carries patch and herd states and exogenous forcing across ten annual cycles and compares discounted private profit and terminal SOC.

while animal selective grazing determines where they graze within that eligible area. The contrast at fixed planned herd target isolates the within-farm redistribution channel of Section B.11. Figure C2 shows the patch layout and the two grazing regimes.

### C.5. Herd nutrition, body weight, and feed accounting

The animal model follows National Research Council relationships (National Academies of Sciences, Engineering, and Medicine 2016). The herd has three classes, cows, replacement heifers, and calves, and uses the planned shares and reference LSU coefficients specified in Appendix C.1. The demographic controller determines realized class counts, which can differ from the planned shares.

Each class carries one representative shrunk body weight that changes from day to day, and the cow class carries one weight for each of its two cohorts. The herd starts on 1 January 2026 with pregnant mature cows of 527 kilograms, body condition score (BCS) 5 on the nine-point scale, and heifers of 255 kilograms, the median 1 January weight of the later heifer cohorts in a preliminary simulation of the model. The feed target per head is the dry-matter intake capacity of cows and heifers, which depends on body weight, lactation, and air temperature, and the solid-feed allowance of nursing calves, which depends on calf age and weight. Calves also drink milk. Milk follows a fixed lactation curve with the Angus peak of 8.0 kilograms per day (National Research Council 2000), and first-calf cows give 0.74 times this amount, the factor for two-year-old dams (National Research Council 2000). Milk is an obligation of the cow, so a cow that is short of energy loses weight but does not give less milk.

On a grazing day, the surrogate state of the occupied patch at the end of the previous day gives the grazeable forage: the dry matter of the plant pools (leaf, stem, and seed) and the standing dead pools above a retained residual of 20 grams per square meter. Equation (B9) sets the hay. The animals eat forage up to the feed target, split between plant and standing dead forage in proportion to their dry matter, and the carbon and nitrogen of the plant pools set the crude protein of the plant forage eaten. When this crude protein is outside the 1st to 99th percentile band of the MEMS training simulations at the site in the same month, it is clipped into the band and the day is counted. Standing dead forage has the median MEMS crude protein of the site, 8.4 percent. The total digestible nutrients of each forage follow its crude protein. The hay has 45 percent carbon and 1.5 percent nitrogen in its dry matter, which is 9.4 percent crude protein, and 53 percent total digestible nutrients.

Off pasture, every management system feeds the same winter program, because winter feeding is outside the scope of the management comparison. Heifers and first-calf cows eat hay to their intake capacity. Mature cows eat the hay that meets their maintenance, pregnancy, and lactation energy and moves them to BCS 5 by the calving date, and they eat

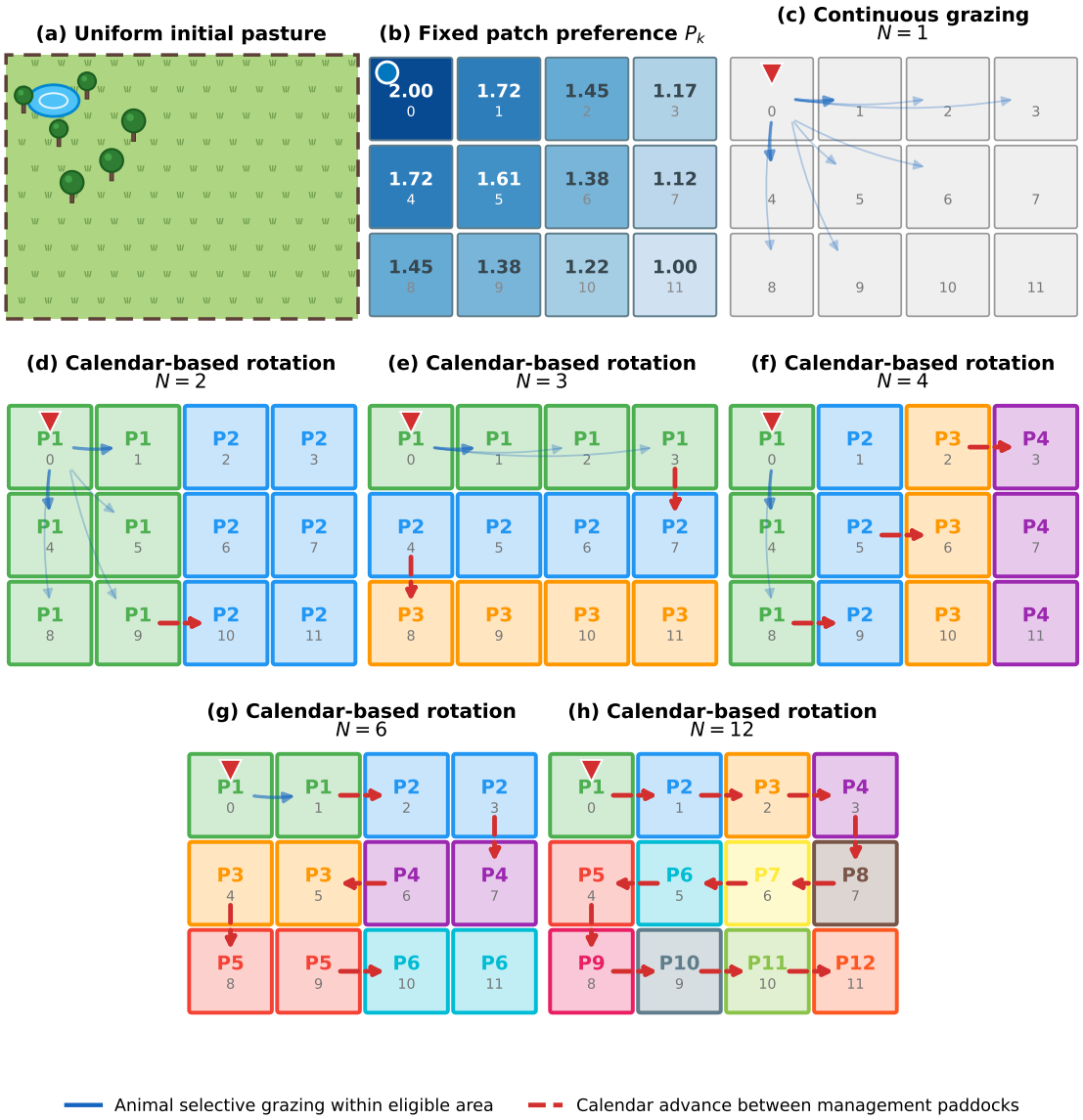



FIGURE C2. **Continuous grazing, calendar-based rotation, and animal selective grazing.** The 60-hectare farm contains $K = 12$ equal 5-hectare biophysical patches. Panel (a) shows their common initial pasture conditions and the fixed corner water-and-shade attractor, and panel (b) shows the resulting patch-preference gradient $P_k$. Under continuous grazing (c), all patches remain eligible throughout the grazing season. This regime is represented by $N = 1$. Under calendar-based rotation (d) to (h), equation (B26) activates one of $N > 1$ management paddocks for gp days. In both regimes, equations (C1) and (B7) select the highest-value eligible biophysical patch. When $N = 12$, each management paddock contains one patch and the calendar fully determines location. Colors identify management-paddock membership, blue arrows show animal selective grazing within the eligible area, and red dashed arrows show calendar advances between management paddocks.

hay to capacity when they are below BCS 5. Heifers receive 1.5 kilograms of dry corn grain per head per day on every off-pasture day, and cows from day 15 to day 119, the last 60 days of gestation to turnout. The corn replaces hay of the same weight and has 88 percent total digestible nutrients (National Research Council 2000). The amount is the upper end of Extension guidance for high-forage cow diets (Drewnoski 2020). Calves before turnout eat hay to their solid-feed allowance.

The realized feed is then allocated among the classes. The allocation first meets the fixed energy obligations of every class when the feed allows it: maintenance for all classes, pregnancy for pregnant cows, and lactation for nursing cows. It then distributes the rest in proportion to metabolic body weight and remaining intake capacity. On pasture, each cow cohort receives a share of the cow feed in proportion to its intake capacity. The energy equations give each class's energy balance and the weight change of equation (B14). The body reserves of a cow supply 0.80 times their tissue energy when the cow loses weight, and a gain stores the full tissue energy, which rises with condition (National Research Council 2000; Fox et al. 2003). Heifers gain by the growth equation of National Research Council (2000), and nursing calves gain by the young-calf retained-energy equation of National Academies of Sciences, Engineering, and Medicine (2021). Cow weight stops rising at 605 kilograms. The body condition score of each cow cohort on the calving date sets its conception rate through equation (B4). A simulation ends as infeasible on the first day when the mature cows reach 385 kilograms (BCS 1.1), the first-calf cows reach 330 kilograms, or a heifer cohort reaches 70 percent of the weight at which it entered the herd. Appendix C.10 gives the equations and constants.

### C.6. Daily coupled simulation

The coupled simulation follows the daily ordering in Figure C1 for every calendar day from 1 January 2026 to 31 December 2035. At the start of day $d$, the demand schedule evaluates the day's class counts from the cohort sizes and fixed event calendar and computes $D_{t,d}$ and $L_{t,d}$ from the current body weights. On a grazing day, the calendar and the preceding patch states determine the occupied patch, its grazeable forage, and the hay. The surrogate then advances the twelve patches together, the occupied patch with the local LSU per hectare and hay and the other patches with zero inputs. The realized feed sets the energy balances and the body weights for day $d + 1$. Off pasture, every patch advances with zero inputs, and the herd eats the winter ration. Current-day animal demand, patch selection, and hay therefore enter the same daily transition, and the next day's demand depends on the outcome. The year closes on 31 December with the herd transition and the annual account. Appendix C.11 records the complete execution order.

### C.7. Economic accounting

Profit is calculated downstream of the biophysical and herd simulation. Within a model run, grazing, SOC, and herd dynamics are therefore unaffected by the accounting result. On a cash basis, annual profit is revenue minus cost:

$$\pi_N(y) = \underbrace{R_{\text{calf}}(y) + R_{\text{cull}}(y)}_{\text{revenue}} - \underbrace{\left[C_{\text{feed}}(y) + C_{\text{herd}}(y) + C_{\text{min}}(y) + C_{\text{labor}}(y) + C_{\text{fence}}(y) + C_{\text{repl}}(y)\right]}_{\text{cost}}. \tag{C2}$$

Each term is evaluated for simulation year $y$. Revenue comes from calf sales $R_{\text{calf}}$ at the simulated weaning weight and cull-cow sales $R_{\text{cull}}$ at the simulated cohort weight on 31 December, priced per hundredweight. Feed cost $C_{\text{feed}}$ prices the winter hay and the grazing-season hay of equation (B13) at the same price and the corn of equation (B12) at its own price. The broad purchased-feed, homegrown-feed, and grazed-feed aggregates of the USDA Economic Research Service (ERS) are excluded to avoid double counting. The remaining costs are herd nonfeed operations $C_{\text{herd}}$, mineral $C_{\text{min}}$, labor $C_{\text{labor}}$ (a per-cow base plus scheduled-move labor), the annual equivalent of perimeter and interior fencing $C_{\text{fence}}$, and replacement-heifer purchases $C_{\text{repl}}$. For a $T$-year simulation beginning in year $y_0$, the reported net present value is

$$\text{NPV}_N^{(T)} = \sum_{\tau=0}^{T-1} \delta^{\tau} \pi_N(y_0 + \tau) + \delta^{T-1} p_{\text{cull}} \sum_g n_g \left(w_g - w_g^5\right), \qquad \delta = (1 + 0.05)^{-1}. \tag{C3}$$

We set $T = 10$ and $y_0 = 2026$, so the first management year has discount factor one. The second term values the condition of the cows that remain at the end: $n_g$ and $w_g$ are the count and weight of cow cohort $g$ on 31 December 2035, $w_g^5$ is its BCS-5 weight, and $p_{\text{cull}}$ is the cull price per kilogram. A herd that ends thinner than BCS 5 therefore has a negative terminal value, so a management system cannot gain NPV by selling the condition of its cows in the last years. Consistent with the cash basis, the private-profit objective carries no carbon revenue, and Section 3.4 adds payments after the run. Terminal SOC is reported as a separate environmental outcome, while the theoretical shadow value $\lambda$ of Section B.9 measures SOC's private continuation value through forage and feed-cost savings.

### C.8. Main numerical assumptions

Six assumptions are especially important for interpreting the numerical results. First, planned herd target, paddock count, grazing period, feed-gap supplementation, and season dates remain constant within each simulation. Dynamic optimization is left for future work. Second, the planned target $H$ represents intended annual herd size, while

realized class counts and body weights set the feed target and the daily demand-equivalent input $L_{t,d}$. Third, body weights respond to the realized feed through published energy relationships, but the complete herd lifecycle has not been validated. We compared the simulated weights with the October weights of a Michigan State University research herd and did not calibrate the model to them (Appendix E.2). Fourth, a protein shortage does not limit gain, so the gains are those that the energy supports. Fifth, the intake capacities that set the feed target and the conversion of 15 kilograms of dry matter per demand-equivalent LSU have not been verified against official nutrient-requirement cases for the three classes. Sixth, every simulation receives the same deterministic exogenous forcing path and the same winter feeding program. Appendix C.12 states the full set of assumptions.

### C.9. Parameterization

Appendix Table C1 summarizes the parameters used in the numerical experiment. The biological and technical parameters fix the 60-hectare, twelve-patch Michigan farm, the grazing season from day of year 120 to 270, the three-class herd with its nutrition, weight, and demographic rules, and the calendar and animal selective-grazing rules described above. These quantities are held constant across management systems. Only the planned stocking rate, installed paddock count, grazing-period length, feed-gap supplementation, and management history vary.

Calves and cull cows are valued at $264 and $115 per hundredweight of their simulated weights. Winter hay is priced at $220 per short ton of consumed dry matter, a rounded hay-equivalent value derived from Michigan hay prices using 85 percent dry matter and 90 percent recovery. Hay fed on pasture has the same price. Corn costs $233.8 per metric ton of dry matter, the mean Michigan price received for the marketing years 2020 to 2024 (USDA-NASS 2026), which is 3.6 percent below the hay price per ton of dry matter. Annual nonfeed herd cost is $280.84 per cow and mineral cost is $54.75 per cow. Labor combines a $27.99 per-cow base, a $19.70 hourly wage, and move time of 30 minutes plus 0.10 minute per cow over a 24-week labor season. Permanent fence costs $2.51 per foot with a 20-year life, temporary fence costs $1.05 per foot with a 10-year life, and replacement heifers cost $1,200 per head. We discount at 5 percent. Calf and cull prices are fixed calibration values, and the replacement-heifer price is treated as a model assumption.

### C.10. Feed, energy, and body weight

Let $n_{c,t,d}$ be the number of animals in class $c$ on day $d$ and $w_{c,t,d}$ their representative shrunk body weight. The feed target per head $D_{c,t,d}$ is the intake capacity of the class. For cows and heifers, it is the dry-matter intake capacity as a fraction of body weight, 2.4 percent for a lactating cow, 2.1 percent for a dry cow, and 2.5 percent for a heifer,

adjusted for air temperature (National Academies of Sciences, Engineering, and Medicine 2016). The nutrient-requirement models have no intake equation for nursing calves, so the model sets an allowance of solid feed for the calf that depends on its age and weight.

On a grazing day, let $C_j$ and $N_j$ be the carbon and nitrogen of forage pool $j$ of the occupied patch at the end of the previous day, in grams per square meter. The five plant pools are live and dead leaf, live and dead stem, and seed of the MEMS crop output, and the three standing dead pools are the leaf, stem, and seed of the MEMS standing dead biomass. The dry matter of a pool is $C_j/f_j$, with $f_j = 0.48$ for the plant seed and $f_j = 0.43$ for every other pool. With $P$ and $S$ the plant and standing dead dry matter, the grazeable forage of the 5-hectare patch is

$$F_{k,t,d-1} = 10A_k \max\{0,\ P + S - 20\} \tag{C4}$$

in kilograms, and the pasture intake $I^{\text{past}}_{t,d} = \min\{D_{t,d}, F_{k_{t,d},t,d-1}\}$ is split between plant and standing dead forage in proportion to the grazeable dry matter of each group. The nitrogen of each group follows its carbon in proportion. The crude protein of the plant group is 6.25 times its nitrogen-to-dry-matter ratio. If it is outside the 1st to 99th percentile band of the MEMS training simulations at the study site in the same month (0.102 to 0.157 in May and 0.082 to 0.094 in September), it is set to the nearest bound and the day is counted. Standing dead forage has the crude protein 0.0838, the median MEMS value at the site. The total digestible nutrients of each forage group are set so that its net energy for maintenance is 0.95 + 3.30 CP megacalories per kilogram of dry matter, bounded to 0.8 to 2.2. The hay of equation (B9) has 45 percent carbon and 1.5 percent nitrogen in its dry matter and 53 percent total digestible nutrients, which give 1.07 megacalories of net energy for maintenance per kilogram. The corn grain has 88 percent total digestible nutrients, 2.18 megacalories of net energy for maintenance and 1.50 for gain per kilogram, and 9.8 percent crude protein (feed 410 of National Research Council 2000).

Off pasture, heifers and first-calf cows eat hay to their intake capacity, with their corn. For mature cows, let $w^5 = 527$ kilograms be the BCS-5 weight, let $\tau_{t,d}$ be the days to the next calving, and let $m$ be the corn per head. A mature cow at or above $w^5$ receives the hay

$$h^{\text{cow}}_{t,d} = \min\Big\{D_{\text{cow},t,d} - m,\ \max\Big[0,\ \frac{O_{t,d} + r^{+}\,(w^5 - w_{\text{cow},t,d})/\tau_{t,d} - m\,\text{NE}^{\text{corn}}_m}{\text{NE}^{\text{hay}}_m}\Big]\Big\}, \tag{C5}$$

where $O_{t,d}$ is the maintenance, pregnancy, and lactation energy and $r^{+}$ is the energy stored per kilogram of gain in equation (C6). Between calving and turnout, the target change is zero. A mature cow below $w^5$ eats hay to capacity. The ration therefore reaches the target exactly when capacity does not bind.

The realized feed is allocated among the classes in two steps. The first step searches

for a split of each feed source among the classes that meets the fixed energy obligations of every class: maintenance for all classes, pregnancy for pregnant cows, and lactation for nursing cows. Milk follows the curve $y(\ell) = \alpha_y \ell^{0.15} e^{-0.003\ell}$ of lactation day $\ell$, with $\alpha_y = 5.169$ so that the peak is 8.0 kilograms per day on day 50 (National Research Council 2000), and it enters the calf diet before solid feed. First-calf cows give $0.74\, y(\ell)$, every calf receives the herd mean milk, and each cohort pays the energy of its own milk. The second step distributes the remaining feed in proportion to $w_{c,t,d}^{0.75}$ times the remaining intake capacity of each class. On pasture, cow cohort $g$ receives the share $n_g w_g / \sum_{g'} n_{g'} w_{g'}$ of the cow feed. Off pasture, each class and cohort eats its own ration. If the feed cannot meet every obligation, the allocation keeps all feed in the diets and the deficit appears in the energy balance.

For an adult class, let $E_{c,t,d}$ be the net energy for maintenance that the diet supplies minus the maintenance, pregnancy, and lactation requirement, in megacalories per head per day. The pregnancy requirement of a pregnant cow on gestation day $t_g$ is $0.576\,\mathrm{CBW}\,(0.4504 - 0.000766\,t_g)\exp[(0.03233 - 0.0000275\,t_g)\,t_g]/1000$ megacalories per day, with calf birth weight CBW = 38 kilograms (National Research Council 2000, Equations 4-4 and 4-7), times the pregnant share of the cohort. The weight change is

$$w_{c,t,d+1} - w_{c,t,d} = \begin{cases} E_{c,t,d}/r_c^-, & E_{c,t,d} < 0, \\ g^E_{c,t,d}, & E_{c,t,d} \geq 0, \end{cases} \tag{C6}$$

where $g^E$ is the gain that the surplus energy supports. For a cow, the tissue energy per kilogram rises with condition, $\mathrm{NE}_{\mathrm{DLW}} = 0.5381\,\mathrm{BCS} + 3.2855$ megacalories per kilogram (Fox et al. 2003, p. 2.15). A loss supplies $r^- = 0.80\,\mathrm{NE}_{\mathrm{DLW}}$ of dietary net energy per kilogram, and a gain needs $r^+ = \mathrm{NE}_{\mathrm{DLW}}$, so $g^E = E/r^+$ (National Research Council 2000, p. 118). A heifer loses $E/2.75$ kilograms per day. Its gain follows the growth equation of the nutrient-requirement model, $g^E = 13.97\,\mathrm{RE}^{0.9116}\,\mathrm{EQSBW}^{-0.6837}$, the exact inverse of the retained-energy equation of the animal model (National Research Council 2000 prints the coefficient as 13.91). Here RE is the surplus net energy for maintenance times the ratio of the net energies for gain and maintenance of the diet, and $\mathrm{EQSBW} = 478\,w/527$ is the equivalent shrunk body weight. The gain that metabolizable protein supports is computed only to count the days on which protein would limit gain. For a nursing calf, the energy balance $E^{\mathrm{ME}}$ is in metabolizable energy. A calf with a deficit changes weight by $k_m E^{\mathrm{ME}}/(1.20 + 0.006\,w)$, with $k_m = 0.719$ on milk only and $k_m = 0.69$ on milk and solid feed. A calf with a surplus retains $\mathrm{RE} = k_g E^{\mathrm{ME}}$, where $k_g$ is the mean of 0.55 for milk (National Academies of Sciences, Engineering, and Medicine 2021) and the ratio of net energy for gain to metabolizable energy of the solid diet (National Research Council 2000), weighted by the shares of milk and solid feed in the metabolizable energy supply of the calf. Its empty-body

gain EBG solves the young-calf equation RE = $\mathrm{EBG}^{1.1}\,\mathrm{EBW}^{0.205}$ (National Academies of Sciences, Engineering, and Medicine 2021), where the empty body weight EBW is 0.94 times the live weight on milk only and 0.93 times on milk and solid feed. The shrunk weight gain is 0.96 EBG/0.94 on milk only and 0.96 EBG/0.91 on milk and solid feed. Calves above 225 kilograms of live weight do not gain. Cow weight does not rise above 605 kilograms, 1.10 times the 550-kilogram mature weight of the animal model.

The body condition score of a cow cohort on the nine-point scale is linear in its weight,

$$\mathrm{BCS} = 5 + \frac{w/w_g^5 - 1}{0.0685}, \tag{C7}$$

so each score is 6.85 percent of the BCS-5 weight (Fox et al. 2003, p. 2.15). For mature cows, $w^5$ = 527 kilograms, so BCS 3, 4, 5, and 6 are 455, 491, 527, and 563 kilograms. The BCS-5 weight of 527 kilograms comes from the mean October weight of 448 cows of the Michigan State University research herd, 568 kilograms of live weight or 545 kilograms of shrunk weight, which we take as BCS 5.5. For first-calf cows, $w^5 = 0.85 \times 527 = 447.95$ kilograms, above the 80 percent of mature weight at first calving that National Research Council (2000, p. 32) assumes. The conception rate ρ of equation (B4) is 0.43, 0.61, 0.86, and 0.93 at BCS 3, 4, 5, and 6 (Kunkle, Sand, and Rae 1994), linear between these points and constant outside them, applied to the score of each cohort on the calving date, day 75. A simulation ends as infeasible, and its management system has no NPV, on the first day when the mature cows reach 385 kilograms (BCS 1.1), the first-calf cows reach 330 kilograms, or a heifer cohort reaches 0.70 times the weight at which it entered the herd. The 330 kilograms is the lower edge of the adult animal model, 0.60 times its mature weight.

### C.11. Coupled execution

The farm loop steps every calendar day from 1 January 2026 to 31 December 2035, 3,652 days. Leap days are off-pasture days. On each day $d$, the loop runs these steps:

1. The herd model applies the calendar events of the day, calving and conception on day 75 and weaning on day 270, and computes $D_{c,t,d}$ and $D_{t,d}$ from the current weights and the air temperature.
2. On a grazing day, the calendar of equation (B26) and the selective-grazing rule of equation (B7), applied to the patch states at the end of day $d - 1$, give the occupied patch. The loop computes $L_{t,d}$, the local density $u_{k,t,d}$, the grazeable forage $F$, and the hay $h_{t,d}$ and $s_{k,t,d}$. Off pasture, it computes the winter ration of each class and cohort.
3. The surrogate advances all twelve patches by one day. The occupied patch receives $(u_{k,t,d}, s_{k,t,d})$, and every other patch receives zero inputs. Off pasture, every patch receives zero inputs.

4. The loop allocates the realized feed among the classes, computes the energy balances, and updates the weights by equation (C6). If a weight reaches its floor, the simulation ends.
5. On 31 December, the herd model applies mortality and culling, sells the culled cows at their cohort weight, and the annual account computes profit.

At weaning, the first-calf cohort joins the mature cohort at the head-weighted mean weight, and the promoted heifers form the new first-calf cohort at their own weight, so the transition conserves live weight. The calves retained for replacement keep their weaning weight. The loop advances the 51 paddock designs of one history, feed-gap supplementation, and stocking rate together in one surrogate batch of 612 patches to save time.

### C.12. Full list of numerical assumptions

The numerical implementation rests on the following assumptions:

1. *Fixed management grid.* Each management system holds its planned herd target, installed paddock count, grazing period, and feed-gap supplementation constant across all years. Grazing-season dates are fixed. The analysis compares these management systems, and the dynamic optimization of equation (B1) remains future work.
2. *Planned and daily LSU are distinct.* The design label $H/A$ uses the reference class coefficients in equation (B2). Daily local pressure instead uses demand-equivalent $L_{t,d}$ from equation (B6).
3. *Consistent demographic accounting.* Pending calves before calving and live calves afterward belong to the same cohort, and the demographic controller's realized cow, heifer, and calf counts drive both daily demand and annual accounting.
4. *Identical start.* The twelve patches share one soil, weather sequence, species mix, and start state. Their within-farm heterogeneity emerges from divergent grazing and hay histories.
5. *Deterministic movement.* The herd follows equation (C1) deterministically, so each simulation yields a single trajectory.
6. *Feed-driven body weight.* Body weights change with the energy balance of the realized diet through equation (C6). A protein shortage does not limit gain, and the days on which it would limit gain are counted. Milk follows a fixed curve and does not respond to cow condition.
7. *One representative weight per class and cohort.* All animals in a class, or in a cow cohort, share one weight, and there is no within-group weight distribution.

8. *Hay rule.* On pasture, the producer feeds the fixed share $\phi$ of the feed gap, with no minimum. Off pasture, every management system uses the same winter program of hay and corn. All hay has one price and no waste.
9. *Conception from cow condition.* The conception rate of a cow cohort depends only on its body condition at calving. All open cows are culled at the year boundary.
10. *Infeasible management.* A simulation whose cows or heifers reach their weight floor ends on that day and is excluded from the NPV ranking. The results report how often this happens.
11. *Surrogate forage quality.* When the surrogate state implies a plant crude protein outside the monthly band of the MEMS training simulations at the site, the crude protein is clipped into the band. Standing dead forage has a fixed crude protein.
12. *Exogenous forcing.* Every simulation receives the same forcing path described in Section 3.1: ten years of historical weather at a constant carbon dioxide concentration.
13. *No economic feedback within a simulation.* Profit is computed after the simulation and never alters biophysical or herd dynamics. The cash basis excludes land rent, the opportunity cost of grazed forage, and carbon payments.
14. *Calendar management-paddock advancement.* The calendar alone determines when the herd advances to the next management paddock. Within the active paddock, previous-day biomass and forage quality affect the daily grazing value and can change which biophysical patch is selected through equation (C1). A forage-triggered advancement rule is reserved for future analysis.

### C.13. Calibration and parameter values

Table C1 reports the parameters needed to interpret and reproduce the management comparison. Appendix D gives the surrogate architecture and validation settings.

The study site is one site of the surrogate training data in Michigan. The 60-hectare farm is a representative-farm design assumption for the numerical experiment. The 5-hectare patch size sets the resolution at which grazing within a patch is approximately uniform, dividing the farm into twelve patches. The soil profile comes from SSURGO, and the five start states come from the MEMS simulations of the site under its five management histories.

Planned class shares and per-head reference livestock-unit coefficients come from the USDA Census class structure together with herd-survey composition, against a 500-kilogram reference animal. They convert $H$ to a planned cow target, and the demographic controller then constructs attrition-consistent opening counts and event-driven daily counts. For example, $H = 60$ gives the planned target vector $(46.15, 9.23, 36.92)$ for cows,

heifers, and calves, but the first opening state contains 40.01 cows, 6.24 heifers, no live calves, and 40.01 pending calves that enter on the calving date. On 1 January 2026, all cows are mature, weigh 527 kilograms, and are pregnant, and heifers weigh 255 kilograms. Calves are born at 38 kilograms on day 75. From then on, every weight follows equation (C6). Calving and weaning dates and the demographic rates in Table C1 are fixed across management systems. The demand-equivalent conversion uses the MEMS value of 15 kilograms of dry matter per LSU per day in equation (B6).

The half-saturation biomass, handling time, and quality exponent of the movement rule (C1) are set from the functional-response and bite-mechanics literature on grazing intake and from the quality-preference literature. The travel-cost coefficients and the nine-hour daily grazing time budget come from grazing-behavior studies and movement-rule calibration, and the preference map encodes a corner-water gradient as the non-forage attractor.

Physical winter and grazing-season hay and winter corn replace the broad ERS feed aggregates, which are excluded to prevent double counting. Both hay streams use one price derived from Michigan hay prices, and corn uses the Michigan price received. Herd nonfeed operations, mineral, labor, fence, and replacement costs remain separate. The discount rate is 5 percent per year, and profit is computed on a cash basis with no carbon revenue, consistent with equation (C2).

TABLE C1. Selected parameter values and model settings for the numerical experiment of 16,575 simulations.

| Quantity | Value | Unit | Source / note |
|---|---|---|---|
| **Site and farm** | | | |
| Latitude; elevation | 43.57; 208 | °N; m | surrogate training site |
| Farm area | 60 | ha | representative-farm design assumption |
| Patch area, count | 5, 12 | ha, — | within-patch uniformity scale |
| Soil texture (sand, clay) | 40 to 48, 21 to 24 | % | SSURGO, 16-layer profile |
| Soil pH | 7.5 to 8.0 | — | SSURGO |
| Grazing season | 120 to 270 | day of year | Michigan off-pasture October to April |
| **Herd composition and weight** | | | |
| Planned class shares (cow / heifer / calf) | 0.50 / 0.10 / 0.40 | — | management-target conversion; USDA Census + herd survey |
| LSU coefficients | 1.00 / 0.70 / 0.20 | — | 500 kg reference weight |

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| Quantity | Value | Unit | Source / note |
|---|---|---|---|
| Planned weighted LSU per head $\bar{\ell}$ | 0.65 | — | derived from target shares |
| Demand capacity $\kappa_{\mathrm{MEMS}}$ | 15 | kg DM LSU$^{-1}$ d$^{-1}$ | MEMS conversion, equation (B6) |
| Calving, weaning | 75, 270 | day of year | fixed herd calendar |
| Initial weight (cow / heifer) | 527 / 255 | kg head$^{-1}$ | herd start on 1 January 2026, cows at BCS 5; heifers at the median of later cohorts in a preliminary simulation |
| Calf birth weight | 38 | kg head$^{-1}$ | animal model default |
| BCS-5 weight (mature / first-calf) | 527 / 447.95 | kg head$^{-1}$ | MSU October cows as BCS 5.5; 0.85 of mature |
| BCS step | 6.85 | % of BCS-5 weight | Fox et al. (2003), equation (C7) |
| Intake capacity (lactating / dry cow / heifer) | 2.4 / 2.1 / 2.5 | % of body weight | times a cold factor |
| Peak milk; first-calf factor | 8.0; 0.74 | kg d$^{-1}$; — | Angus, National Research Council (2000) |
| Cow tissue energy $\mathrm{NE}_{\mathrm{DLW}}$ | 0.5381 BCS + 3.2855 | Mcal kg$^{-1}$ | Fox et al. (2003); loss 0.80, gain 1.00 times, National Research Council (2000) |
| Heifer weight change | loss $E$/2.75; gain NRC growth | kg d$^{-1}$ | National Research Council (2000) |
| Calf weight gain | young-calf RE = $\mathrm{EBG}^{1.1}\mathrm{EBW}^{0.205}$; milk $k_g$ 0.55 | kg d$^{-1}$ | National Academies of Sciences, Engineering, and Medicine (2021); solid $k_g$ from National Research Council (2000) |
| Pregnancy energy | NRC Eq. 4-4, 4-7 | Mcal d$^{-1}$ | National Research Council (2000), CBW 38 kg |
| Cow weight cap | 605 | kg head$^{-1}$ | 1.10 times 550 kg mature weight |
| Weight floor (mature / first-calf / heifer) | 385 / 330 / 0.70 entry | kg head$^{-1}$ | reaching it ends the simulation |
| Conception at BCS 3 / 4 / 5 / 6 | 0.43 / 0.61 / 0.86 / 0.93 | fraction | Kunkle, Sand, and Rae (1994); linear between points |
| Mortality (cow / heifer / calf) | 0.015 / 0.015 / 0.030 | fraction yr$^{-1}$ | demographic controller |
| Cow culling | all open cows, at least 0.12 | fraction yr$^{-1}$ | open cows first |
| Female-calf ratio; heifer retention rate | 0.50; 0.15 | fraction | demographic controller |

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| Quantity | Value | Unit | Source / note |
|---|---|---|---|
| **Movement rule (equation C1)** | | | |
| Half-saturation biomass $B_{1/2}$ | 1,200 | kg DM ha$^{-1}$ | intake-response calibration |
| Handling time $h$ | 1.5 | s g$^{-1}$ DM | bite-mechanics literature |
| Quality exponent $\gamma$ | 2 | — | quality-preference literature |
| Travel cost ($\alpha_0$, $\alpha$) | 0.25, 0.02 | —, per 100 m | movement-rule calibration |
| Grazing time budget | 9 | h d$^{-1}$ | grazing-behavior literature |
| Preference map $P_k$ | corner-water gradient | — | non-forage attractor (water) |
| **Feed and economics (cash basis, equation C2)** | | | |
| Calf price | 264 | \$ cwt$^{-1}$ | fixed calibration value; realized weaning weight |
| Cull-cow price | 115 | \$ cwt$^{-1}$ | fixed calibration value |
| Cull-cow sale weight | cohort weight on 31 December | kg head$^{-1}$ | simulated |
| Hay price (pasture and winter) | 220 | \$ short ton$^{-1}$ consumed DM | Michigan hay price, 85% DM, 90% recovery; rounded |
| Hay composition (C, N; TDN) | 0.45, 0.015; 0.53 | fraction of DM | 9.4% crude protein |
| Corn (cows day 15 to 119; heifers off pasture) | 1.5 | kg DM head$^{-1}$ d$^{-1}$ | Drewnoski (2020), upper end |
| Corn price; TDN | 233.8; 0.88 | \$ t$^{-1}$ DM; fraction | USDA-NASS (2026); National Research Council (2000) feed 410 |
| Retained forage residual | 20 | g DM m$^{-2}$ | grazeable forage, equation (C4) |
| Plant crude protein band | monthly p1 to p99 | fraction of DM | MEMS at the study site; surrogate values outside are clipped |
| Standing dead crude protein | 0.0838 | fraction of DM | MEMS median at the study site |
| Forage net energy for maintenance | 0.95 + 3.30 CP | Mcal kg$^{-1}$ DM | bounded to 0.8 to 2.2 |
| Herd nonfeed cost | 280.84 | \$ cow$^{-1}$ yr$^{-1}$ | broad ERS feed aggregates excluded |
| Mineral cost | 54.75 | \$ cow$^{-1}$ yr$^{-1}$ | separate annual account |
| Labor base, wage | 27.99, 19.70 | \$ cow$^{-1}$, \$ h$^{-1}$ | farm labor parameterization |
| Move time | $30 + 0.10 n_{\text{cow}}$ | min move$^{-1}$ | 24-week labor season |

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| Quantity | Value | Unit | Source / note |
|---|---|---|---|
| Fence (permanent, temporary) | 2.51 / 20, 1.05 / 10 | \$ ft$^{-1}$ / yr life | Missouri Extension costs; NRCS/product-life assumptions |
| Replacement heifer | 1,200 | \$ head$^{-1}$ | model assumption |
| Discount rate | 5 | % yr$^{-1}$ | analysis convention |
| **Design grid** | | | |
| Planned stocking rate $H/A$ | 0.8 to 3.2 by 0.2 | LSU ha$^{-1}$ | 48 to 192 planned LSU on 60 ha |
| Paddock count $N$ | 1, 2, 3, 4, 6, 12 | — | continuous grazing ($N$ = 1); calendar-based rotation ($N$ > 1) |
| Grazing period, $N$ = 2, 3, 4 | 1 to 7, 10, 15, 20, 25, 30 | d | 12 levels |
| Grazing period, $N$ = 6, 12 | 1 to 7 | d | 7 levels |
| Scheduled management-paddock rest | $(N-1)$gp | d | calendar identity |
| Rotation start | advances by year | management paddock | fixed rotation; balances incomplete seasonal cycles |
| Feed-gap supplementation $\phi$ | 0, 0.25, 0.5, 0.75, 1 | — | share of the feed gap, equation (B9) |
| Management history | 5 | — | ungrazed, low-pressure continuous, low-pressure rotational, heavy continuous with supplement, intensive rotational with supplement |
| Start state | MEMS state at the end of 2025 | per history | all twelve patches equal |
| Forcing path | gridMET 1996 to 2005 | weather year | replayed for 2026 to 2035; $CO_2$ 427 ppm |
| Management years; SOC endpoints | 2026 to 2035 | calendar yr | SOC on 1 January 2026 and 31 December 2035 |
| Biophysical transition engine | surrogate of MEMS | — | advances all twelve patches one day at a time |
| Surrogate management inputs | local grazing density; hay | LSU ha$^{-1}$; g DM m$^{-2}$ d$^{-1}$ | occupied patch only |
| Feed allocation | maintenance first, then $w^{0.75}\times$ remaining capacity | — | Appendix C.10 |
| Simulations | 16,575 | — | 13 SR × 51 paddock designs × 5 levels of feed-gap supplementation × 5 histories |

## Appendix D. Surrogate architecture and validation

This appendix reports the input and output specification, the architecture, the training objective and data, and the validation details of the surrogate described in Section 3.2.

### D.1. Recurrence and management inputs

The 7.89-million-parameter model combines a convolutional encoder over the 16 soil layers with an encoder for the non-layered state, site, weather, calendar, and management inputs. Its management vector is $\mathbf{a}_{k,t,d} = (u_{k,t,d}, s_{k,t,d})$: the local grazing density in demand-equivalent LSU per hectare and the hay fed on the patch in grams of dry matter per square meter per day. In MEMS, one demand-equivalent LSU has a daily dry-matter demand of 15 kilograms. The fused representation enters a two-layer gated recurrent unit (GRU) with 768 hidden features.

For patch $k$ on day $d$, its recursion can be written compactly as

$$
\begin{aligned}
e_{k,t,d} &= \Phi_\theta(\widehat{X}_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}),\\
h_{k,t,d} &= \mathrm{GRU}_\theta(e_{k,t,d}, h_{k,t,d-1}),\\
\widehat{X}_{k,t,d} &= \mathcal{R}_{\mathrm{phys}}\Big(\widehat{X}_{k,t,d-1}, \Psi_\theta(h_{k,t,d})\Big).
\end{aligned}
\tag{D1}
$$

Here $\theta$ collects learned parameters, $\Phi_\theta$ encodes the state and daily inputs into feature vector $e_{k,t,d}$, $h_{k,t,d}$ is the GRU memory carried between days, and $\Psi_\theta$ maps that memory to the output heads. The farm-management layer constructs $u_{k,t,d}$ causally before the transition, and $\mathcal{R}_{\mathrm{phys}}$ reconstructs the next physical state while applying the output links. The heads of 409 state values predict their daily change, and the heads of the 16 soil-temperature layers predict their level. The GRU therefore gives the recursive emulator a learned memory of recent weather, vegetation, soil, and management history in addition to the explicit preceding state.

For farm simulation, the calendar and animal selective-grazing layers determine $u_{k,t,d}$ before the transition. The management-responsive approximation is

$$
\begin{aligned}
&F_{\mathrm{MEMS}}\Big(X_{k,t,d-1}, z_k, w_{t,d}, \mathcal{A}(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d})\Big)\\
&\qquad \simeq \widehat{F}_\theta\Big(X_{k,t,d-1}, z_k, w_{t,d}, u_{k,t,d}, s_{k,t,d}\Big),
\end{aligned}
\tag{D2}
$$

where $\mathcal{A}$ denotes the internal action construction used by MEMS. Paddock count, farm geometry, calendar scheduling, and selective movement remain outside the neural model in the farm-management layer. The hay input $s$ follows the feed-gap supplementation of each management system (Section 3.3). Appendix D.2 gives the complete management input and output specification.

### D.2. Input and output specification

The recurrent physical state contains 425 values from nine MEMS output families: soil organic matter, crop, soil nutrients, soil temperature, soil water content, surface organic matter, surface feces, standing dead biomass, and water-storage states. Its 23 layered variables keep the 16-layer soil profile, and 57 values are not layered. Each daily transition also receives 177 fixed site values (11 soil profiles over 16 layers, and the elevation) and 14 weather and calendar values of the target day: latitude, maximum and minimum air temperature, precipitation, solar radiation, relative humidity, wind speed, mean air temperature, daily growing degree days, the seven-day precipitation sum, the atmospheric carbon dioxide concentration, the sine and cosine of the day of year, and the photoperiod. The neural model predicts the complete next state. The forage available to grazing is computed after the prediction, as the dry matter of the plant and standing dead pools above a retained residual of 20 grams per square meter.

The surrogate receives two management inputs for the target day: the local grazing density in demand-equivalent LSU per hectare and the hay fed on the patch in grams of dry matter per square meter per day. It learns the direct mapping on the right side of equation (D2). MEMS constructs the corresponding removal, trampling, fecal, and urinary fluxes internally on the left side. MEMS lets the animals eat forage before hay, so the hay changes the patch state through the excreta of the animals that eat it. Paddock count, geometry, calendar scheduling, and selective movement stay in the farm-management layer.

### D.3. Architecture, training objective, and training data

The structured recurrent model has 7,886,698 parameters. A one-dimensional convolution over the 16 soil layers reads the 23 layered state variables, the 11 site profiles, and an eight-value depth embedding with 64 channels. Three residual depth blocks follow, and a projection gives 256 profile features. A two-layer multilayer perceptron encodes the 57 non-layered state values, the elevation, the 14 weather and calendar values, and the two management inputs into 256 features. A fusion layer combines the two representations into 512 features before the two-layer, 768-feature GRU in equation (D1). An output head with 512 hidden features maps the GRU state to the 425 daily outputs. The heads of 409 state values predict the daily change, which is added to the preceding state. The heads of the 16 soil-temperature layers predict the level. The output links clamp the 411 state values whose training minimum is not negative at zero and clamp every state value to the range of the training data.

Let $\mathcal{D}$ be the set of days in a training window and let $x_{j,d}$ and $\widehat{x}_{j,d}$ be the target and prediction for state value $j$ on day $d$ in the native representation of its head: a daily change

or a level. Let $\rho_1(v)$ denote the smooth-L1 function with unit threshold, let $m_j$ identify the state values with a loss, let $\sigma_j$ be the daily-change scale of a change head or the level scale of a level head, and let $w_j$ be the channel weight. The principal state-transition loss is

$$\mathcal{L}_{\text{trans}} = \frac{1}{\sum_j w_j m_j} \sum_j w_j m_j \left[ \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \rho_1 \left( \frac{\widehat{x}_{j,d} - x_{j,d}}{\sigma_j} \right) \right]. \tag{D3}$$

The channel weights come from a fixed recursive rollout on training simulations, repeated every 300 updates: $w_j = \min\{10, \max[0.1, (r_j/0.2912)^{0.5}]\}$, where $r_j$ is the rollout MAE of state value $j$ divided by its persistence MAE, the error of holding the first state of each window constant. A value that drifts more than its own change over the horizon therefore receives more weight. On the eight forage carbon pools, each training sample also receives the level weight $[(L_s + 0.1R)/R]^{-1}$, normalized to a batch mean of one, where $L_s$ is the mean true forage carbon of the sample window and $R$ is the forage carbon of the normalizer reference state. With this weight, the forage loss is close to a relative error, so simulations with little forage are not neglected.

Two auxiliary terms address failure modes beyond the one-day transition loss. The drift term $\mathcal{L}_{\text{drift}}$ penalizes the mean error over a training window of the 409 change heads, divided by the daily-change scale, so that small daily biases do not accumulate recursively. For the seven decision outputs $\mathcal{J}_7$ (total forage, live leaf, live stem, and dead plus seed dry matter, total SOC, mineral-associated organic matter, and particulate organic matter), the level term is

$$\mathcal{L}_{\text{level}} = \frac{1}{7|\mathcal{D}|} \sum_{j \in \mathcal{J}_7} \sum_{d \in \mathcal{D}} \rho_1 \left( \frac{\widehat{y}_{j,d} - y_{j,d}}{|y_{j,d}| + 0.15\, \bar{y}_j} \right), \tag{D4}$$

where $y_{j,d}$ and $\widehat{y}_{j,d}$ are the true and predicted physical levels and $\bar{y}_j$ is the value of output $j$ at the normalizer reference state. This term makes persistent errors visible during low-biomass or nearly stationary periods. The objectives are

$$\mathcal{L}^{\text{teacher}} = \mathcal{L}_{\text{trans}} + 0.30\, \mathcal{L}_{\text{drift}}, \qquad \mathcal{L}^{\text{push}} = \mathcal{L}^{\text{teacher}} + 0.20\, \mathcal{L}_{\text{level}}. \tag{D5}$$

Teacher-forced windows train against true preceding states. Pushed windows feed the surrogate's own preceding prediction back through equation (D1), and each pushed day carries a loss with probability 0.25. The horizon curriculum increases from one day to 1,095 days while the pushed-window probability rises to 0.90. In pushed windows, the gradient passes through 60 days of the physical state.

Training is a chain of five parts with 59,826 optimizer steps on one graphics processor (NVIDIA A100), about 52 hours in all, with one random seed. Each part starts from the weights of the one before. The final part runs 5,000 steps with batches of 4,096 pushed

windows of 1,095 days. It uses AdamW with a peak learning rate of $1.2 \times 10^{-4}$ after a 500-step warmup, a cosine decay to $1.2 \times 10^{-5}$ at step 2,700 and a constant rate after it, weight decay $10^{-4}$, gradient-norm clipping at 1.0, and bfloat16 mixed precision.

The training data are the training split of a set of MEMS simulations: 1,667 simulations with 18,252,901 daily transitions. Each simulation starts on 1 January 2026 and runs 30 years with gridMET weather of 1996 to 2025 (Abatzoglou 2013), repeated for 2026 to 2055, and a constant carbon dioxide concentration of 427 parts per million. The simulations cover 39 sites: 36 in Michigan, 2 in Ohio, and 1 in Wisconsin. Each site has five histories of management before 2026: ungrazed, low-pressure continuous, low-pressure rotational, heavy continuous with supplement, and intensive rotational with supplement. Each history gives one MEMS state at the end of 2025, and nine grazing schedules continue from it. The schedules come from 90 templates, 72 rotational and 18 continuous, with daily local densities from 0.5 to 36 LSU per hectare between day of year 120 and 270. On 56 percent of the grazing days, the schedule also feeds hay, with a median of 2.45 and a maximum of 54 grams of dry matter per square meter per day. Two simulations in which the forage of MEMS collapses remain in the training data.

### D.4. Validation protocol and detailed results

The test split holds the 18 test simulations, which training never uses: the nine schedules of test site 1 after the low-pressure rotational history and the nine schedules of test site 2 after the history of heavy continuous grazing with supplement. Training sees these two sites only after their other histories, so the test withholds combinations of site and history, not whole sites. Each test forecast starts from the MEMS state at the end of 31 December 2025 and predicts recursively for 3,652 days in 32-bit floating point, with the recorded weather and management of the simulation. We chose the final checkpoint from the late checkpoints of training by a selection score on these 18 test simulations. The test results are therefore optimistic by the noise of that choice. The last two checkpoints differ by up to 3.7 percentage points on the worst of these selection scores.

For output $j$ of the pooled test days, the relative error is

$$\mathrm{RE}_j = 100 \, \frac{\sum_{r,d} \left| \widehat{y}_{rjd} - y_{rjd} \right|}{\sum_{r,d} \left| y_{rjd} \right|}, \tag{D6}$$

summed over test simulations $r$ and days $d$. The skill score against a reference forecast is $1 - \mathrm{MAE}/\mathrm{MAE}_{\mathrm{ref}}$.

Figure 2 shows one test simulation at each test site, the one with the median plant C error. Figure D1 shows that the plant errors do not grow with the forecast horizon: the plant C MAE is 27.5 to 45.7 kilograms per hectare in each year. The SOC error grows from

TABLE D1. Ten-year accuracy of the surrogate against MEMS, 18 test simulations. The plant rows count the days of the grazing season (day of year 120 to 270), 27,180 days in all. The total SOC row counts one value per test simulation, the stock at the end of the 10 years. The relative error is the sum of the absolute errors divided by the sum of the absolute MEMS values. The skill score is 1 – MAE/$MAE_{ref}$. Persistence holds the MEMS value at the end of 1 January 2026, and Figure D1 defines the site climatology. The number in parentheses is the number of test simulations in which the surrogate MAE is lower than the reference MAE.

| | MEMS | | | Relative | Skill score against | |
|---|---|---|---|---|---|---|
| Output | mean | MAE | Bias | error (%) | Persistence | Site climatology |
| Plant C (kg C $ha^{-1}$) | 459 | 36.5 | +1.8 | 7.95 | 0.92 (18) | 0.94 (18) |
| Plant N (kg N $ha^{-1}$) | 17.9 | 1.58 | +0.17 | 8.84 | 0.91 (18) | 0.93 (18) |
| Total SOC in year 10 (Mg C $ha^{-1}$) | 119 | 0.48 | −0.41 | 0.403 | 0.98 (18) | 0.99 (18) |

0.043 Mg C $ha^{-1}$ in the first year to 0.65 Mg C $ha^{-1}$ in the seventh, and it stays at 1.3 to 2.3 percent of the site-climatology error in every year (Appendix Table D2). On the first forecast day, one step from the MEMS state, the relative error is 1.26 percent for plant C, 0.654 percent for plant N, and 0.00131 percent for total SOC (Appendix Table D3). On the second day, the plant errors are already 6.40 and 9.05 percent, so the first-day values do not measure the accuracy of a longer forecast. All 18 test forecasts are free of non-finite values and of negative values in the 411 nonnegative state values.

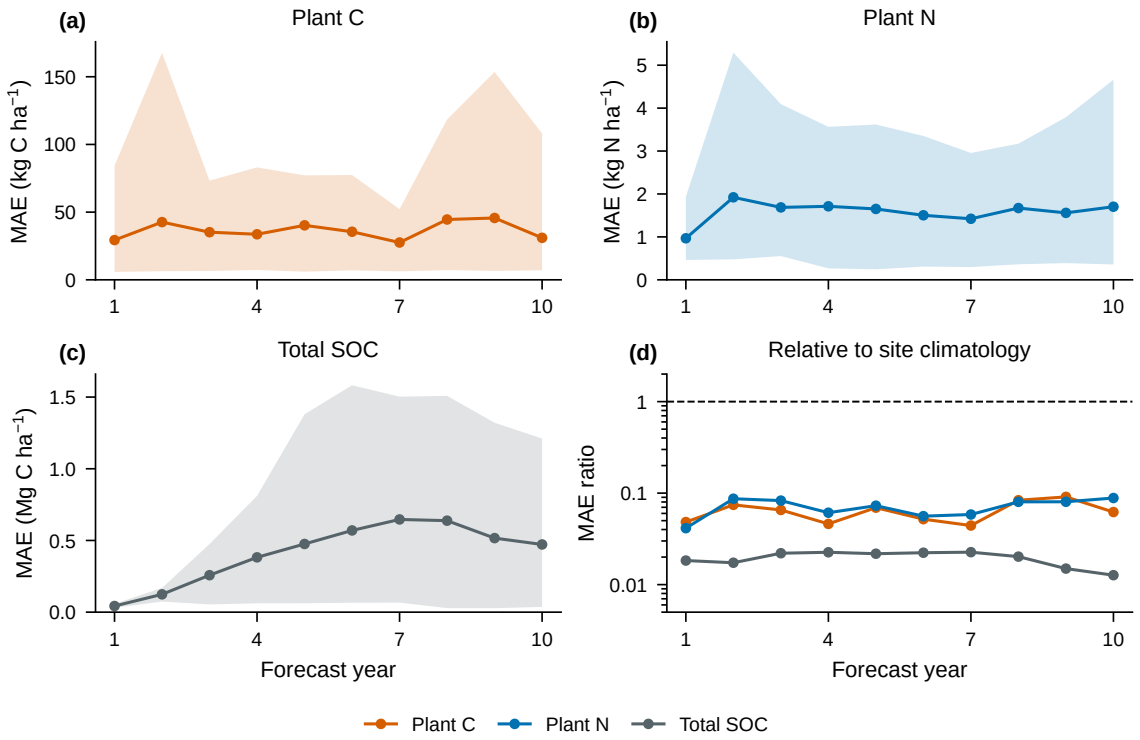



FIGURE D1. **Surrogate error in each forecast year, 18 test simulations.** Panels (a) to (c) show the mean absolute error (MAE) of each output in each year, pooled over the test simulations. The shaded band spans the 10th to 90th percentiles of the MAE of the test simulations. Plant C and N count the days of the grazing season, and total SOC counts all days. Panel (d) divides the surrogate MAE by the MAE of the site climatology in the same year, on a logarithmic axis. Below the dashed line, the surrogate is more accurate than the site climatology. The site climatology is the mean MEMS value of each day of the year over 2036 to 2055, of all training simulations at the test site. The training simulations start from other SOC stocks, so for total SOC the site climatology adds the mean change of these simulations to the MEMS value of the test simulation at the end of 1 January 2026.

Table D2 gives the MAE of each forecast year, and Table D3 gives the accuracy on the first forecast day. Figure D2 compares the total SOC at the end of the tenth year on the 18 test simulations and on 417 training simulations. The largest test error is 2.71 Mg C $ha^{-1}$. On the seven checkpoints from step 2,500 to step 5,000 of the final training part, the ten-year MAE is 36.5 to 37.9 kilograms per hectare for plant C, 1.58 to 1.69 kilograms per hectare for plant N, and 0.43 to 0.51 Mg C $ha^{-1}$ for total SOC. This range is the only available measure of the training variation, because the model has one random seed. The model was trained on the training split only.

TABLE D2. Surrogate MAE in each forecast year, pooled over the 18 test simulations. The plant columns count the days of the grazing season. Total SOC counts all days of the year. Figure D1 shows the spread over the test simulations.

| Forecast year | Plant C (kg C ha$^{-1}$) | Plant N (kg N ha$^{-1}$) | Total SOC (Mg C ha$^{-1}$) |
|---|---|---|---|
| 1 | 29.3 | 0.967 | 0.0428 |
| 2 | 42.6 | 1.92 | 0.124 |
| 3 | 35.2 | 1.69 | 0.258 |
| 4 | 33.6 | 1.71 | 0.382 |
| 5 | 40.2 | 1.65 | 0.475 |
| 6 | 35.5 | 1.50 | 0.570 |
| 7 | 27.5 | 1.42 | 0.647 |
| 8 | 44.6 | 1.67 | 0.638 |
| 9 | 45.7 | 1.56 | 0.516 |
| 10 | 31.0 | 1.70 | 0.472 |

TABLE D3. Accuracy of the surrogate against MEMS on the first forecast day (1 January 2026), 18 test simulations. The surrogate makes one step from the MEMS state at the end of 31 December 2025. Each output counts one value per test simulation. All test simulations of one site start from the same state, so this day has only 2 distinct MEMS values, one per site. The relative error is defined as in Table D1.

| Output | MEMS mean | MAE | Relative error (%) |
|---|---|---|---|
| Plant C (kg C ha$^{-1}$) | 29.2 | 0.369 | 1.26 |
| Plant N (kg N ha$^{-1}$) | 0.882 | 0.00577 | 0.654 |
| Total SOC (Mg C ha$^{-1}$) | 136 | 0.00179 | 0.00131 |

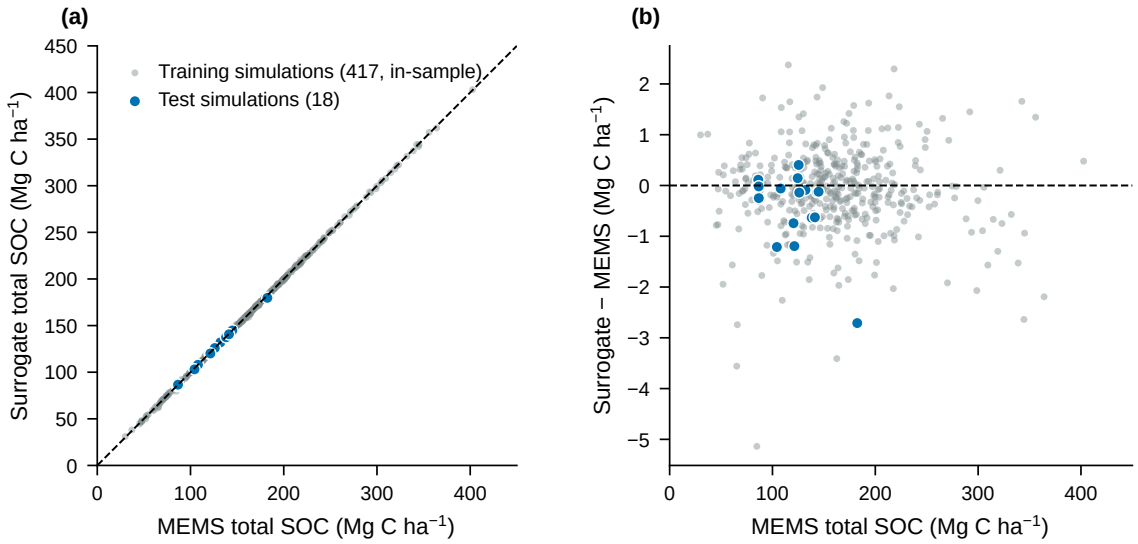



FIGURE D2. **Total SOC at the end of the tenth year, surrogate against MEMS.** Each point is one MEMS simulation. Panel (a) compares the two stocks, and the dashed line is the 1:1 line. Panel (b) shows the surrogate error against the MEMS stock. The training simulations are in-sample.

## Appendix E. Additional results and limits

This appendix reports the results that Section 5 summarizes or leaves out, and the limits of the model. Table E1 gives the highest-NPV management system of each start history at each planned stocking rate.

TABLE E1. **The highest-NPV management system at each planned stocking rate.** Each start history has 255 management systems at each planned stocking rate (SR, LSU ha$^{-1}$): 51 paddock designs at each of five levels of feed-gap supplementation $\phi$. A simulation is complete when it runs the ten years 2026 to 2035 without reaching an animal weight floor. For each start history and SR, the table gives the number of complete simulations and the complete management system with the highest ten-year NPV. CG is continuous grazing on one paddock, which has no grazing period. Feed-gap supplementation $\phi$ is the share of each grazing-season day's feed gap that hay on pasture fills. NPV is the ten-year net present value of the 60-hectare farm in $ thousands, at a 5 percent discount rate. $\Delta$SOC is the change of farm mean total SOC from 1 January 2026 to 31 December 2035. Bold rows give the profit optimum of each start history, the complete management system with the highest NPV over all SR. Every SR has at least one complete simulation in every start history.

| SR (LSU ha$^{-1}$) | Complete simulations | Paddocks | Grazing period (d) | $\phi$ | NPV ($ thousand) | $\Delta$SOC (Mg C ha$^{-1}$) |
|---|---|---|---|---|---|---|
| *Ungrazed* | | | | | | |
| 0.8 | 255 | 1 | CG | 1 | 37.9 | −9.15 |
| 1.0 | 255 | 1 | CG | 0.75 | 50.2 | −12.12 |
| 1.2 | 255 | 1 | CG | 1 | 60.8 | −14.72 |
| 1.4 | 253 | 2 | 10 | 1 | 70.0 | −16.16 |
| 1.6 | 253 | 2 | 5 | 1 | 80.7 | −17.72 |
| 1.8 | 248 | 2 | 10 | 1 | 91.7 | −21.54 |
| 2.0 | 240 | 2 | 5 | 1 | 101.4 | −23.71 |
| **2.2** | **234** | **2** | **4** | **1** | **108.9** | **−27.38** |
| 2.4 | 195 | 2 | 4 | 1 | 107.3 | −31.78 |
| 2.6 | 89 | 4 | 7 | 1 | 82.4 | −36.52 |
| 2.8 | 49 | 6 | 6 | 1 | 41.2 | −38.58 |
| 3.0 | 40 | 12 | 4 | 1 | 2.6 | −35.84 |
| 3.2 | 25 | 12 | 6 | 1 | −37.6 | −33.13 |
| *Low-pressure continuous* | | | | | | |
| 0.8 | 253 | 1 | CG | 1 | 36.3 | −0.49 |
| 1.0 | 254 | 2 | 25 | 1 | 46.9 | −2.32 |
| 1.2 | 254 | 1 | CG | 1 | 59.3 | −5.35 |

*Continued on the next page*

*Table E1, continued*

| SR (LSU ha$^{-1}$) | Complete simulations | Paddocks | Grazing period (d) | ϕ | NPV ($ thousand) | ΔSOC (Mg C ha$^{-1}$) |
|---|---|---|---|---|---|---|
| 1.4 | 253 | 1 | CG | 1 | 70.1 | −7.87 |
| 1.6 | 252 | 2 | 5 | 1 | 80.1 | −8.27 |
| 1.8 | 245 | 2 | 5 | 1 | 91.3 | −10.40 |
| 2.0 | 239 | 2 | 5 | 1 | 99.7 | −13.20 |
| **2.2** | **229** | **2** | **5** | **1** | **106.9** | **−16.55** |
| 2.4 | 176 | 4 | 3 | 1 | 105.8 | −21.84 |
| 2.6 | 83 | 2 | 20 | 1 | 69.6 | −27.34 |
| 2.8 | 48 | 12 | 4 | 1 | 36.3 | −22.10 |
| 3.0 | 32 | 12 | 4 | 1 | −6.7 | −25.02 |
| 3.2 | 21 | 12 | 6 | 1 | −42.1 | −21.82 |
| *Low-pressure rotational* | | | | | | |
| 0.8 | 255 | 1 | CG | 1 | 38.3 | −6.36 |
| 1.0 | 253 | 1 | CG | 1 | 47.9 | −9.20 |
| 1.2 | 254 | 1 | CG | 0.75 | 60.5 | −12.32 |
| 1.4 | 255 | 1 | CG | 1 | 70.5 | −14.87 |
| 1.6 | 251 | 2 | 6 | 1 | 81.1 | −15.04 |
| 1.8 | 249 | 2 | 5 | 1 | 91.9 | −17.12 |
| 2.0 | 242 | 2 | 10 | 1 | 102.5 | −21.12 |
| **2.2** | **230** | **2** | **15** | **1** | **109.2** | **−24.48** |
| 2.4 | 197 | 2 | 6 | 1 | 108.3 | −27.90 |
| 2.6 | 87 | 4 | 7 | 1 | 82.1 | −33.36 |
| 2.8 | 50 | 12 | 4 | 1 | 41.5 | −29.43 |
| 3.0 | 37 | 12 | 4 | 1 | 1.1 | −32.45 |
| 3.2 | 22 | 12 | 6 | 1 | −38.6 | −29.63 |
| *Heavy continuous* | | | | | | |
| 0.8 | 156 | 1 | CG | 1 | 33.4 | 45.31 |
| 1.0 | 149 | 2 | 30 | 1 | 42.4 | 43.74 |
| 1.2 | 143 | 1 | CG | 1 | 54.6 | 40.88 |
| 1.4 | 135 | 2 | 10 | 1 | 61.7 | 40.27 |
| 1.6 | 129 | 2 | 15 | 1 | 71.3 | 37.57 |
| 1.8 | 103 | 2 | 10 | 1 | 79.6 | 33.90 |
| **2.0** | **93** | **2** | **6** | **1** | **81.7** | **30.82** |
| 2.2 | 78 | 2 | 10 | 1 | 70.5 | 26.91 |
| 2.4 | 54 | 2 | 20 | 1 | 41.1 | 23.05 |

*Continued on the next page*

*Table E1, continued*

| SR (LSU ha$^{-1}$) | Complete simulations | Paddocks | Grazing period (d) | ϕ | NPV ($ thousand) | ΔSOC (Mg C ha$^{-1}$) |
|---|---|---|---|---|---|---|
| 2.6 | 51 | 2 | 25 | 1 | 13.1 | 20.55 |
| 2.8 | 48 | 12 | 4 | 1 | −21.5 | 23.46 |
| 3.0 | 46 | 12 | 6 | 1 | −52.9 | 25.58 |
| 3.2 | 34 | 12 | 6 | 1 | −87.4 | 23.56 |
| *Intensive rotational* | | | | | | |
| 0.8 | 255 | 1 | CG | 1 | 32.1 | 7.31 |
| 1.0 | 255 | 1 | CG | 1 | 45.1 | 5.47 |
| 1.2 | 255 | 1 | CG | 1 | 57.5 | 3.60 |
| 1.4 | 251 | 2 | 7 | 1 | 63.8 | 2.10 |
| 1.6 | 238 | 2 | 10 | 1 | 73.9 | 0.80 |
| 1.8 | 230 | 2 | 10 | 1 | 82.1 | −0.82 |
| **2.0** | **207** | **2** | **5** | **1** | **87.7** | **−3.38** |
| 2.2 | 129 | 4 | 4 | 1 | 82.3 | −4.94 |
| 2.4 | 75 | 4 | 6 | 1 | 69.8 | −7.00 |
| 2.6 | 48 | 6 | 6 | 1 | 37.0 | −8.12 |
| 2.8 | 47 | 6 | 7 | 1 | −1.7 | −9.32 |
| 3.0 | 30 | 12 | 4 | 1 | −44.9 | −8.65 |
| 3.2 | 22 | 12 | 6 | 1 | −82.8 | −5.02 |

## E.1. Which simulations complete

Figure E1 shows the share of complete simulations for each history, feed-gap supplementation, and stocking rate. Completion falls with stocking rate, from 92.1 percent of the simulations at 0.8 LSU per hectare to 9.7 percent at 3.2 LSU per hectare, and it rises with the feed-gap supplementation. In the ungrazed history without hay on pasture ($\phi = 0$), all 51 paddock designs complete up to 1.2 LSU per hectare, and at least one paddock design completes up to 2.4 LSU per hectare. With $\phi = 1$, all paddock designs complete up to 2.4 LSU per hectare, and at least one completes up to 3.2 LSU per hectare. The low-pressure continuous and rotational histories behave like the ungrazed history, and the intensive rotational history completes less often.

The heavy continuous history needs hay. No simulation of this history completes at $\phi = 0$, 7 simulations complete at $\phi = 0.25$, and the highest stocking rate with a complete simulation is 1.0, 1.8, 2.4, and 3.2 LSU per hectare at $\phi = 0.25$, 0.5, 0.75, and 1. This history starts with 84.1 Mg C per hectare of SOC, against 158.5 to 218.9 in the other histories, and its patches offer less forage. With twelve paddocks, a five-day grazing period, $\phi = 1$, and

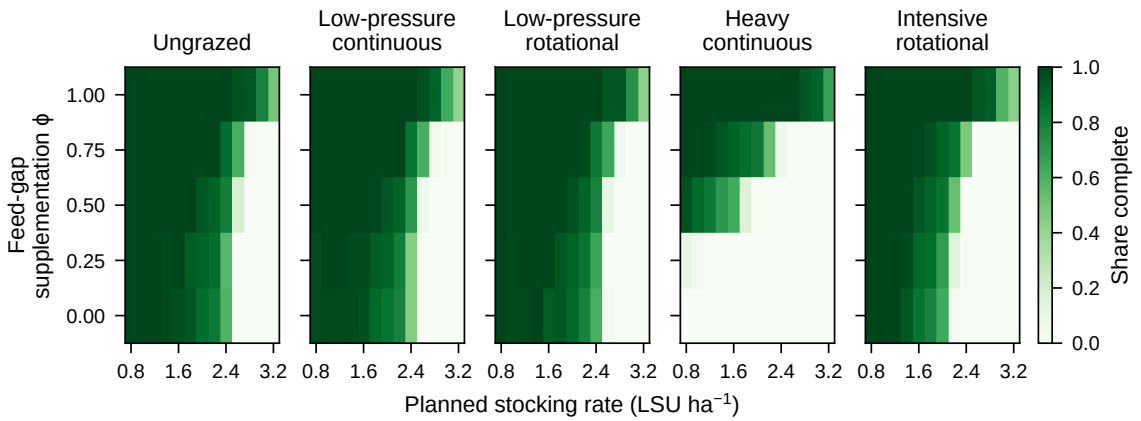



FIGURE E1. **Share of complete simulations.** Each tile is one planned stocking rate and level of feed-gap supplementation $\phi$ of one start history and contains the 51 paddock designs. The shade gives the share of these simulations that complete the ten years. A simulation ends early on the first day when the mature cows reach 385 kilograms, the first-calf cows reach 330 kilograms, or a heifer cohort reaches 70 percent of its entry weight.

0.8 LSU per hectare, the occupied patch has a mean of 152 grams per square meter of grazeable forage on the grazing days of 2026, against 370 grams per square meter in the ungrazed history. Without hay at 0.8 LSU per hectare, every simulation of this history ends by 2027. In the 29 of these 51 simulations that reach weaning in 2026, the mature cows fall from 527 kilograms to 419 to 454 kilograms, and the heifers reach only 321 to 346 kilograms.

Simulations that fail end early. Of the failed simulations, 91.9 percent end in 2026 to 2029, and 99.0 percent of the failed simulations of the heavy continuous history end in 2026 and 2027. The first-calf cohort is the limit in 91 percent of the failed simulations. In 4,317 simulations, the first-calf cows reach their floor of 330 kilograms, and in 1,320 simulations the bred heifers join the first-calf cohort at weaning below that floor. The mature cows reach their floor of 385 kilograms in 565 simulations, 556 of them in the heavy continuous history. The heifer floor ends no simulation. On day 270 of the complete simulations, the median bred heifer is 88 kilograms above the first-calf floor when it joins the cow herd, and the median mature cow is 187 kilograms above its own floor. A shortage of forage therefore reaches the first-calf floor first.

The selective-grazing rule contributes to these failures. Without hay on pasture, 54 percent of the continuous-grazing simulations of the ungrazed history complete, against 63 percent of its calendar-based-rotation simulations. The rule of equation (C1) scores a patch by its biomass and its live-leaf share, but the animals eat only the forage above the 20-gram residual. A patch that is grazed to the residual and then grows new leaves has a high live-leaf share. Near the water attractor, such a patch can keep the highest

score while it supplies no grazeable forage. In the continuous-grazing simulation of the ungrazed history at 1.4 LSU per hectare without hay, the herd stays on one patch for 30 days from day of year 239 of 2031. From day 259, the patch has no grazeable forage on each of the last 10 days of the stay, and four of the twelve patches are not occupied in that year. The mature and first-calf cows lose 3.0 and 2.5 kilograms per day on these days, and the simulation ends on 5 April 2032, when the first-calf cows reach their floor. Continuous grazing without hay completes at 1.2 and 1.6 LSU per hectare in this history. Over the experiment, the longest run of grazing days without grazeable forage on the occupied patch has a median of 23 days in the failed continuous-grazing simulations and 6 days in the complete ones, and 75 percent of the failed continuous-grazing simulations have a run of 10 days or more. Calendar-based rotation moves the herd at the end of each grazing period, which limits such runs. The median longest run in the failed simulations is 14 days with two paddocks and 4 to 8 days with three or more paddocks. Part of the completion difference between the grazing regimes therefore comes from this rule (Appendix E.9).

### E.2. Animal weights and condition

Figure E2 compares the simulated animals with the October weights of the Michigan State University research herd, converted to shrunk weight with a 4 percent shrink: 545.5 kilograms for 448 mature cows, 207.8 kilograms for 554 calves at weaning, and 462.4 kilograms for 143 two-year-old first-calf heifers. We selected the calf growth equation with the weaning weight of this herd as the target, but we fitted no parameter to these weights. Over the complete simulations, the median mature cow weighs 572.1 kilograms on day 270 (5th to 95th percentile 551.2 to 576.0), 4.9 percent above the research herd. The median calf weighs 201.7 kilograms at weaning (193.5 to 203.6), 2.9 percent below the research herd. The median first-calf cow weighs 474.5 kilograms at its first weaning (412.9 to 487.7), 2.6 percent above the two-year-old heifers of the herd. The weights fall with stocking rate above 2.0 LSU per hectare.

The calf weight responds little to management. In the model, milk is an obligation of the cow, so a shortage of feed lowers the cow weight and not the milk (Appendix C.5). The median calf weight at weaning falls from 203.1 kilograms at 0.8 LSU per hectare to 194.6 kilograms at 3.2 LSU per hectare. Management therefore changes calf revenue through the number of calves, which follows the conception rate, and through a small change in calf weight. The winter ration returns the mature cows to BCS 5 before calving. Their median score at calving is 4.99 (4.95 to 4.99), and their median conception rate is 0.858. The first-calf cows carry the effect of the grazing season into the next year. Their median score at calving is 4.40, the 5th percentile is 3.47, and their conception rate is 0.714 at the median and 0.529 at the 5th percentile. Between BCS 4 and 5, the conception rate of equation (B4) rises by 0.25 per score, against 0.07 between BCS 5 and 6. The mature conception rate

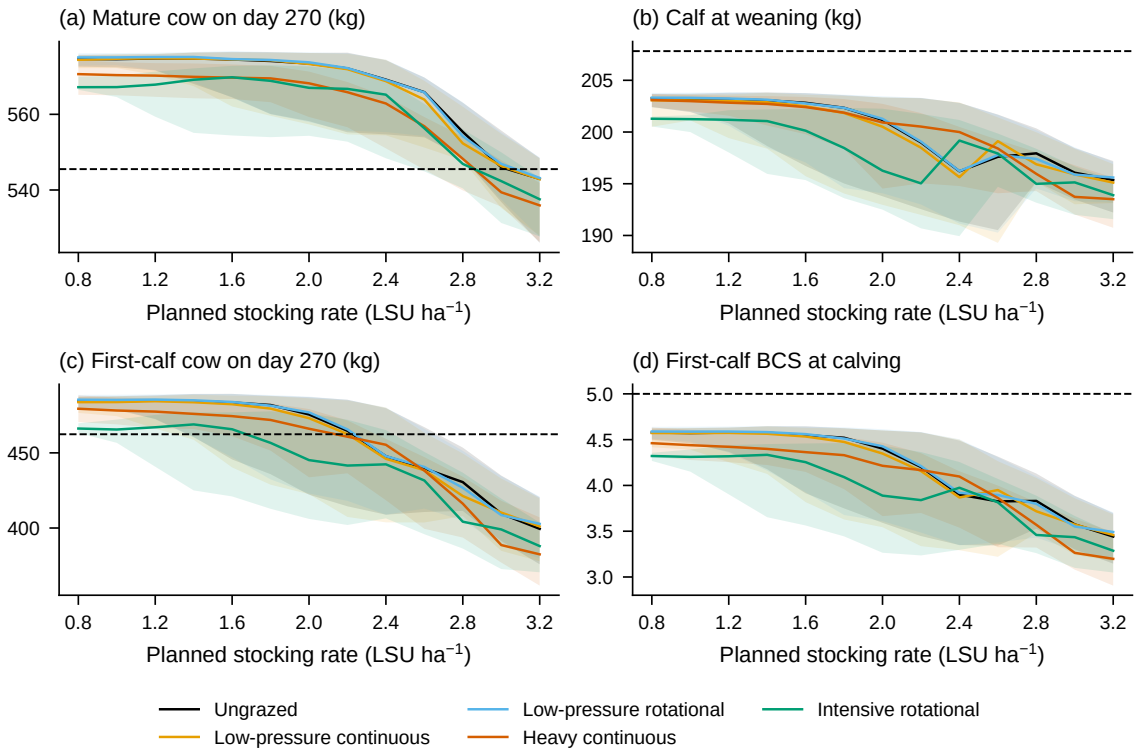



FIGURE E2. **Simulated animals against the Michigan State University research herd.** Lines are medians over the complete simulations of each start history at each planned stocking rate, and bands span the 5th to 95th percentiles. Each simulation enters as its mean over the ten years. Panels (a) to (c) give shrunk body weights on day 270, the weaning date. Panel (c) gives the first-calf cows before they join the mature cows. The dashed lines give the October weights of the research herd with a 4 percent shrink, and in panel (c) the weight of its two-year-old first-calf heifers. Panel (d) gives the body condition score (BCS) of the first-calf cows on the calving date, day 75, and the dashed line marks BCS 5.

varies only from 0.848 to 0.858 (5th to 95th percentile), so the condition of the first-calf cows sets most of the difference in the calf crop between management systems. Panel (d) of Figure E2 shows that the first-calf score falls with stocking rate in every history.

### E.3. Private-profit and soil-carbon trade-offs

Table E2 and Figure E3 report the highest-NPV management systems. Among the complete simulations, the ten-year NPV ranges from a loss of $243,026 to a gain of $109,205. The highest NPV in the grid, $109,205, uses the low-pressure rotational history, 2.2 LSU per hectare, two paddocks, a 15-day grazing period, and $\phi = 1$. The ungrazed history reaches $108,906 at the same stocking rate and feed-gap supplementation with two paddocks and a four-day grazing period. In the ungrazed and the two low-pressure histories, the highest NPV occurs at 2.2 LSU per hectare and $\phi = 1$. In the two histories with supplement, it

occurs at 2.0 LSU per hectare and $\phi = 1$, and it is \$19,171 to \$27,467 lower than in the low-pressure histories.

TABLE E2. Highest-NPV management system of each start history. SR is the planned stocking rate in LSU per hectare, NPV is in US dollars, and ΔSOC is the ten-year change of farm mean total SOC in Mg C ha$^{-1}$. Conception is the mean over the ten calving dates of both cow cohorts. A management system is nondominated when no other complete management system of the same history has both a higher NPV and a smaller SOC loss.

| History | SR | $\phi$ | Paddock design | NPV | ΔSOC | Conception | Complete simulations | Nondominated systems |
|---|---|---|---|---|---|---|---|---|
| Ungrazed | 2.2 | 1 | $N = 2$, gp= 4 | 108,906 | −27.38 | 0.842 | 2,391 | 128 |
| Low-pressure continuous | 2.2 | 1 | $N = 2$, gp= 5 | 106,878 | −16.55 | 0.843 | 2,339 | 124 |
| Low-pressure rotational | 2.2 | 1 | $N = 2$, gp= 15 | 109,205 | −24.48 | 0.841 | 2,382 | 124 |
| Heavy continuous | 2.0 | 1 | $N = 2$, gp= 6 | 81,738 | 30.82 | 0.839 | 1,219 | 54 |
| Intensive rotational | 2.0 | 1 | $N = 2$, gp= 5 | 87,707 | −3.38 | 0.837 | 2,042 | 53 |

Up to the peak, NPV rises with stocking rate because calf and cull sales grow faster than the costs (Figure E3a). Winter feeding dominates the costs. In the highest-NPV management system, the undiscounted ten-year sales are \$812,096 for calves and \$186,449 for cull cows. The feed cost is \$535,795, of which winter hay is \$483,671, corn \$43,243, and hay on pasture \$8,881. Above the peak, NPV falls because fewer simulations complete, the first-calf condition and the conception rate fall, and the simulations that complete need more hay on pasture (Section E.4).

Panel (b) of Figure E3 shows that the SOC change of the highest-NPV management system falls with stocking rate up to 2.6 LSU per hectare in every history. Within each history, the management system with the least SOC loss uses 0.8 LSU per hectare, and in four histories it uses twelve paddocks. Panel (c) shows the nondominated management systems of each history: no other complete management system of the same history has both a higher NPV and a smaller SOC loss. Each history has 53 to 128 nondominated management systems, which run from the least-loss management system to the highest-NPV one. In the ungrazed history, the frontier runs from an NPV of \$20,879 and an SOC change of −5.11 Mg C per hectare (twelve paddocks, three-day grazing period, $\phi = 0$) to \$108,906 and −27.38 Mg C per hectare. Along this frontier, the highest-NPV end gains \$88,027 of NPV and loses 22.3 Mg C per hectare more SOC than the least-loss end. On the 60-hectare farm, this is 1,336 Mg C, or \$66 of NPV per Mg C on average.

### E.4. Feed-gap supplementation

Figure E4 shows the ungrazed history at the five levels of feed-gap supplementation. Hay on pasture raises the highest NPV in every history, and the highest NPV of every history uses $\phi = 1$. The highest ungrazed NPV is \$93,659 at $\phi = 0$ and \$108,906 at $\phi = 1$, 16.3 percent

higher. In the four histories that complete without hay on pasture, the best management system at $\phi = 1$ earns 15 to 28 percent more than the best one at $\phi = 0$. Most of the gain comes from the first-calf cows. We pair each management system of these four histories at $\phi = 1$ with the management system of the same history, stocking rate, paddock count, and grazing period at $\phi = 0$. Up to 1.0 LSU per hectare, there is little feed gap to fill, and the median NPV difference is zero. From 1.6 to 2.4 LSU per hectare, $\phi = 1$ has the higher NPV in 95 to 100 percent of the pairs, and the median gain rises from $4,541 to $23,874. At 2.2 LSU per hectare, the median pair at $\phi = 1$ pays $12,848 for hay on pasture over ten years, raises the first-calf score at calving by 0.52 and the first-calf conception rate by 0.117, and sells $61,359 more calves and $21,608 fewer cull cows. Calf revenue is the number of calves sold times their weight and price. About two thirds of the added calf revenue come from 32 more calves sold over ten years, as the farm culls 17 fewer cows and its herd holds 1.6 more cows. The other third comes from calves that wean 5.6 kilograms heavier. Without hay on pasture, the first-calf cows calve below BCS 5, where the conception rate rises steeply with condition (Appendix E.2), so a small amount of hay keeps more cows in the herd. At 2.6 LSU per hectare, no simulation of the ungrazed history completes without hay, and the median complete simulation at $\phi = 1$ feeds 33.9 tonnes of hay per year on pasture. The heavy continuous history depends on hay: its highest NPV is $19,751 at $\phi = 0.25$, $70,102 at $\phi = 0.5$, $73,949 at $\phi = 0.75$, and $81,738 at $\phi = 1$.

### E.5. When calendar-based rotation pays

Appendix Table E4 compares the highest-NPV continuous grazing with the highest-NPV calendar-based rotation at each stocking rate of the ungrazed history, each over the five levels of feed-gap supplementation. Over all five histories, at 0.8 to 1.4 LSU per hectare, the NPV of the best rotation minus that of the best continuous grazing is −$4,390 to $4,800, and continuous grazing has the higher NPV in 15 of these 20 history and stocking-rate pairs. From 1.6 to 2.4 LSU per hectare, the best rotation has the higher NPV in all 25 pairs, by $318 to $21,854. From 2.6 LSU per hectare upward, it has the higher NPV in all 16 pairs in which a continuous-grazing simulation completes, by $24,197 to $124,760. At 3.2 LSU per hectare, continuous grazing completes only in the heavy continuous history.

Three stocking rates of the ungrazed history show the mechanism. At 1.0 LSU per hectare, the two have almost the same sales, and the rotation pays $2,211 more for fencing and $933 more for labor over ten years, so continuous grazing leads by $2,867. At 2.2 LSU per hectare, the best management systems of both regimes use $\phi = 1$. The rotation with two paddocks and a four-day grazing period feeds 3.0 tonnes of hay per year on pasture, against 8.1 tonnes under continuous grazing, which saves $12,312 over ten years. It raises the first-calf score at calving from 4.52 to 4.58, its calf sales are $5,877 higher, and its cull sales are $2,108 lower. After $2,211 more fencing and $5,379 more labor, the rotation leads

by $7,100. At 2.6 LSU per hectare, continuous grazing completes only with $\phi = 1$, and its best management system feeds 36.6 tonnes of hay per year on pasture at a ten-year cost of $88,828. The best rotation, four paddocks with a seven-day grazing period, feeds 21.1 tonnes per year at a cost of $51,202, raises the first-calf score at calving from 4.12 to 4.36, sells $27,068 more calves, and leads by $31,848. In the ungrazed history, the best rotation also loses less SOC than the best continuous grazing at every stocking rate at which both regimes complete, by 0.40 to 2.48 Mg C per hectare up to 2.4 LSU per hectare and by 5.90 to 13.75 Mg C per hectare from 2.6 LSU per hectare upward.

### E.6. How calendar-based rotation redistributes patch use

Figure E5 quantifies one spatial pathway of calendar-based rotation. On each grazing-season day, the whole herd occupies one 5-hectare patch, so every complete simulation allocates 1,510 patch-occupancy days to the 12 patches. Patch-use imbalance is the largest patch count divided by the smallest, and a ratio of one means that all patches are occupied equally often. The statistic counts days of occupancy and contains no information about cattle-days or the grazed area within a patch.

The paddock counts use different grazing periods: two to four paddocks use 1 to 30 days, and six and twelve paddocks use 1 to 7 days. The share of complete simulations also differs by stocking rate. A summary over all complete simulations would therefore compare different sets of management systems. Figure E5 instead holds the history, the feed-gap supplementation, and the grazing period fixed, so each point is one simulation. The stocking rate sets how unevenly continuous grazing uses the patches. In the figure, the imbalance of continuous grazing falls from 68.2 at 0.8 LSU per hectare to 2.52 at 2.2 LSU per hectare. In the score of equation (C1), the intake response rises with the biomass of the patch. At a low stocking rate, a grazed patch therefore keeps a high score and the herd stays on the preferred patches, while at a high stocking rate the herd depletes each patch faster and moves on. Every rotation is more even than continuous grazing up to 2.4 LSU per hectare, and more paddocks give more even use. At 0.8 LSU per hectare, the imbalance is 36.5 with two paddocks, 19.4 with four, 8.69 with six, and 1.05 with twelve. Three paddocks are less even than two in this figure, 54.4 at 0.8 LSU per hectare. From 2.6 LSU per hectare, every regime stays below 2.

Matched comparisons over all complete simulations give the same pattern. Against continuous grazing with the same history, feed-gap supplementation, and stocking rate, the median ratio of the imbalance is 0.41 with two paddocks, 0.55 with three, 0.27 with four, 0.19 with six, and 0.10 with twelve, and the rotation is more even in 93 to 100 percent of the pairs. The feed-gap supplementation has almost no effect: between $\phi = 1$ and $\phi = 0$ with all else equal, the median ratio is 1.00, and 80 percent of the pairs lie between 0.84 and 1.02. A 7-day grazing period gives a median of 1.38 times the imbalance of a 1-day

grazing period. The order of two and three paddocks depends on the grazing period and the stocking rate. Three paddocks are more even than two in 16 percent of the matched pairs, and in 33 percent of the pairs at 0.8 to 1.4 LSU per hectare with grazing periods of 10 to 30 days. Three complete simulations, each with three paddocks, a 30-day grazing period, and 0.8 LSU per hectare, never occupy one of their patches, so their ratio is infinite, and the comparisons leave them out. Every other complete simulation occupies every patch at least once. Because the outcomes combine forage, SOC, animal, and cost channels, the figure cannot isolate the effect of the redistribution, and the simulated cattle locations await field validation.

### E.7. Payment designs in detail

Table E3 and Figures E6 and E7 report the payment experiment of Section 3.4. With a two-sided outcome payment, the baseline decides who receives the payment but not the management of a participant. The qualifying rule decides whether a practice payment or a premium changes management.

With the own baseline, the outcome payment moves every farm at a rate of $14 per Mg C (heavy continuous history) to $44 per Mg C (intensive rotational history). At $50 per Mg C, the ungrazed and the two low-pressure farms move from 2.2 to 2.0 LSU per hectare with four paddocks and a three-day grazing period, and the heavy continuous farm moves from 2.0 to 1.6 LSU per hectare. The mean farm adds 301 Mg C for a payment of $13,232, or $44 per additional Mg C. At $128 per Mg C, every farm moves to 0.8 to 1.0 LSU per hectare and adds 1,063 Mg C on average for $118,576 ($112 per Mg C). The profit optimum earns nothing under this baseline, so no payment is windfall.

The start-stock baselines pay mainly the heavy continuous farm, whose SOC rises in every complete simulation (Section 5). With annual gains, the heavy continuous farm receives $46,256 at $28 per Mg C for a move from 2.0 to 1.8 LSU per hectare, and 90 percent of the payment over the five farms is windfall. The ungrazed and the two low-pressure farms lose SOC in their profit optima. Under the annual-gain rule, they do not change below $106 per Mg C (low-pressure continuous) or $376 to $394 per Mg C (the other two), and under the net rule they never join up to $400 per Mg C. The intensive rotational farm changes under the annual-gain rule from $110 per Mg C and joins the net rule only from $150 per Mg C. Over the band of observed prices, the start-stock designs cost $197 to $402 per additional Mg C. A farm that joins the net rule chooses the same management system as under the own baseline at every rate, as equation (B24) implies: the two baselines differ by a lump sum.

A practice payment for rotation alone changes no farm up to $300 per hectare per year, because every profit optimum already uses two paddocks. At the Michigan EQIP rates, it transfers $37,240 to $85,539 per farm and adds no SOC. With the cap of 2.0 LSU

per hectare, the ungrazed and the two low-pressure farms move from 2.2 to 2.0 LSU per hectare at $14 to $16 per hectare per year and add 201 to 220 Mg C each. The two farms with supplement already qualify, so 40 percent of the payment is windfall. With the cap of 1.6 LSU per hectare, all five farms move to 1.6 LSU per hectare at $22 to $58 per hectare per year and add 251 to 580 Mg C each. At the EQIP Pasture Standard rate, this costs $81 per additional Mg C with no windfall, and the Pasture Intensive rate buys the same SOC for $186 per Mg C. Above the switch prices, a higher rate adds only transfer.

A premium for rotation alone is a pure transfer at 0.3 percent of the calf price, $1,905 per farm, and raises the stocking rate at higher shares. From 1.65 to 2.85 percent, the ungrazed and the two low-pressure farms move from 2.2 to 2.4 LSU per hectare, because the premium rewards more calves, and they lose 205 to 317 Mg C more SOC per farm. With the cap of 2.0 LSU per hectare, the premium moves the same three farms as the practice payment at 1.15 to 1.25 percent, and at 3.0 percent the buyer pays $144 per additional Mg C. The cap of 1.6 LSU per hectare needs 2.2 to 5.85 percent, so within the observed band it moves only the heavy continuous and intensive rotational farms. The outcome-verified premium changes no farm up to 10 percent. The profit optimum qualifies by definition, and a move to a qualifying management system with more calf revenue would need a premium of at least 18 percent of the calf price.

At the equal budget of $29,527 per farm, with one level for all five farms, the practice payment with the cap of 1.6 LSU per hectare buys the most additional SOC, 460 Mg C per farm at $58 per hectare per year. The outcome payment with the own baseline buys 423 Mg C at $72 per Mg C, the two start-stock outcome payments 134 Mg C, and the practice payment with the cap of 2.0 LSU per hectare 125 Mg C. The premium with the cap of 1.6 LSU per hectare also buys 460 Mg C, at 5.85 percent of the calf price, above the observed band, and the buyer of the calves pays it. The practice payment and the premium for rotation alone buy none. The capped practice payment buys a discrete change of management and only has to cover its private cost, while the outcome payment pays the same rate on every additional Mg C, including the Mg C that the farm would add for less. Appendix Table E5 adds the threshold design of Rouhi Rad et al. (2026). At $50 per Mg C, a threshold at the no-payment change of each farm buys 1,130 Mg C over the five farms for $36,420 ($32 per Mg C), and the best uniform threshold for a similar target, at $150 per Mg C, buys 1,047 Mg C for $62,504 ($60 per Mg C). For about 1,600 Mg C, the farm-specific threshold costs $48 per Mg C and the best uniform threshold $223 per Mg C. At $100 per Mg C, the farm-specific threshold buys 427 Mg C per farm for $27,541 per farm, within the equal budget.

TABLE E3. Payment designs at the low and high ends of the observed payment levels. Each start history is one 60-hectare farm that chooses the complete management system with the highest NPV plus payment. The no-payment management system of a farm is its profit optimum, and the own baseline is the yearly SOC change of that management system. Changed is the number of the five farms that leave their no-payment management system, and SR is the planned stocking rate in LSU per hectare, as the range over the chosen management systems. Additional SOC is the ten-year SOC change of the chosen management system minus that of the no-payment management system, in Mg C per farm, and payment is the present value per farm in US dollars, both as means over the five farms. Cost is the payment per additional Mg C (n/a when no SOC is added). Windfall is the share of the payment that the farm would receive without changing its management. Outcome rates are in dollars per Mg C, practice rates in dollars per hectare per year for ten years, and premiums in percent of the calf price. The buyer of the calves pays the premium.

| Design | Level | Changed | SR | Additional SOC | Payment | Cost per Mg C | Windfall (%) |
|---|---|---|---|---|---|---|---|
| Outcome, own baseline | 28 | 2 | 1.8 to 2.2 | 61 | 1,572 | 26 | 0 |
| | 128 | 5 | 0.8 to 1.0 | 1,063 | 118,576 | 112 | 0 |
| Outcome, start stock, annual gains | 28 | 1 | 1.8 to 2.2 | 37 | 9,896 | 268 | 90 |
| | 128 | 3 | 0.8 to 2.2 | 238 | 66,945 | 282 | 61 |
| Outcome, start stock, net change | 28 | 1 | 1.8 to 2.2 | 37 | 9,251 | 251 | 90 |
| | 128 | 1 | 0.8 to 2.2 | 209 | 62,233 | 297 | 61 |
| Practice, rotation | 76.55 | 0 | 2.0 to 2.2 | 0 | 37,240 | n/a | 100 |
| | 175.84 | 0 | 2.0 to 2.2 | 0 | 85,539 | n/a | 100 |
| Practice, rotation and SR ≤ 2.0 | 76.55 | 3 | 2.0 | 125 | 37,240 | 299 | 40 |
| | 175.84 | 3 | 2.0 | 125 | 85,539 | 687 | 40 |
| Practice, rotation and SR ≤ 1.6 | 76.55 | 5 | 1.6 | 460 | 37,240 | 81 | 0 |
| | 175.84 | 5 | 1.6 | 460 | 85,539 | 186 | 0 |
| Premium, rotation | 0.3% | 0 | 2.0 to 2.2 | 0 | 1,905 | n/a | 100 |
| | 3.0% | 3 | 2.0 to 2.4 | −157 | 20,071 | n/a | 95 |
| Premium, rotation and SR ≤ 2.0 | 0.3% | 0 | 2.0 to 2.2 | 0 | 713 | n/a | 100 |
| | 3.0% | 3 | 2.0 | 125 | 17,987 | 144 | 40 |
| Premium, rotation and SR ≤ 1.6 | 0.3% | 0 | 2.0 to 2.2 | 0 | 0 | n/a | n/a |
| | 3.0% | 2 | 1.6 to 2.2 | 131 | 5,713 | 44 | 0 |
| Premium, SOC not below | 0.3% | 0 | 2.0 to 2.2 | 0 | 1,905 | n/a | 100 |
| profit optimum | 3.0% | 0 | 2.0 to 2.2 | 0 | 19,049 | n/a | 100 |

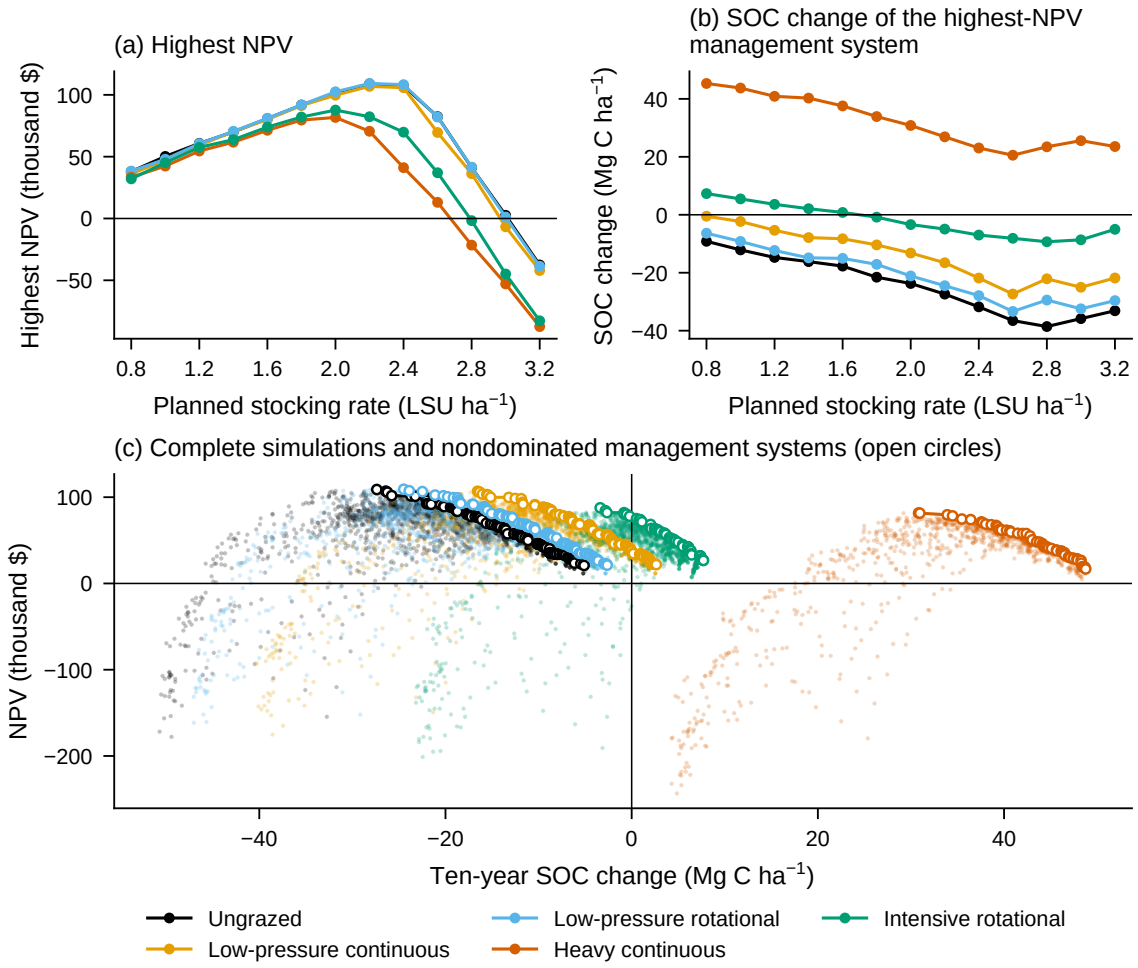



FIGURE E3. **Private NPV and soil-carbon change by start history.** Panel (a) gives the highest ten-year NPV among the complete simulations of each history at each planned stocking rate, over all paddock counts, grazing periods, and levels of feed-gap supplementation. Panel (b) gives the ten-year change of farm mean total SOC of the same management systems. Panel (c) plots every complete simulation. Open circles and lines mark the nondominated management systems of each history, for which no other complete management system of the same history has both a higher NPV and a smaller SOC loss. The histories start from different SOC stocks, so SOC is compared within a history.

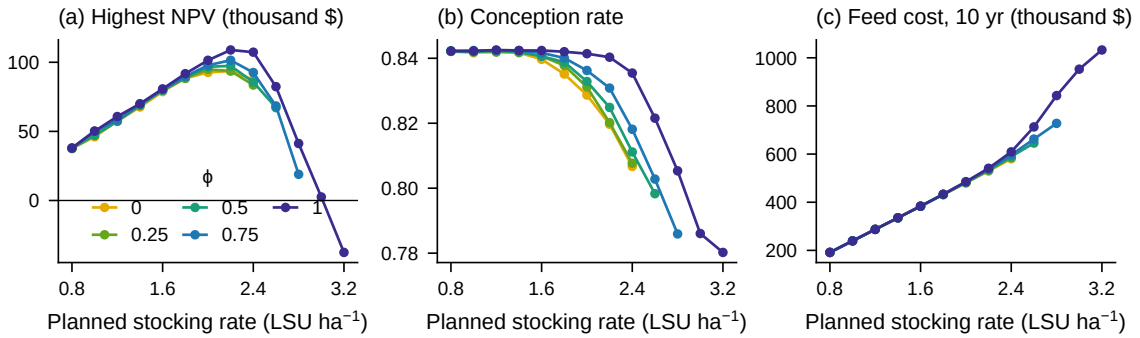



FIGURE E4. **Effect of the feed-gap supplementation, ungrazed start history.** Panel (a) gives the highest ten-year NPV among the complete simulations at each planned stocking rate and feed-gap supplementation $\phi$. Panels (b) and (c) give the medians over the complete simulations of the mean conception rate and of the undiscounted ten-year feed cost (winter hay, hay on pasture, and corn). A line ends at the highest stocking rate with a complete simulation.

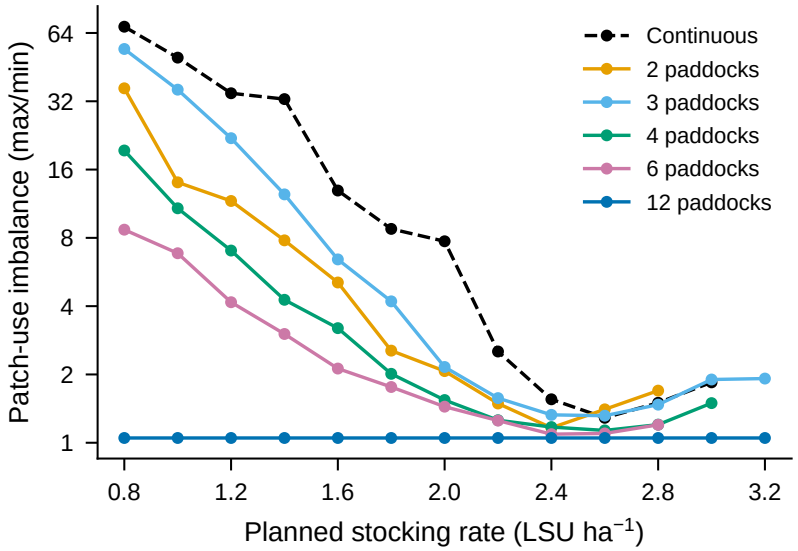



FIGURE E5. **Patch-use imbalance by stocking rate and paddock count.** Each point is one complete simulation of the ungrazed history with $\phi = 1$. Every rotation uses a 3-day grazing period, which all paddock counts share. The outcome is the ratio of the largest to the smallest ten-year patch-occupancy count, and one denotes equal use. The vertical axis is base-2 logarithmic. A missing point is a simulation that ends before ten years. Under calendar-based rotation, the calendar sets the active management paddock, and under both regimes the selective-grazing rule sets the occupied patch within the eligible area.

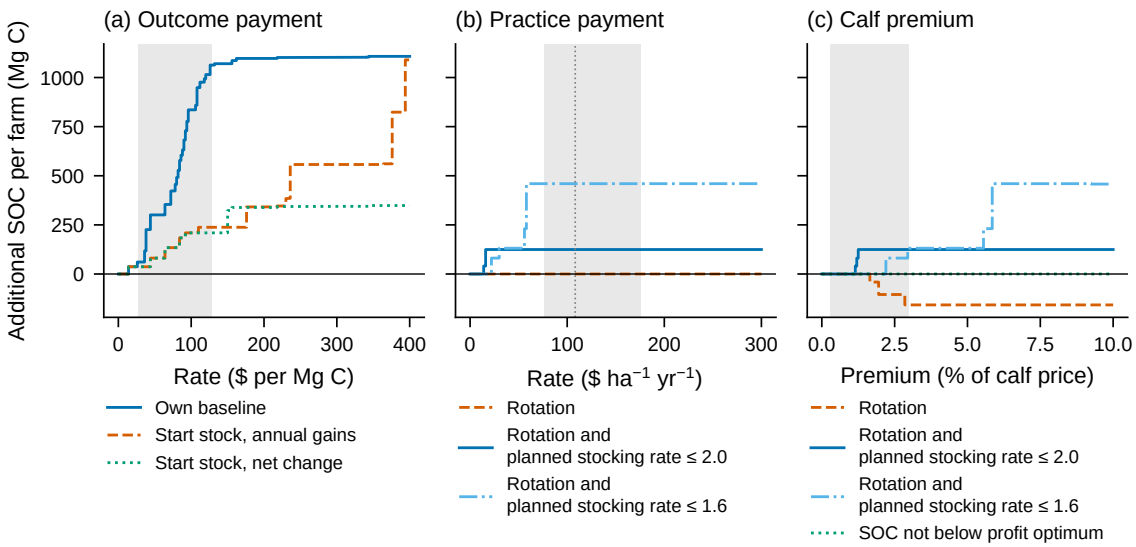



FIGURE E6. **Response of the five farms to the payment designs.** Each start history is one 60-hectare farm that chooses the complete management system with the highest NPV plus the present value of the payment. Lines give the mean additional SOC per farm: the ten-year SOC change of the chosen management system minus that of the farm's profit optimum, times 60 hectares. Gray bands mark the observed payment levels: $28 to $128 per Mg C in panel (a), the Michigan EQIP rates of $76.55 to $175.84 per hectare per year in panel (b), and 0.3 to 3.0 percent of the calf price in panel (c). The dotted line in panel (b) marks the CSP enhancement E528R rate of $108.25 per hectare per year.

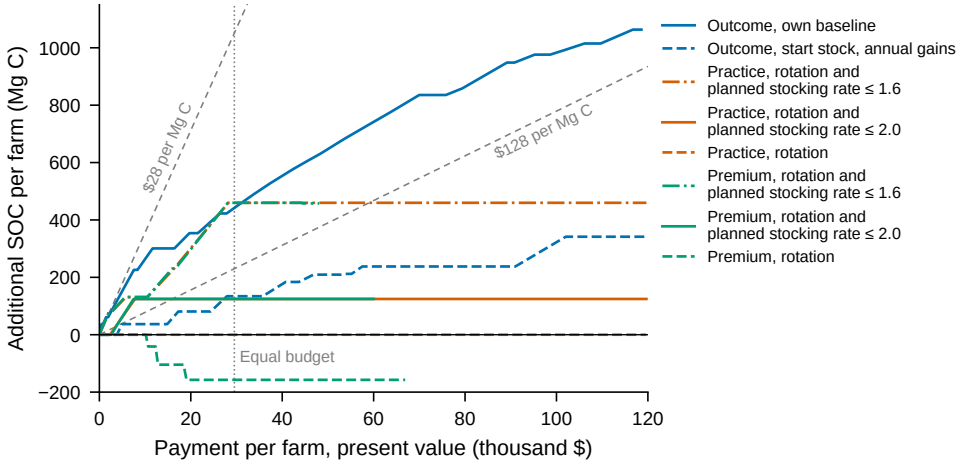



FIGURE E7. **Additional SOC against the payment.** Each line traces one design of Figure E6 as its level rises, with the mean present value of the payment per farm on the horizontal axis. The figure omits two designs: the outcome payment on the net change from the start stock and the outcome-verified premium. Dashed gray rays mark costs of $28 and $128 per additional Mg C, and the dotted line marks the equal budget of $29,527 per farm. The buyer of the calves pays the premium. With the same stocking-rate cap, the practice payment and the premium move the same farms to the same management systems, so their lines nearly overlap.

## E.8. Comparison tables

Table E4 compares continuous grazing with the highest-NPV calendar-based rotation at every stocking rate of the ungrazed start history. "CG" denotes continuous grazing. Each regime takes its highest NPV over the five levels of feed-gap supplementation, and the table gives the feed-gap supplementation $\phi$ of that management system. The calendar-based rotation is reported by paddock count $N$ and grazing period gp in days. The difference column is the rotation NPV minus the CG NPV and is negative when continuous grazing has the higher NPV.

TABLE E4. Highest-NPV continuous grazing (CG) and highest-NPV calendar-based rotation by planned stocking rate, ungrazed start history, each over the five levels of feed-gap supplementation $\phi$. SR is the planned stocking rate in LSU per hectare, NPV is in US dollars, and $\Delta$SOC is the ten-year change of farm mean total SOC in Mg C ha$^{-1}$. The difference is the rotation NPV minus the CG NPV. n/a marks a stocking rate at which no continuous-grazing simulation completes.

| | Continuous grazing (CG) | | | Highest-NPV rotation | | | | |
|---|---|---|---|---|---|---|---|---|
| SR | $\phi$ | NPV | $\Delta$SOC | Paddock design | $\phi$ | NPV | $\Delta$SOC | Difference |
| 0.8 | 1 | 37,949 | −9.15 | $N$ = 2, gp= 30 | 0.75 | 35,256 | −8.75 | −2,693 |
| 1.0 | 0.75 | 50,201 | −12.12 | $N$ = 2, gp= 20 | 1 | 47,335 | −11.43 | −2,867 |
| 1.2 | 1 | 60,752 | −14.72 | $N$ = 2, gp= 30 | 0.75 | 58,023 | −13.92 | −2,729 |
| 1.4 | 1 | 65,431 | −16.64 | $N$ = 2, gp= 10 | 1 | 70,011 | −16.16 | 4,581 |
| 1.6 | 0.75 | 79,248 | −19.51 | $N$ = 2, gp= 5 | 1 | 80,750 | −17.72 | 1,502 |
| 1.8 | 1 | 91,423 | −22.44 | $N$ = 2, gp= 10 | 1 | 91,741 | −21.54 | 318 |
| 2.0 | 1 | 94,006 | −25.32 | $N$ = 2, gp= 5 | 1 | 101,444 | −23.71 | 7,438 |
| 2.2 | 1 | 101,806 | −28.49 | $N$ = 2, gp= 4 | 1 | 108,906 | −27.38 | 7,100 |
| 2.4 | 1 | 93,462 | −34.26 | $N$ = 2, gp= 4 | 1 | 107,310 | −31.78 | 13,847 |
| 2.6 | 1 | 50,523 | −42.42 | $N$ = 4, gp= 7 | 1 | 82,370 | −36.52 | 31,848 |
| 2.8 | 1 | −19,883 | −47.13 | $N$ = 6, gp= 6 | 1 | 41,228 | −38.58 | 61,111 |
| 3.0 | 1 | −108,622 | −49.59 | $N$ = 12, gp= 4 | 1 | 2,556 | −35.84 | 111,178 |
| 3.2 | n/a | n/a | n/a | $N$ = 12, gp= 6 | 1 | −37,585 | −33.13 | n/a |

Table E5 reports the threshold variant of the outcome payment (Appendix B.13). The payment is $pA\max(0, \Delta\text{SOC} - \theta)$ for the ten-year SOC change, paid at the end of 2035. We evaluate uniform thresholds from −30 to 40 Mg C per hectare in steps of 2.5, thresholds at the no-payment change of each farm plus $k \in \{-5, -2.5, 0, 2.5, 5\}$ Mg C per hectare, and the rates \$10, \$25, \$50, \$75, \$100, \$150, \$200, \$300, and \$400 per Mg C. For each band of total additional SOC over the five farms, the table gives the design of each kind with the smallest payment. At small targets, a high uniform threshold reaches only the heavy continuous farm and costs as little as the farm-specific threshold. At larger targets, a uniform threshold either pays the heavy continuous farm for SOC that it gains without a change or sets a bar that the farms with SOC loss cannot reach.

TABLE E5. Outcome designs with a threshold on the ten-year SOC change, paid at the end of the contract: the design of each kind with the smallest payment for each band of total additional SOC over the five farms. A uniform threshold θ applies to every farm, and a history threshold is the no-payment SOC change of the farm plus $k$, both in Mg C ha$^{-1}$. Payment is the present value over the five farms in US dollars, and cost is the payment per additional Mg C.

| Threshold | θ or $k$ | Rate ($/Mg C) | Additional SOC (Mg C) | Payment | Cost per Mg C |
|---|---|---|---|---|---|
| History | 0.0 | 25 | 184 | 2,973 | 16 |
| Uniform | 32.5 | 50 | 184 | 2,705 | 15 |
| History | 5.0 | 100 | 694 | 25,383 | 37 |
| Uniform | 35.0 | 100 | 694 | 28,570 | 41 |
| History | 0.0 | 50 | 1,130 | 36,420 | 32 |
| Uniform | 37.5 | 150 | 1,047 | 62,504 | 60 |
| History | 0.0 | 75 | 1,599 | 77,296 | 48 |
| Uniform | −27.5 | 75 | 1,599 | 357,304 | 223 |
| History | 5.0 | 150 | 4,417 | 311,098 | 70 |
| Uniform | −10.0 | 200 | 2,711 | 665,792 | 246 |

### E.9. Limits of the animal, grazing, and payment models

The animal side has four limits. First, the animal-demand rules and the herd lifecycle are not validated. The simulated weights were compared with, but not calibrated to, a Michigan research herd: at weaning, the mature cows are 4.9 percent heavier and the calves 2.9 percent lighter than the herd, and the first-calf cows are 2.6 percent heavier than its two-year-old heifers. The model calves are born on day 75 and weaned at 195 days of age, about 20 days older than the calves of the herd at their October weighing, so they gain about 0.84 kilograms of shrunk weight per day against 0.97 kilograms per day in the herd. Second, every management system feeds the same winter program, which returns the mature cows to BCS 5 by calving, so the management comparison covers only the grazing season. Third, gain follows the energy supply, and a protein shortage does not limit it. Fourth, milk is an obligation of the cow, so the calf weight at weaning responds little to management, and a shortage of feed falls on the cows.

The grazing side has three limits. First, the selective-grazing rule scores a patch by its biomass and live-leaf share and not by the forage above the grazing residual. The herd can therefore stay on a grazed-down patch with no grazeable forage while other patches have forage. This rule contributes to the lower completion of continuous grazing and of long grazing periods. A rule that scores the grazeable forage, or that moves the herd when the occupied patch has none, would show how much of the comparison between the grazing regimes depends on it. Second, the crude protein of the plant forage eaten falls outside the training band of the surrogate on 24.4 percent of the grazing days and is clipped into it. Third, the patch-use measure counts only the days on which each patch is the herd's location. It omits the number of animals on that patch, the area grazed within it, behavioral differences among cattle, and forage-triggered movement between management paddocks, which follow the fixed calendar.

The payment experiment has three limits. First, the own baseline assumes that the program knows the no-payment management of each farm, so it is a full-information benchmark. A real program must estimate the no-payment path with a model and its error. Second, the profit optima lie at 2.0 to 2.2 LSU per hectare. A check that reruns 1,095 farm simulations of an earlier specification of the experiment in MEMS with the same daily actions shows that the surrogate overpredicts the forage of the grazed patch and understates the SOC loss, and that the error grows with the stocking rate: at 2.0 to 2.2 LSU per hectare, the surrogate's farm SOC at the end of 2035 exceeds the MEMS value by a median of 3.9 Mg C per hectare. The median difference is 2.3 Mg C per hectare at 1.6 to 1.8 LSU per hectare, where the cap of 1.6 lies, and −0.3 Mg C per hectare at 0.8 to 1.0. We conjecture that the surrogate therefore understates the additional SOC of a move to a lower stocking rate and overstates its private cost, which would make the switch prices too high, but we have not tested this. Third, the experiment has no transaction, measurement,

or verification costs, no demand for qualifying calves, and no SOC loss after the contract ends.